DEGREE PROJECT

AI Driven Threat Detection for Building Management Industrial Control Systems



Chalaka Haripriya Jayaweera
Harsha Chaturanga Thilakaratne Mudiyanselage

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Luleå University of Technology
Department of Computer Science, Electrical and Space Engineering

LULEÅ UNIVERSITY OF TECHNOLOGY

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AI-Driven Threat Detection for Building Management Industrial Control Systems

Master's Research

Chalaka Haripriya Jayaweera (chajay-4)

Harsha Chaturanga Thilakaratne Mudiyanselage (harthi-4)

Supervisor:	Saguna Saguna
Examiner:	Christer Åhlund
Credits:	30 credits

Abstract

Building Management Systems (BMS) control critical building operations such as heating, ventilation, access control, and energy management. As these systems adopt IP networking, IoT devices, and cloud services, their exposure to cyber threats grows significantly. Yet dedicated anomaly detection tools that are technically effective, explainable, and practical for BMS environments remain scarce in the literature. This thesis presents the design, implementation, and evaluation of the BMS Threat Intelligence and Detection Dashboard (BMS-TIDD), an explainable hybrid anomaly detection system built using a Design Science Research methodology. BMS-TIDD combines a Variational Autoencoder (VAE) for file-level structural anomaly detection, a Long Short-Term Memory (LSTM) network for temporal sequence detection, and an explainability layer using SHAP for global feature attribution and LIME for per-instance explanation. The system was evaluated on real BACnet network traffic from a live Siemens Desigo CC building management environment, covering seven simulated attack scenarios: BACnet WriteProperty injection, Ping Sweep, ARP spoofing, port scanning, SSH brute force, BACnet flooding, and coordinated DDoS.

At the file level, the hybrid VAE-LSTM system achieved perfect binary classification (precision, recall, and F1-score of 1.0000) across all twenty evaluation captures, with no labelled attack data required during training. At the window level, the LSTM component achieved a precision of 0.4242, recall of 0.4577, and F1-score of 0.4404, reflecting the inherent difficulty of classifying short temporal windows within intermittent anomaly captures. The full detection and explainability pipeline executed in under one second on standard CPU hardware, confirming suitability for real-time BMS deployment. SHAP analysis identified byte rate (byte_rate_bps) as the primary anomaly driver for the VAE component, and packet volume per bucket (n_pkts_bucket) as the primary driver for the LSTM component, reflecting each model's distinct detection perspective. LIME explanations produced instance-level attributions consistent with known BMS attack signatures. The BMS-TIDD dashboard presents anomaly alerts with plain-language feature explanations through a role-based interface aligned with NIST SP 800-82 guidelines.

The work contributes an explainable hybrid detection architecture tailored to BMS environments, a novel BACnet PCAP dataset with seven labelled attack scenarios, and empirical evidence for VAE-based one-class detection in operational BACnet networks. Key limitations include evaluation on a single BMS site, reduced LSTM window-level recall on intermittent anomaly traffic, and the exclusion of process-level sensor data and the non-completion of the formal usability evaluation, participant recruitment constraints precluded the domain-expert assessment sessions within the submission timeline (see Section 7.7). Future work should focus on multi-site generalisation, adaptive threshold calibration, and integration with enterprise OT security operations platforms.

Keywords: Building Management Systems (BMS); Anomaly Detection; Explainable Artificial Intelligence (XAI); Variational Autoencoder (VAE); Long Short-Term Memory (LSTM); Cyber-Physical Systems Security; Industrial Control Systems (ICS); Design Science Research

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Abbreviations

Abbreviation	Description
APT	Advanced Persistent Threat
BACnet	Building Automation and Control Networks
BAS	Building Automation System
BMS	Building Management System
DCS	Distributed Control System
DSR	Design Science Research
HMI	Human–Machine Interface
HVAC	Heating, Ventilation, and Air Conditioning
ICS	Industrial Control System
IDS	Intrusion Detection System
IoT	Internet of Things
IT	Information Technology
LIME	Local Interpretable Model-Agnostic Explanations
LSTM	Long Short-Term Memory
NIST	National Institute of Standards and Technology
OT	Operational Technology
PCAP	Packet Capture
PLC	Programmable Logic Controller
SCADA	Supervisory Control and Data Acquisition
SHAP	SHapley Additive exPlanations
VAE	Variational Autoencoder
XAI	Explainable Artificial Intelligence

1. Introduction

1.1 Background and Context

Building Management Systems (BMS), also referred to as Building Automation Systems (BAS), are cyber-physical infrastructures used to monitor and control essential building operations such as heating, ventilation, and air conditioning (HVAC), lighting, energy management, and access control. These systems integrate sensors, controllers, communication networks, and supervisory interfaces to maintain stable environmental conditions while supporting efficient building operation (Fisk, 2012; Peacock & Johnstone, 2014). In modern commercial and institutional facilities, BMS technologies play a critical role in maintaining occupant comfort, operational efficiency, and energy optimisation.

Traditionally, building automation environments were engineered primarily with reliability and operational continuity in mind rather than cybersecurity resilience. Early BMS architectures assumed limited connectivity and relied on physical network isolation to protect system integrity (Novak & Gerstinger, 2010; Peacock & Johnstone, 2014). Under these assumptions, many automation protocols and control devices were designed without strong authentication, encryption, or monitoring capabilities.

However, the rapid development of smart building technologies has significantly altered the technological landscape of building automation. Modern BMS environments increasingly integrate Internet Protocol (IP) networks, cloud-based analytics, Internet of Things (IoT) devices, and remote management interfaces to improve operational efficiency and enable advanced automation capabilities (Minoli, et al., 2017; Mazhar, et al., 2022). While these developments provide important benefits such as predictive maintenance, energy optimisation, and real-time monitoring, they simultaneously expand the attack surface of building systems.

The convergence of operational technology (OT) and enterprise information technology (IT) has therefore introduced new cybersecurity challenges. Building automation networks that were once isolated are now interconnected with corporate networks and external services. This integration exposes building systems to cyber threats that may exploit vulnerabilities in communication protocols, misconfigured network interfaces, or weak authentication mechanisms (Peacock & Johnstone, 2014; Li, et al., 2023).

Building Management Systems operate as cyber-physical systems; cybersecurity incidents may produce physical consequences rather than purely digital effects. Manipulation of sensor data, control commands, or network communication can disrupt environmental control processes and influence building operations. Such incidents may result in service disruption, equipment malfunction,

energy inefficiency, or compromised occupant comfort (Fisk, 2012; Li, et al., 2023).

These developments highlight the growing importance of effective monitoring and anomaly detection mechanisms capable of identifying abnormal behaviour within building automation environments. As building infrastructures continue to become more connected and complex, traditional preventive security measures alone are no longer sufficient. Instead, continuous monitoring and intelligent detection approaches are required to identify deviations from normal system behaviour and respond to potential threats.

1.2 Research Field and Motivation

The cybersecurity of Building Management Systems represents an emerging research field at the intersection of industrial control system security, cyber-physical systems, and intelligent infrastructure management. As smart building technologies become increasingly integrated with enterprise networks and cloud platforms, the need for effective monitoring and threat detection mechanisms has become more pressing.

Previous research has demonstrated that BMS infrastructures share many characteristics with Industrial Control Systems (ICS), including long operational lifecycles, deterministic control processes, and strict availability requirements (Stouffer, et al., 2014). These characteristics create unique security challenges that differ significantly from those encountered in conventional IT systems. In building environments, disruptions to digital infrastructure may directly affect physical processes such as environmental regulation, ventilation, or access control.

Several studies have shown that vulnerabilities within building automation networks can enable attackers to manipulate building operations. For example, unauthorised modification of HVAC setpoints may lead to energy inefficiency or occupant discomfort, while disruption of ventilation or safety-related systems may affect building functionality (Fisk, 2012; Peacock & Johnstone, 2014). These examples illustrate how cyber incidents in building automation systems can produce tangible operational consequences.

To address these risks, research has increasingly explored the use of machine learning techniques for detecting abnormal behaviour in cyber-physical environments. Machine learning models can analyse large volumes of operational data generated by sensors, controllers, and network traffic in order to identify deviations from expected system behaviour (Madupati, 2020; Khan, et al., 2022). In environments where labelled attack data is limited, unsupervised or semi-supervised approaches have been proposed to model normal behaviour and detect anomalies.

Despite their potential, many machine learning based anomaly detection systems operate as black box models that provide limited explanation of their decision

processes. In operational environments, this lack of transparency can reduce operator trust and hinder effective decision-making (Khan, et al., 2022). Building system operators must distinguish between anomalies caused by equipment faults, configuration changes, maintenance activities, or potential cyber incidents. Without interpretable explanations, anomaly alerts may be difficult to evaluate.

Another important challenge relates to the operational constraints of Building Management Systems. These systems operate continuously and must maintain stable control of physical processes. Detection mechanisms must therefore function with low latency and minimal computational overhead to avoid interfering with operational performance (Stouffer, et al., 2014).

Taken together, these challenges highlight the need for anomaly detection solutions that combine technical effectiveness with interpretability and operational feasibility. This research is motivated by the need to design and evaluate an explainable anomaly detection framework that supports both automated detection and human decision-making within Building Management System environments.

1.3 Problem Statement

Building Management Systems operate as complex cyber-physical environments responsible for controlling critical building infrastructure. As these systems increasingly integrate network connectivity, cloud services, and IoT devices, their exposure to cyber threats continues to grow (Minoli, et al., 2017; Li, et al., 2023). Despite this increasing exposure, many existing security mechanisms within BMS environments remain limited in their ability to detect and interpret abnormal system behaviour.

Traditional security approaches, including rule-based monitoring and signature-based intrusion detection, primarily rely on predefined patterns of known threats. While these methods can detect previously documented attacks, they are less effective against novel or evolving attack strategies. In addition, it may struggle to adapt to the dynamic operational behaviour of building systems (Gupta, 2025).

Machine learning techniques have therefore been proposed as a promising solution for anomaly detection in cyber-physical environments. These techniques enable the modelling of normal operational behaviour and the identification of deviations that may indicate faults or malicious activity (Madupati, 2020). However, the practical implementation of machine learning-based detection systems in BMS environments presents several challenges.

First, labelled cybersecurity datasets are often unavailable in operational building environments. Cyber incidents are relatively rare, and collecting realistic attack data may be difficult due to confidentiality and operational constraints (Moosavi, et al., 2024). As a result, detection systems must operate effectively on largely unlabeled datasets.

Second, many advanced machine learning models function as opaque systems that provide limited explanation of their predictions. In cyber-physical environments where human operators must interpret system alerts, lack of transparency can reduce trust and limit practical usability (Khan, et al., 2022).

Third, Building Management Systems operate under strict performance and reliability requirements. Detection systems must therefore operate with minimal latency and computational overhead to avoid disrupting building operations (Stouffer, et al., 2014).

Consequently, there is a need for an integrated approach that combines effective anomaly detection with explainable artificial intelligence techniques and operationally feasible system design. Addressing this gap requires a research methodology that supports both technological development and practical evaluation in realistic operational contexts.

2. Research Concepts, Questions, and Theoretical Framework

2.1 Key Concepts and Definitions

Understanding the key concepts used in this research is essential for establishing a common conceptual foundation for the study. Building Management Systems operate within complex cyber-physical environments where digital control mechanisms interact directly with physical infrastructure. Concepts such as Industrial Control Systems, anomaly detection, explainable artificial intelligence, and design science research, therefore, provide important theoretical grounding for the development of the proposed detection framework.

2.1.1 Industrial Control Systems (ICS)

Industrial Control Systems (ICS) refer to integrated technological infrastructures used to monitor and control physical processes across a wide range of critical sectors, including manufacturing, energy production, transportation, water treatment, and building automation (Stouffer, et al., 2014). These systems combine hardware devices, control software, and communication networks to enable automated monitoring and control of physical operations.

ICS environments typically consist of several hierarchical layers. At the supervisory level, Human–Machine Interfaces (HMIs) and supervisory control servers provide monitoring and management capabilities for system operators. The control layer includes devices such as Programmable Logic Controllers (PLCs) and Distributed Control Systems (DCS), which execute control logic and regulate system behaviour. At the field layer, sensors and actuators interact directly with physical processes by collecting environmental data and executing control commands (Stouffer, et al., 2014).

Building Management Systems represent a specialised implementation of Industrial Control Systems designed specifically for managing building infrastructure such as HVAC systems, lighting, energy management, and access control (Fisk, 2012; Peacock & Johnstone, 2014). Similar to other ICS environments, BMS infrastructures involve continuous monitoring of environmental variables and automated control of physical devices.

One defining characteristic of ICS environments is their cyber-physical nature. Digital control commands influence physical processes, while physical system states generate digital data that informs system behaviour. This bidirectional interaction introduces unique security challenges because cyber incidents may directly affect physical operations (Stouffer, et al., 2014).

Another distinguishing characteristic of ICS environments is the emphasis on **operational availability and reliability**. Many control systems operate continuously for extended periods and must maintain stable system behaviour even under abnormal conditions. Consequently, security mechanisms must be carefully designed to avoid interfering with real-time control processes or introducing additional system latency (Stouffer, et al., 2014; Novak & Gerstinger, 2010).

These characteristics are particularly relevant for Building Management Systems, where operational disruptions may affect occupant comfort, building safety, and energy efficiency.

2.1.2 Anomaly Detection in ICS

Anomaly detection refers to the process of identifying patterns in system behaviour that deviate from expected operational conditions. In cyber-physical environments such as ICS and BMS infrastructures, anomaly detection involves analysing system data streams including sensor readings, network traffic, and control commands to identify irregular behaviour that may indicate system faults or cyber intrusions (Madupati, 2020).

Traditional cybersecurity mechanisms in control environments have often relied on rule-based monitoring or signature-based intrusion detection systems. These approaches operate by comparing system activity against predefined rules or known attack signatures. While such methods can effectively detect previously identified attack patterns, they may struggle to identify novel or evolving threats that do not match existing signatures (Gupta, 2025).

Machine learning techniques have therefore emerged as a promising approach for anomaly detection in cyber-physical environments. Machine learning models can analyse large volumes of operational data to learn patterns of normal system behaviour and detect deviations from these patterns. This capability is particularly valuable in Building Management Systems, where operational behaviour may vary depending on environmental conditions, occupancy patterns, and system configurations.

Machine learning-based anomaly detection can be broadly classified into three categories. Supervised approaches rely on labelled datasets containing examples of both normal and malicious behaviour. Unsupervised approaches learn patterns of normal behaviour without requiring labelled attack data. Semi-supervised approaches combine elements of both, using limited labelled data alongside larger volumes of unlabelled operational data.

In many real-world BMS environments, labelled attack datasets are scarce because cybersecurity incidents occur relatively infrequently, and collecting realistic attack data may be difficult due to operational constraints. As a result, unsupervised anomaly detection techniques have been explored as an alternative

given constraints on labelled data availability. (Khan, et al., 2022; Elnour, et al., 2021).

However, anomaly detection in cyber-physical systems presents several challenges. Operational data may be highly heterogeneous, containing both network traffic information and sensor measurements from multiple devices. In addition, anomalies may arise from a variety of sources, including equipment faults, configuration errors, maintenance activities, or malicious manipulation. Effective anomaly detection systems must therefore distinguish between benign operational variation and potential security threats.

2.1.3 Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence (XAI) refers to a set of techniques designed to improve the interpretability and transparency of machine learning models. As machine learning systems become increasingly complex, particularly with the adoption of deep learning architectures, understanding the reasoning behind model predictions becomes more difficult.

Many advanced machine learning models operate as **black-box systems**, meaning that their internal decision processes are not easily interpretable by human users. While such models may achieve high predictive accuracy, their lack of transparency can create challenges in operational environments where human decision-makers must interpret model outputs (Khan, et al., 2022).

In cyber-physical environments such as Building Management Systems, explainability plays a particularly important role. Operators responsible for managing building infrastructure must evaluate anomaly alerts and determine appropriate response actions. Without clear explanations of why a system behaviour was classified as anomalous, operators may find it difficult to assess the relevance or severity of alerts.

Explainable AI techniques aim to address this challenge by providing insights into the factors contributing to model predictions. Methods such as Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) analyse the influence of individual input variables on model outputs, enabling users to understand which features contributed most strongly to a particular prediction (Khan, et al., 2022).

Integrating explainability into anomaly detection systems provides several operational benefits. It improves operator trust by making model reasoning transparent, enhances situational awareness by highlighting the key variables influencing anomalous behaviour, and supports incident investigation by enabling operators to identify potential root causes of detected anomalies.

In Building Management Systems, where operational context is critical, explainable detection mechanisms can therefore improve both the usability and effectiveness of anomaly detection systems (Khan, et al., 2022).

2.1.4 Design Science Research (DSR)

Design Science Research (DSR) is a research methodology that focuses on the development and evaluation of technological artefacts designed to address real-world problems. Unlike purely descriptive research approaches, DSR emphasises the creation of innovative solutions and the generation of knowledge through artefact design and evaluation (Jatkiewicz, 2025).

The design science paradigm is particularly well-suited for cybersecurity research, where practical technological solutions are often required to address complex operational challenges. Rather than analysing problems in isolation, DSR seeks to develop artefacts, such as frameworks, models, or systems, that improve existing conditions.

Design Science Research typically follows an iterative multi stage process: problem identification and motivation, where the research problem and its significance are established; design and development, where a technological artefact is constructed to address the problem; and evaluation, where the artefact is assessed for both technical effectiveness and practical usefulness (Peffers, et al., 2007).

In this study, the proposed artefact is the **Building Management System Threat Intelligence and Detection Dashboard (BMS-TIDD)**. This system integrates machine learning-based anomaly detection with explainability techniques and provides visualisation support for system operators.

By adopting a Design Science Research approach, the study aims to contribute both **practical solutions for BMS cybersecurity** and **theoretical insights into explainable anomaly detection in cyber-physical environments**.

2.2 Theoretical Framework

This research is grounded in several complementary theoretical perspectives that together support the design, development, and evaluation of the proposed anomaly detection artefact. The theoretical framework integrates Design Science Research methodology with conceptual perspectives from systems thinking and resilience engineering.

These perspectives provide a foundation for understanding how anomaly detection technologies can support improved monitoring, decision-making, and operational resilience in Building Management Systems.

2.2.1 SWOT Analysis

SWOT analysis provides a structured method for evaluating the internal and external factors influencing the adoption of technological solutions. The framework examines four dimensions: strengths, weaknesses, opportunities, and threats.

Within the context of Building Management System security, SWOT analysis can help evaluate the potential benefits and limitations of AI-based anomaly detection technologies.

Strengths include the ability of machine learning techniques to identify previously unseen anomalies and analyse complex operational data. Weaknesses may involve challenges related to data quality, model interpretability, and computational requirements.

Opportunities arise from the increasing availability of operational data in smart building environments and the growing adoption of intelligent infrastructure technologies. These developments create favourable conditions for applying machine learning techniques to cybersecurity monitoring.

Threats include evolving cyber threats targeting automation environments and the possibility of adversarial manipulation of machine learning systems.

Applying SWOT analysis in this research helps ensure that both technological and organisational factors are considered when designing the proposed detection framework.

2.2.2 Design Science Research (DSR)

Design Science Research forms the methodological foundation of this study. The DSR approach emphasises the development of innovative artefacts that address real-world problems while contributing to scientific knowledge (Jatkiewicz, 2025).

In the context of this research, DSR guides the development of the BMS-TIDD framework through iterative cycles of design, implementation, and evaluation. The artefact integrates machine learning models for anomaly detection with explainability mechanisms and an operator-oriented monitoring dashboard.

The evaluation phase assesses the artefact in terms of several criteria, including detection performance, interpretability, and operational usability. Through this structured process, DSR supports both rigorous academic investigation and practical technological development.

2.2.3 Systems Thinking

Systems Thinking provides a conceptual perspective that emphasises understanding complex systems as interconnected networks of interacting components. Rather than analysing individual components in isolation, this perspective considers relationships between system elements and the broader environment in which they operate.

Building Management Systems represent complex socio-technical systems involving technological infrastructure, human operators, organisational processes, and environmental factors (Ige, et al., 2024). Applying a systems thinking perspective helps ensure that anomaly detection solutions consider interactions between these elements.

For example, anomaly detection alerts must not only identify abnormal system behaviour but also provide information that can be interpreted by human operators responsible for managing building operations. Systems Thinking, therefore, supports the integration of technical detection mechanisms with human decision-making processes.

2.2.4 Resilience Engineering

Resilience engineering focuses on the ability of complex systems to maintain safe and stable operation under unexpected conditions. Rather than preventing all possible failures, resilience emphasises the capacity of systems to adapt, respond, and recover from disturbances.

In Building Management Systems, resilience involves the ability to detect anomalies, respond to potential threats, and maintain stable building operation despite unexpected events. Early detection of abnormal behaviour plays a critical role in enabling effective incident response.

Explainable anomaly detection systems can contribute to resilience by improving situational awareness and supporting informed decision-making by system operators (Khan, et al., 2022). When operators can understand the causes of anomalies and evaluate their potential impact, they are better equipped to implement corrective actions.

2.2.5 Framework Integration

The theoretical perspectives described above collectively form the conceptual foundation of this research.

Design Science Research provides the methodological structure guiding artefact development and evaluation. Systems Thinking ensures that the proposed solution considers interactions between technological infrastructure and human operators. Resilience Engineering links anomaly detection capability with the broader goal of maintaining reliable and secure building operations.

By integrating these perspectives, the research aims to develop an anomaly detection framework that not only identifies abnormal behaviour but also supports operational decision-making and enhances system resilience in Building Management Systems.

3. Research Questions

Building on the security challenges and methodological limitations identified in the preceding sections, this research is guided by one main research question and four sub-questions. Together, these questions address anomaly detection under data scarcity, integration of explainability, real-time operational feasibility, and evaluation of the proposed artefact in building management environments.

3.1 Main Research Question

How can an explainable AI-based anomaly detection approach be designed and evaluated to effectively support secure and reliable operation of Building Management Systems under real-world conditions?

This main research question focuses on the development of a practical anomaly detection framework that combines machine learning-based detection techniques with explainability mechanisms and operator-oriented system design. The objective is not only to detect anomalous behaviour but also to ensure that detection outputs are interpretable and useful for system operators responsible for managing building infrastructure.

3.2 Sub-Research Questions

To address the main research question, the study is guided by the following sub-research questions.

1. **How can anomaly detection be performed effectively in Building Management Systems where labelled attack data is limited or unavailable?**

Building Management Systems generate diverse operational data from multiple sources, including sensors, controllers, and network communication channels. This data often varies significantly depending on building configuration, operational schedules, and environmental conditions. In addition, labelled cybersecurity datasets are rarely available in real-world building environments. This research question, therefore, investigates how anomaly detection models can be designed to operate effectively in environments characterised by heterogeneous data and limited labelled examples (Madupati, 2020; Moosavi, et al., 2024).

2. **How can explainable artificial intelligence techniques be integrated into anomaly detection models to improve transparency and operator understanding?**

Machine learning models capable of detecting subtle behavioural anomalies may produce outputs that are difficult for human operators to interpret. In operational environments, anomaly alerts must be accompanied by clear explanations that enable operators to understand the factors contributing to detected anomalies. This research question explores how explainable artificial intelligence techniques, such as SHAP and LIME, can be integrated with anomaly detection models to improve transparency and usability (Khan, et al., 2022).

3. **How can anomaly detection systems be designed to operate within the real-time performance constraints of Building Management Systems?**

Building Management Systems are expected to operate continuously with minimal disruption to building operations. Detection mechanisms must therefore function with low latency and limited computational overhead. This research question focuses on the design considerations required to ensure that anomaly detection systems remain operationally feasible within the performance constraints of BMS infrastructures (Stouffer, et al., 2014).

4. **How can the usability and operational value of an explainable anomaly detection system be evaluated in building management environments?**

Beyond detection accuracy, anomaly detection systems must also provide practical value in real operational environments. This includes usability, interpretability, and support for operator decision-making. This research question investigates how the proposed anomaly detection artefact can be evaluated using both technical performance metrics and qualitative assessments of usability and interpretability (Benjamin, et al., 2024; Jatkiewicz, 2025; Khan, et al., 2022).

3.3 Summary

Together, these research questions guide the design, implementation, and evaluation of the proposed anomaly detection framework for Building Management Systems. By addressing both technical detection capability and human-centered interpretability, the research aims to develop a solution that improves situational awareness and operational decision-making in cyber-physical building environments.

4. Literature Review

4.1 Introduction to the Literature Review

The cybersecurity landscape of Building Management Systems (BMS) has undergone a significant transformation due to the digital convergence of operational technology (OT), enterprise information technology (IT), and Internet of Things (IoT) infrastructures. Historically, building automation systems operated as isolated control environments designed primarily to regulate heating, ventilation, air conditioning (HVAC), lighting, access control, and energy optimisation. These systems were developed with emphasis on availability, operational continuity, and deterministic performance rather than cybersecurity resilience (Stouffer, et al., 2014). However, increasing connectivity to enterprise networks, remote management interfaces, and cloud-based services has fundamentally altered their exposure profile, introducing new cybersecurity risks that were not anticipated during their original design.

Peacock & Johnstone (2014) observe that many Building Automation Systems (BAS), which form the operational backbone of BMS infrastructures, are now connected to corporate IT networks and, in some cases, directly accessible from the internet. While this connectivity enhances operational efficiency, enabling centralised monitoring, predictive maintenance, and energy optimisation, it simultaneously expands the attack surface. As building systems become integrated with broader digital ecosystems, vulnerabilities arising from legacy protocols, inadequate authentication mechanisms, and insufficient network segmentation become increasingly exploitable.

The architectural and operational foundations of BMS are rooted in Industrial Control Systems (ICS), which manage physical processes through distributed sensing, control logic, and actuator response. ICS environments differ fundamentally from conventional IT systems due to their cyber-physical characteristics. Digital commands influence physical processes such as temperature regulation and environmental control, while physical state changes generate digital telemetry data that informs system behaviour. These bidirectional interactions create complex dependencies where cybersecurity incidents may produce real-world physical consequences rather than solely informational compromise (Stouffer, et al., 2014).

The emergence of smart building technologies has further intensified this complexity. Minoli et al. (2017) describe next-generation smart buildings as integrated cyber-physical environments combining IoT sensors, distributed controllers, and cloud-based analytics. While these technologies enhance operational efficiency and sustainability, they introduce heterogeneous communication channels and expand potential points of compromise. The

convergence of IT and OT domains, therefore, represents both a functional advancement and a structural security challenge (Siam, et al., 2025).

Traditional security mechanisms in building automation environments have relied heavily on preventive measures such as network segmentation, access control, and perimeter defence. However, these mechanisms are increasingly insufficient in environments where attackers may exploit unknown vulnerabilities or operate within authorised network boundaries. Consequently, behavioural anomaly detection has emerged as a promising approach for identifying deviations from normal system operation without relying solely on predefined attack signatures (Stouffer, et al., 2014; Peacock & Johnstone, 2014).

At the same time, the increasing adoption of machine learning and deep learning techniques for anomaly detection introduces new challenges related to interpretability, operational trust, and governance alignment. Detection models capable of identifying subtle behavioural anomalies may operate as opaque systems, limiting their practical usability in operational environments. Explainable Artificial Intelligence (XAI) techniques seek to address this limitation by providing interpretable insights into model decisions, thereby improving operator trust and facilitating informed incident response (Khan, et al., 2022).

This literature review synthesises research across six interconnected domains: ICS and BMS architecture, cyber-physical threat evolution, anomaly detection methodologies, explainable artificial intelligence, cybersecurity governance frameworks, and integrated conceptual models. Rather than merely summarising prior work, this review critically analyses relationships between technological developments, threat patterns, and detection strategies. The objective is to identify gaps in existing research and establish a theoretical foundation for the development of an explainable hybrid anomaly detection framework tailored to Building Management Systems.

A structured literature review provides a systematic foundation for identifying research gaps and supporting the development of theoretical and methodological frameworks for subsequent research (Leedy & Ormrod, 2015).

4.2 Industrial Control Systems and Building Management Systems: Architecture and Security Context

4.2.1 Architectural Foundations of ICS and BMS

Building Management Systems operate as specialised implementations of Industrial Control Systems adapted for smart building environments. ICS architectures typically follow a hierarchical layered model comprising supervisory, control, and field levels. This structure reflects functional separation between monitoring, decision-making, and physical interaction components.

At the supervisory layer, servers and Human–Machine Interfaces (HMIs) aggregate system data and enable administrators to monitor and configure building operations. These interfaces often connect with enterprise IT infrastructure to support reporting, analytics, and remote management. Gasser et al. (2017) demonstrate that publicly accessible building automation interfaces remain discoverable through internet scanning techniques, highlighting how inadequate network segmentation can expose critical control interfaces.

The control layer consists of programmable logic controllers (PLCs) and building controllers responsible for executing real-time control logic. These devices process sensor input and issue commands to actuators, maintaining environmental stability. Due to strict real-time constraints, introducing computationally intensive security mechanisms at this layer may interfere with operational performance, necessitating careful architectural placement of monitoring solutions (Novak & Gerstinger, 2010).

The field layer includes sensors and actuators that directly interact with physical processes. Sensors measure environmental variables such as temperature, humidity, and occupancy, while actuators adjust physical systems accordingly. Communication between field devices and controllers often relies on specialised protocols such as BACnet and KNX. Granzer et al. (2010) note that many building automation protocols were originally designed with limited security considerations, relying on assumptions of network isolation and trusted environments. However, these sensors represent primary sources of operational data; their integrity is critical for reliable system behaviour. Research has shown that analysing sensor data using machine learning-based anomaly detection techniques can help identify abnormal patterns or tampering attempts within IoT-enabled cyber-physical environments (Pathak, et al., 2021).

These architectural characteristics create inherent security challenges. Vulnerabilities at any layer may propagate across the system due to cyber-physical interdependencies. Compromise of supervisory interfaces may enable manipulation of control logic, while exploitation of field-level communication protocols may allow injection of falsified sensor data.

4.2.2 Cyber-Physical Characteristics and Operational Constraints

A defining feature of BMS infrastructures is their cyber-physical nature. Digital control systems regulate physical processes, and physical state changes influence digital system behaviour. This bidirectional interaction creates complex dependencies where cybersecurity incidents may directly affect environmental conditions and operational safety.

Fisk (2012) emphasises that intelligent building systems must balance operational efficiency with resilience against digital threats. Unlike conventional

IT systems, temporary system disruption may result in physical consequences such as equipment damage, energy inefficiency, or occupant discomfort.

Operational constraints further complicate cybersecurity implementation. ICS environments prioritise availability and reliability, often operating continuously for extended periods. Stouffer et al. (2014) highlight that security mechanisms must avoid introducing latency or instability that could interfere with control processes. This requirement limits the feasibility of deploying computationally intensive detection mechanisms directly within resource-constrained field devices.

Additionally, building automation systems typically have long operational lifecycles, often exceeding several decades. Legacy devices may lack modern security capabilities, and replacing or upgrading infrastructure may be prohibitively expensive. As a result, many BMS environments consist of heterogeneous components with varying security capabilities, creating uneven protection levels. Early Industrial Control System cybersecurity research emphasised the importance of balancing security implementation with system performance and operational reliability, highlighting challenges associated with integrating security mechanisms into operational control environments (Falco, et al., 2004).

4.2.3 IT–OT Convergence and Expanding Attack Surface

The convergence of IT and OT domains represents one of the most significant structural changes affecting modern BMS security. Enterprise IT systems increasingly integrate with building automation platforms to support remote monitoring, energy optimisation, and predictive maintenance. While this integration enhances functionality, it introduces new pathways for cyber intrusion.

Minoli et al. (2017) describe smart buildings as interconnected cyber-physical ecosystems combining IoT sensors, cloud analytics, and enterprise integration. These architectures enable advanced automation but simultaneously increase system complexity and attack surface.

Attackers may exploit vulnerabilities in enterprise IT systems to gain access to operational networks through lateral movement. Conversely, vulnerabilities in operational devices may expose enterprise systems. Risk analyses of smart home environments demonstrate that interconnected device ecosystems amplify overall vulnerability through component interdependencies (Jacobsson, et al., 2016), a pattern that scales to larger smart building deployments.

Public exposure of building automation systems remains a persistent concern. Gasser et al. (2017) identified numerous publicly accessible BMS interfaces, illustrating how misconfiguration and inadequate security controls can expose critical infrastructure.

These developments highlight a fundamental shift from isolated control environments to interconnected cyber-physical systems. Traditional perimeter-based security models are insufficient in environments where internal compromise

and lateral movement are plausible attack strategies. Continuous monitoring and behavioural anomaly detection, therefore, become essential components of modern BMS cybersecurity architectures.

4.3 Evolution of Cyber-Physical Threats in ICS and Building Management Systems

4.3.1 From Isolated Systems to Persistent Threat Environments

Historically, ICS and BMS environments were assumed to operate within physically secure and isolated networks. This assumption, commonly referred to as air-gapping, provided implicit protection against external cyber threats. However, increasing connectivity has eroded this protection, exposing building automation systems to a broader range of cyber adversaries (Stouffer, et al., 2014).

Peacock & Johnstone (2014) demonstrate that many building automation systems now operate over IP-based networks and integrate with enterprise infrastructure. As a result, attackers may exploit conventional network vulnerabilities to gain access to operational systems.

Early cyber incidents affecting ICS environments often involved denial-of-service (DoS) attacks, malware propagation, and unauthorised access. These attacks typically produced observable system disruption, enabling relatively straightforward detection. However, modern attack strategies increasingly prioritise stealth and persistence rather than immediate disruption (Stouffer, et al., 2014).

Advanced Persistent Threats (APTs) represent a significant evolution in adversarial capability. Rather than disrupting systems immediately, attackers may establish a long-term presence, conduct reconnaissance, and manipulate system behaviour gradually. Barrère et al. (2023) demonstrate that cyber-physical attack paths may propagate across interconnected system layers, combining digital and physical vulnerabilities.

4.3.2 Integrity Attacks and False Data Injection

Integrity attacks represent a particularly concerning threat category in cyber-physical systems. Instead of disrupting system availability, attackers manipulate data to alter system perception and behaviour. False data injection attacks involve modifying sensor readings or control commands to produce unintended system behaviour (Li, et al., 2023).

Granzer et al. (2010) highlight structural weaknesses in building automation protocols that may facilitate such attacks. Limited authentication and encryption mechanisms allow attackers to intercept and modify communication.

False data injection may enable subtle manipulation of environmental control systems. For example, falsified temperature readings may cause HVAC systems to

operate inefficiently or fail to maintain appropriate environmental conditions. Because manipulated values may remain within plausible ranges, detection using static thresholds becomes difficult.

Li et al. (2023) emphasise that cyber-physical attacks may exploit architectural and protocol-level vulnerabilities across multiple system layers. Detection strategies must therefore analyse behavioural patterns rather than relying solely on static rules.

4.3.3 Availability and Network-Based Attacks

Availability remains a critical requirement in ICS environments. Disruption of communication between controllers and field devices may affect system stability and environmental control. Denial-of-service attacks targeting network infrastructure may prevent controllers from communicating with sensors and actuators (Stouffer, et al., 2014).

Gupta (2025) identifies denial-of-service attacks, malware, and spoofing as persistent threats affecting ICS environments. Machine learning-based detection methods have been specifically applied to identify distributed denial-of-service attacks in building management systems, demonstrating the effectiveness of AI-based detection approaches in cyber-physical environments (Avci & Koca, 2023). While availability attacks may be more visible than integrity attacks, they can still disrupt critical building functions.

However, modern attackers increasingly prioritise stealth over disruption. Gradual manipulation of system parameters may produce long-term operational impact without triggering immediate alarms.

4.3.4 Insider Threats and System Complexity

Insider threats represent an additional risk factor in BMS environments. Authorised personnel may inadvertently misconfigure systems or intentionally exploit access privileges. Governance weaknesses may amplify technical vulnerabilities, particularly in environments lacking structured cybersecurity policies (Peacock & Johnstone, 2014).

Increasing system complexity further complicates security. Smart buildings integrate heterogeneous devices, proprietary protocols, and cloud services (Mazhar, et al., 2022). Minoli et al. (2017) note that this complexity introduces fragmented security architectures, increasing the likelihood of configuration gaps and unaddressed security risks.

These developments highlight limitations of traditional security mechanisms. Signature-based detection systems rely on predefined attack patterns and may fail to detect novel or adaptive threats. Consequently, alternative approaches such as behavioural anomaly detection have gained attention, particularly in cyber-physical

systems where attacks may manifest as subtle deviations in operational behaviour rather than clearly identifiable signatures.

4.4 Anomaly Detection in Industrial Control Systems and Building Management Systems

4.4.1 Detection Paradigms in ICS Security

The evolution of cyber-physical threats in Industrial Control Systems (ICS) and Building Management Systems (BMS) has necessitated corresponding advancements in detection strategies. Traditional cybersecurity approaches relied primarily on preventive mechanisms such as network segmentation, access control, and signature-based intrusion detection. While these methods remain relevant, they are increasingly insufficient in environments characterised by dynamic system behaviour, heterogeneous devices, and evolving threat techniques (Stouffer, et al., 2014).

Detection approaches in ICS security can be broadly categorised into three paradigms: signature-based detection, rule-based monitoring, and behavioural anomaly detection. Each paradigm reflects distinct assumptions regarding system behaviour and threat identification.

Signature-based detection operates by comparing observed activity against predefined attack signatures. This method provides reliable detection for known threats but cannot identify novel or previously unseen attack patterns. Gupta (2025) notes that signature-based systems are inherently reactive, relying on prior knowledge of attack behaviour. In environments where attackers employ adaptive or stealthy techniques, reliance on static signatures introduces critical detection gaps.

Rule-based monitoring extends this approach by defining acceptable operational thresholds. Alerts are generated when system parameters exceed predefined limits. While computationally efficient and interpretable, rule-based systems struggle to accommodate legitimate operational variability. Building systems operate under dynamic environmental conditions influenced by weather, occupancy, and equipment ageing. Static thresholds may generate excessive false positives or fail to detect gradual manipulation (Gupta, 2025).

Behavioural anomaly detection represents a shift toward modelling baseline system behaviour and identifying deviations from learned patterns. Rather than relying solely on predefined signatures or thresholds, anomaly detection systems analyse relationships between system variables to identify irregular behaviour (Shang, et al., 2024; Zakariah, et al., 2025). In IoT-enabled cyber-physical environments, machine learning algorithms can analyse continuous sensor data streams to detect deviations from expected operational patterns, allowing the identification of sensor manipulation or abnormal device behaviour (Pathak, et al.,

2021). Li et al. (2023) emphasise that behavioural anomaly detection is increasingly recognised as a critical component of ICS cybersecurity due to its ability to identify previously unknown attack patterns.

This paradigm is particularly relevant in BMS environments, where cyber-physical attacks may manifest as subtle manipulation of sensor data or control logic. Behavioural anomaly detection enables the identification of deviations that may not correspond to known attack signatures but nevertheless indicate compromise (Li, et al., 2023).

4.4.2 Machine Learning Approaches to Anomaly Detection

The increasing availability of operational data in smart building environments has facilitated the adoption of machine learning techniques for anomaly detection (Mazhar, et al., 2022; Avci & Bidollahkhani, 2023). Building systems generate continuous streams of telemetry data from sensors, controllers, and network infrastructure. These data streams exhibit complex temporal and spatial relationships that are difficult to model using traditional rule-based approaches.

Machine learning techniques enable automated modelling of baseline behaviour and identification of deviations indicative of anomalies. These techniques are particularly valuable in environments where large volumes of sensor data are continuously generated, as machine learning models can detect subtle behavioural deviations that may indicate sensor tampering or malicious interference in IoT-enabled cyber-physical systems (Pathak, et al., 2021). Gupta (2025) identifies several machine learning algorithms applied in ICS anomaly detection, including Support Vector Machines, Random Forests, Convolutional Neural Networks, and Long Short-Term Memory networks. These models can capture non-linear relationships and identify patterns that may not be apparent through manual analysis. Machine learning techniques have been increasingly adopted in Industrial Control System cybersecurity to enable proactive anomaly detection, behavioural modelling, and automated threat identification, improving detection capability compared to traditional rule-based approaches (Madupati, 2020).

Machine learning approaches can be categorised into supervised and unsupervised methods. Supervised learning models require labelled datasets containing examples of both normal and anomalous behaviour. While these models may achieve high classification accuracy, obtaining representative labelled datasets in operational BMS environments is challenging. Cybersecurity incidents are relatively rare, and generating realistic attack data in operational systems may be impractical or unsafe (Moosavi, et al., 2024).

Unsupervised learning approaches address this limitation by modelling normal system behaviour without requiring labelled attack data. These models identify anomalies based on deviations from learned behavioural patterns. This approach is particularly suitable for BMS environments, where normal operational data is

abundant, and attack patterns may be unknown or evolving (Ren, et al., 2023; Shang, et al., 2024).

Despite their advantages, machine learning models introduce challenges related to interpretability, data quality, and operational integration. Detection systems must operate within the constraints of cyber-physical environments and support human decision-making processes.

Machine learning-based analysis of network traffic data has demonstrated effectiveness in detecting abnormal communication patterns and identifying potential cybersecurity threats in operational environments (Graftceva, 2024).

4.4.3 Time-Series Modelling Using Long Short-Term Memory Networks

Building Management Systems generate time-dependent data reflecting continuous system operation. Sensor readings, control commands, and network activity evolve over time, influenced by environmental conditions, operational schedules, and system interactions. Effective anomaly detection, therefore, requires modelling temporal dependencies rather than analysing individual data points in isolation. Continuous monitoring of such time-series sensor data provides an opportunity for anomaly detection models to identify behavioural deviations that may signal cyber-physical attacks or manipulation of sensor readings in smart building infrastructures (Moosavi, et al., 2024).

Long Short-Term Memory (LSTM) networks provide a specialised neural network architecture designed to model sequential data and capture temporal dependencies within cyber-physical systems. These networks enable modelling of evolving operational behaviour and identification of anomalies that develop gradually over time. In Building Management System environments, temporal modelling is essential because sensor readings, controller commands, and environmental conditions exhibit structured sequential patterns influenced by operational schedules and system interactions, making them particularly suited for detecting anomalies that develop gradually over time, thereby enabling the modelling of long-term dependencies and sequential relationships (Gupta, 2025) (Fährmann, et al., 2022; Ogunseyi & Thiyagarajan, 2025; Wei, et al., 2022).

In BMS environments, temporal modelling is essential for detecting gradual anomalies that may not be apparent at individual time steps. For example, incremental manipulation of temperature setpoints may remain within acceptable ranges at each observation but exhibit anomalous trends over time. LSTM networks can identify such temporal patterns by analysing sequences of observations rather than isolated values (Wei, et al., 2022; Barrère, et al., 2023).

Gupta (2025) highlights the effectiveness of LSTM networks in ICS anomaly detection due to their ability to capture temporal dependencies. Barrère et al. (2023)

further demonstrate that cyber-physical attack sequences often unfold across multiple stages, reinforcing the importance of sequential modelling.

However, while LSTM networks provide strong temporal modelling capability, they primarily focus on sequence relationships and may not fully capture complex multi-variable structural relationships present in high-dimensional building system data. This limitation motivates integration with complementary modelling approaches. Despite their strength in modelling temporal dependency, LSTM-based approaches are not universally optimal in cyber-physical anomaly detection. Their performance may be affected by training data quality, sequence length selection, and sensitivity to changing operational regimes. In BMS contexts, where occupancy patterns, seasonal effects, and equipment behaviour may shift over time, a model trained on one operational pattern may generalise poorly to another. LSTM models also provide limited native interpretability, which reduces their transparency for operators who must assess whether temporal deviations reflect genuine compromise or benign operational fluctuation (Fährmann, et al., 2022).

4.4.4 Variational Autoencoders for Anomaly Detection

Autoencoders represent a class of neural networks designed to learn compressed representations of input data through reconstruction. During training, the model learns to reconstruct normal system behaviour with minimal error. Anomalies are subsequently detected based on reconstruction error, with higher errors indicating deviations from learned patterns.

Variational Autoencoders (VAEs) extend traditional autoencoder architectures by incorporating probabilistic latent space modelling, enabling representation of complex system behaviour and variability (Liu, et al., 2024; Ren, et al., 2023). This probabilistic approach enhances the ability to model high-dimensional operational data and detect deviations from normal system behaviour. In smart building environments, VAEs provide effective modelling of multi-variable relationships across sensor and control data, supporting anomaly detection in cyber-physical systems (Liu, et al., 2024; Ren, et al., 2023). Rather than learning deterministic representations, VAEs model latent variables as probability distributions. This probabilistic framework enhances generalisation and enables modelling of complex data distributions.

In the context of BMS anomaly detection, VAEs offer several advantages. Building systems generate high-dimensional data involving multiple interdependent variables, including temperature, humidity, airflow, occupancy, and energy consumption. VAEs can learn latent representations capturing relationships between these variables, enabling identification of anomalies that deviate from normal system behaviour (Liu, et al., 2024; Ren, et al., 2023).

Probabilistic latent modelling enables VAEs to capture variability in normal system behaviour while remaining sensitive to structural deviations.

However, VAEs typically operate on fixed input windows and do not inherently model temporal dependencies across extended sequences. As cyber-physical anomalies may develop gradually over time, relying solely on structural modelling may limit detection capability (Fährmann, et al., 2022).

This limitation highlights the complementary strengths of VAE and LSTM architectures. VAEs provide structural modelling capability, while LSTM networks capture temporal dependencies.

4.4.5 Hybrid VAE–LSTM Architectures for Cyber-Physical Anomaly Detection

The usefulness of VAEs depends on the assumption that anomalous events can be distinguished through deviations in latent reconstruction behaviour. In complex BMS environments, this assumption may not always hold consistently, especially where normal operating conditions are themselves highly variable. As a result, reconstruction-based methods may produce false positives or ambiguous results when unusual but legitimate states are encountered. This means that VAEs offer strong structural modelling capability, but their effectiveness remains dependent on the representativeness and stability of the training distribution.

Together, these limitations highlight the complementary strengths of VAE and LSTM architectures, motivating their integration into a hybrid detection approach suited to the structural and temporal complexity of Building Management Systems.

Hybrid anomaly detection architectures combining Variational Autoencoders and Long Short-Term Memory networks integrate structural and temporal modelling capabilities. This approach addresses limitations inherent in individual models and provides enhanced detection capability (Fährmann, et al., 2022).

In hybrid architectures, the VAE component learns compressed probabilistic representations of input data and generates reconstruction-based anomaly scores. The LSTM component models temporal relationships across sequences of latent representations or reconstruction errors. This integration enables the detection of both structural anomalies and temporal deviations (Fährmann, et al., 2022).

Hybrid architectures are particularly suited to BMS environments due to the cyber-physical nature of building systems. Operational behaviour involves both complex multi-variable relationships and time-dependent dynamics. Structural modelling captures relationships between system variables, while temporal modelling identifies sequential anomalies (Fährmann, et al., 2022).

Gupta (2025) highlights the potential of hybrid deep learning architectures for ICS anomaly detection. Barrère et al. (2023) demonstrate that cyber-physical attacks may unfold across multiple stages, reinforcing the importance of modelling sequential dependencies.

Despite their theoretical advantages, hybrid VAE–LSTM architectures remain underexplored in BMS-specific contexts. Much existing research focuses on

industrial manufacturing datasets or network intrusion detection rather than building automation systems. Li et al. (2023) note that BAS security research often lacks integration between detection models and operational environments.

Furthermore, deep learning models introduce interpretability challenges. Detection systems must provide actionable insights to support operational decision-making. Without interpretability, anomaly detection models may generate alerts that operators cannot effectively evaluate.

This limitation highlights the importance of integrating explainable artificial intelligence techniques with anomaly detection models to improve transparency and operational usability.

Although hybrid VAE–LSTM architectures are attractive because they combine structural and temporal modelling, this integration also increases architectural complexity, training demands, and tuning difficulty. In operational BMS settings, the added detection capability must be weighed against practical concerns such as computational overhead, explainability, maintainability, and deployment feasibility. A hybrid model may therefore be theoretically stronger than a single-model approach, but its real value depends on whether this added complexity results in interpretable and operationally usable detection outcomes.

4.4.6 Evaluation Considerations in Cyber-Physical Anomaly Detection

Evaluation of anomaly detection models in ICS environments extends beyond statistical performance metrics. While accuracy, precision, recall, and F1-score provide useful benchmarks, operational considerations must also be addressed (Stouffer, et al., 2014).

False positives may generate excessive alerts, reducing operator trust and system usability. False negatives may allow undetected compromise, potentially resulting in operational disruption or safety risks. Detection latency is particularly critical in cyber-physical systems, where delayed detection may allow continued execution of malicious commands (Stouffer, et al., 2014).

Operational usability and interpretability, therefore, represent essential evaluation criteria. Detection systems must provide interpretable outputs that enable operators to understand anomaly causes and respond appropriately.

These requirements motivate the integration of explainable artificial intelligence techniques with anomaly detection models. Network traffic analysis and packet-level monitoring play a critical role in identifying abnormal communication patterns in IoT and smart building environments (Setiawan, et al., 2025).

4.4.7 Datasets Used in ICS and BMS Anomaly Detection Research

The evaluation of anomaly detection models in ICS and BMS research depends heavily on the availability of representative datasets. In practice, however, realistic cyber-physical security datasets remain limited, fragmented, and often poorly aligned with the operational characteristics of live building environments. This creates a persistent challenge for the development, training, and validation of machine learning-based detection models (Moosavi, et al., 2024; Elnour, et al., 2021).

Many studies in ICS cybersecurity rely on benchmark datasets collected from laboratory environments or simulated industrial processes. While such datasets support controlled experimentation, they may not adequately reflect the heterogeneity, noise, device diversity, and contextual variability present in operational BMS environments. Models trained on narrowly scoped datasets may therefore perform well in experimental settings while exhibiting weaker generalisability in real deployments.

The problem is more pronounced in building management contexts, where publicly available cybersecurity datasets are comparatively scarce (Elnour, et al., 2021). BMS environments combine HVAC, lighting, access control, sensors, actuators, and management platforms across infrastructures that vary significantly by building type and vendor ecosystem. As a result, the availability of labelled datasets capturing both normal operation and realistic attack scenarios remains limited. Moosavi et al. (2024) highlight the challenge of obtaining representative BACnet-related data suitable for security evaluation, reinforcing the broader problem of data scarcity in building automation research.

At the same time, this limitation also highlights the importance of studies that incorporate operational data from real building environments. In response to this limitation, as detailed in Chapter 5, the present study draws on real-world data collected from a municipality in Sweden, together with freely available datasets to support the construction and evaluation of malicious scenarios (Habte-Medhanie, 2025). This combined approach contributes to reducing the gap between controlled experimental analysis and operational realism, while also addressing the practical difficulty of obtaining sufficiently diverse labelled attack data in live BMS settings. However, the use of partially constructed malicious scenarios also means that external validity must still be considered carefully when interpreting model performance across different operational environments.

Dataset limitations also affect explainability and governance. Where training and evaluation data are incomplete, overly synthetic, or insufficiently diverse, the resulting anomaly explanations may reflect narrow experimental assumptions rather than operationally valid system behaviour. This undermines confidence in both detection performance and interpretive reliability. Accordingly, the dataset

problem is not merely a technical limitation in model development, but a broader constraint on the practical deployment and trustworthiness of anomaly detection in BMS environments.

4.5 Explainable Artificial Intelligence in ICS and Building Management System Security

4.5.1 The Interpretability Challenge in Deep Learning-Based Detection

The adoption of machine learning and deep learning techniques has significantly enhanced anomaly detection capability in Industrial Control Systems (ICS) and Building Management Systems (BMS). Models such as Variational Autoencoders and Long Short-Term Memory networks can capture complex relationships within cyber-physical environments and identify subtle deviations indicative of potential compromise. However, these models introduce a fundamental challenge: a lack of interpretability.

Deep learning models typically operate as opaque systems, often described as "black boxes," where internal decision processes are not directly observable. While these models may achieve high detection accuracy, their predictions may not provide sufficient explanation regarding why specific behaviour is classified as anomalous. This limitation presents operational and governance challenges in cyber-physical environments (Khan, et al., 2022).

Stouffer et al. (2014) emphasise that ICS security mechanisms must align with operational reliability and support human oversight. In building environments, anomaly detection alerts may require evaluation by facility managers, engineers, and security personnel. Without interpretable explanations, operators may struggle to distinguish between legitimate operational variation, equipment malfunction, and malicious activity.

Lack of interpretability may also undermine trust in automated detection systems. If anomaly alerts cannot be explained or validated, operators may disregard alerts or hesitate to take corrective action. This limitation reduces the practical effectiveness of anomaly detection models despite strong analytical performance.

Interpretability therefore, represents a critical requirement for operational deployment of machine learning-based anomaly detection in BMS environments.

The interpretability problem is therefore not only technical but also epistemic, because operators may be unable to assess whether a model output reflects a meaningful cyber-physical anomaly or an opaque computational artefact.

4.5.2 Importance of Explainability in Cyber-Physical Systems

Explainability assumes heightened importance in cyber-physical environments due to the direct relationship between digital system behaviour and physical processes. In conventional IT environments, incorrect anomaly classification may result in inconvenience or unnecessary investigation. In contrast, in BMS environments, incorrect responses to anomaly alerts may affect environmental stability, energy efficiency, or occupant safety (Khan, et al., 2022; Ige, et al., 2024).

Explainable anomaly detection supports several operational objectives. First, it enables operators to identify which system variables contributed to anomaly detection, facilitating investigation and root cause analysis. Second, explainability enhances situational awareness by providing contextual information regarding system behaviour. Third, explainable models support governance and compliance requirements by enabling auditing and validation of automated decisions. For instance, Khan et al. (2022) demonstrate that explainable models improve transparency in HVAC attack detection, directly supporting operator understanding and response.

Li et al. (2023) highlight that many anomaly detection frameworks proposed for Building Automation Systems lack integration with operational workflows. Detection models that generate anomaly scores without explanation may not provide sufficient support for operational decision-making.

Governance frameworks such as the NIST Cybersecurity Framework (Barrett, 2018) emphasise continuous monitoring and incident response capability. Stouffer et al. (2014) further provide ICS-specific guidance on security monitoring and risk management. Explainable anomaly detection supports these objectives by enabling operators to evaluate alerts and determine appropriate response actions.

Research has demonstrated that explainable artificial intelligence techniques enhance cybersecurity detection systems by improving interpretability and supporting effective incident response in cyber-physical environments (Khan, et al., 2022; Ogunseyi & Thiyagarajan, 2025).

4.5.3 Explainable Artificial Intelligence Techniques

Explainable Artificial Intelligence (XAI) techniques are designed to make model outputs more interpretable by showing how a prediction or anomaly score was produced. In cyber-physical security settings, this is especially important because operators must understand whether an alert reflects a plausible system fault, malicious manipulation, or benign operational variation. For anomaly detection in BMS and ICS environments, technical explainability methods are valuable not only for transparency but also for operational verification and post-incident analysis (Khan, et al., 2022).

One widely used technique is SHAP (Shapley Additive Explanations), which estimates the contribution of each input feature to a model output. SHAP is useful

in anomaly detection because it can identify which variables, such as temperature, pressure, occupancy or actuator state, contributed most strongly to an anomalous classification. In cyber-physical environments, this type of explanation is particularly valuable because operators often need to trace an alert back to the physical variables influencing the decision. SHAP also supports consistent feature-level interpretation across model outputs, making it useful for auditability and governance-oriented review (Khan, et al., 2022; Hermosilla, et al., 2025).

Another important method is LIME (Local Interpretable Model-Agnostic Explanations), which explains individual predictions by approximating the behaviour of a complex model locally around a specific instance. LIME is particularly relevant when analysts need to understand why one specific event or sequence was flagged as anomalous. In security operations, this can help distinguish between unusual but legitimate behaviour and suspicious activity requiring escalation. However, because LIME is local and approximation-based, its explanations may vary depending on the sampled neighbourhood, which can reduce stability in highly dynamic cyber-physical environments (Khan, et al., 2022; Hermosilla, et al., 2025; Muhammad, et al., 2025).

Attention mechanisms provide another form of interpretability, particularly in sequential and time-series models. When integrated into neural architectures, attention can highlight the time steps or features that most influenced a model's output. This is especially relevant for BMS anomaly detection because system behaviour unfolds over time and anomalies may emerge through temporal patterns rather than isolated data points. Attention-based explanations may therefore be better suited than purely static feature explanations when analysing evolving cyber-physical states (Muhammad, et al., 2025).

Some XAI methods are more suitable for structured tabular and time-series data than others, meaning that technique selection must reflect the characteristics of BMS telemetry and the operational context in which explanations are required (Khan, et al., 2022).

In comparative terms, SHAP provides strong feature-level interpretability and supports governance-oriented explanation, while LIME offers instance-level local insight into individual decisions. Attention mechanisms are particularly relevant for time-series anomaly detection because they can expose temporally significant inputs. For a study centred on hybrid VAE-LSTM anomaly detection in BMS environments, SHAP, LIME, and temporally aware attention-based explanation are especially relevant because they align more closely with sequential cyber-physical data and operational decision requirements. The specific explainability mechanism adopted in the research design should therefore be selected based on its suitability for sequential anomaly detection, interpretive stability, and operational usability (Khan, et al., 2022; Hermosilla, et al., 2025).

4.5.4 Explainability and Operational Trust

Operational trust represents a critical factor influencing the adoption of anomaly detection systems in building environments. Facility managers and operational personnel must be confident in detection systems to rely on anomaly alerts for decision-making.

Explainability enhances trust by providing transparency into model behaviour. When anomaly alerts are accompanied by explanations identifying contributing variables and behavioural deviations, operators can validate alerts and determine appropriate response actions.

Explainability also supports differentiation between benign anomalies and malicious activity. Building systems frequently experience legitimate operational variation due to environmental changes, occupancy patterns, and maintenance activities. Explainable models enable operators to determine whether anomaly alerts reflect normal variation or potential compromise.

Without explainability, detection systems may generate excessive false positives that reduce operator confidence. Stouffer et al. (2014) emphasise the importance of aligning security mechanisms with operational usability requirements.

Explainability, therefore, represents a key factor enabling practical deployment of anomaly detection models in cyber-physical environments.

In operational environments, explainability strengthens trust not by eliminating uncertainty, but by enabling personnel to judge whether an alert is credible, actionable, and consistent with observed system conditions.

4.5.5 Explainability and Cybersecurity Governance

Cybersecurity governance frameworks increasingly emphasise accountability, transparency, and auditability. Explainable anomaly detection supports these governance objectives by enabling validation and documentation of automated detection decisions.

Ige et al. (2024) emphasise the importance of integrating cybersecurity governance into smart building management. Detection systems must support audit processes and provide evidence supporting incident investigation.

Explainable models enable documentation of anomaly detection decisions by identifying contributing system variables and behavioural deviations. This capability supports compliance with cybersecurity standards and facilitates forensic analysis.

Integration of explainable artificial intelligence with anomaly detection therefore, enhances both technical detection capability and governance alignment.

4.5.6 Challenges and Limitations of Explainable AI in ICS Environments

Despite its benefits, integrating explainability into anomaly detection systems presents technical and operational challenges. Explainability techniques may introduce computational overhead, particularly when applied to large datasets or real-time monitoring environments. ICS systems operate under strict performance constraints, requiring careful optimisation to avoid interference with operational processes (Stouffer, et al., 2014).

Additionally, explanation outputs must be presented in an accessible and meaningful format. Excessively complex explanations may increase cognitive load and reduce usability. Effective explainability mechanisms must balance transparency with operational clarity.

Furthermore, explanation quality depends on model architecture and data representation. Poorly calibrated explanation methods may produce misleading interpretations. Careful design and evaluation are therefore required to ensure reliable and meaningful explainability.

Despite these challenges, explainable artificial intelligence represents a critical component of anomaly detection systems in cyber-physical environments. By enhancing interpretability, explainability improves operational usability, trust, and governance alignment.

This capability is particularly important in hybrid anomaly detection architectures, where integration of structural and temporal modelling increases analytical complexity.

4.6 Standards, Governance, and Resilience in ICS and BMS Security

4.6.1 Cybersecurity Governance Frameworks

Cybersecurity in Industrial Control Systems and Building Management Systems must operate within structured governance frameworks that define risk management, monitoring, and incident response processes. Unlike conventional IT environments, ICS infrastructures manage physical processes, making cybersecurity governance essential for maintaining operational safety and reliability (Stouffer, et al., 2014; Stouffer, et al., 2023).

The National Institute of Standards and Technology (NIST) Cybersecurity Framework provides a widely adopted structure for managing cybersecurity risk across critical infrastructure sectors. The framework is organised around five core functions: Identify, Protect, Detect, Respond, and Recover. These functions collectively support continuous risk management and resilience. ICS-specific implementation guidance is provided by NIST SP 800-82, with the most current

version (Stouffer, et al., 2023) superseding the earlier 2014 edition and extending coverage to operational technology environments more broadly.

Anomaly detection systems contribute primarily to the Detect function by enabling the identification of abnormal system behaviour indicative of potential compromise. However, detection mechanisms must operate in conjunction with preventive and response controls. Detection without a structured response capability provides limited operational benefit.

ICS environments impose additional constraints due to deterministic timing requirements and operational continuity. Security mechanisms must avoid introducing latency or instability. These constraints influence architectural design and the placement of anomaly detection systems (Stouffer, et al., 2014).

Cybersecurity monitoring and decision-support systems have been developed to enhance operational visibility and improve cybersecurity management in complex infrastructure environments (Stouffer, et al., 2023).

4.6.2 Industrial Automation Security Standards

IEC 62443 represents an internationally recognised standard specifically addressing cybersecurity for industrial automation and control systems. The standard promotes a defence-in-depth approach, incorporating layered security controls across system architecture.

Cindrić et al. (2025) emphasise that IEC 62443 promotes network segmentation, secure device configuration, and continuous monitoring. Anomaly detection systems complement these controls by identifying behavioural deviations that may indicate compromise.

However, regulatory standards primarily focus on preventive controls and risk management processes. They provide limited guidance regarding the implementation of machine learning-based detection mechanisms. This creates an important limitation for anomaly detection research. While standards such as IEC-oriented industrial security guidance provide valuable governance structure, they do not in themselves explain how machine learning-based detection models should be validated, interpreted, or operationalised within live BMS environments. There is, therefore, a persistent gap between compliance-oriented control frameworks and the practical design of explainable anomaly detection systems suitable for cyber-physical building operations.

4.6.3 Resilience Engineering in Cyber-Physical Systems

Resilience engineering focuses on the ability of cyber-physical systems to maintain functionality despite disruption. In BMS environments, resilience includes detection, response, and recovery capability.

Novak & Gerstinger (2010) highlight the importance of maintaining safety-critical functionality in building automation systems. Early anomaly detection

contributes to resilience by enabling the identification of system compromise before significant operational impact occurs.

However, detection systems must integrate with operational workflows and governance processes to support an effective response. Detection capability alone is insufficient without mechanisms supporting investigation and mitigation.

Explainable anomaly detection enhances resilience by providing interpretable insights supporting incident response and operational decision-making.

4.7 Synthesis of Literature and Identified Research Gaps

The literature reviewed in this chapter highlights significant advances in cybersecurity research for ICS and BMS environments. However, several critical gaps remain that limit the practical deployment and effectiveness of anomaly detection systems. Design Science Research has been widely applied in cybersecurity to develop and evaluate detection artefacts that improve threat detection capability and organisational cybersecurity resilience (Benjamin, et al., 2024; Eyilmez, et al., 2022).

Cybersecurity research has emphasised the importance of developing technological artefacts and detection mechanisms that address practical operational challenges and improve system resilience (Jatkiewicz, 2025).

Gap 1: Fragmentation Between Detection Models and Operational Context

Many anomaly detection models are developed and evaluated using laboratory datasets without consideration of operational deployment constraints. ICS environments impose strict performance, reliability, and usability requirements that may not be addressed in experimental research (Stouffer, et al., 2014). In cyber-physical systems such as IoT-enabled building environments, sensor integrity represents a critical security concern because manipulated sensor readings may lead to incorrect control decisions.

This fragmentation limits the operational relevance of technically strong detection models in live BMS environments.

Gap 2: Overemphasis on Detection Accuracy Without Interpretability

Research frequently prioritises detection accuracy metrics such as precision and recall while neglecting interpretability. Deep learning models often operate as black boxes, limiting operator trust and usability (Khan, et al., 2022).

This gap reduces operator trust and limits the practical adoption of ML-based detection in BMS settings.

Gap 3: Limited Focus on Building Management System Environments

Much anomaly detection research focuses on industrial manufacturing or network intrusion detection rather than building automation systems. BMS environments present unique challenges due to cyber-physical interdependencies and heterogeneous infrastructure (Minoli, et al., 2017; Siam, et al., 2025).

This limits the transferability of findings from industrial or network intrusion datasets to building automation contexts.

Gap 4: Insufficient Integration with Governance and Compliance Frameworks

Cybersecurity standards emphasise governance, accountability, and auditability. However, many anomaly detection models are developed independently of governance frameworks, limiting practical adoption (Ige, et al., 2024).

This gap makes it difficult to deploy and justify AI-based detection systems within regulated operational environments.

Gap 5: Dataset Limitations and Generalisability Challenges

The preceding review shows that anomaly detection research in ICS often depends on controlled benchmark datasets, while BMS-specific cybersecurity datasets remain limited in availability, realism, and diversity. Obtaining representative operational data from live building environments is difficult because of privacy, safety, proprietary infrastructure, and limited access to labelled attack scenarios (Moosavi, et al., 2024; Elnour, et al., 2021). These constraints restrict both model training and external validity, making it difficult to determine whether results obtained in experimental settings will generalise across different BMS architectures and operational contexts. Research is therefore needed to develop anomaly detection approaches that remain robust under dataset scarcity and are better suited to heterogeneous real-world environments.

Gap 6: Limited Development of Integrated and Explainable Hybrid Detection Systems

Hybrid architectures combining structural and temporal modelling show theoretical promise but remain underexplored in BMS contexts. Additionally, integration of explainability mechanisms within hybrid architectures remains limited.

This gap represents a significant opportunity for research to develop explainable hybrid anomaly detection frameworks tailored to BMS environments.

4.8 Chapter Summary

This chapter examined the cybersecurity landscape of Industrial Control Systems and Building Management Systems, focusing on architectural vulnerabilities, evolving cyber-physical threats, anomaly detection methodologies, explainable artificial intelligence, and governance frameworks. Building Management Systems have evolved from isolated control environments into interconnected cyber-physical infrastructures integrated with enterprise networks and IoT technologies. This transformation has significantly expanded the attack surface while legacy protocols and long equipment lifecycles continue to introduce structural vulnerabilities.

The threat landscape has correspondingly evolved from overt disruption toward stealthier attack strategies, including false data injection, persistent compromise, and manipulation of system behaviour. These threats exploit structural weaknesses and cyber-physical interdependencies within building automation systems. Traditional detection approaches relying on predefined signatures and static thresholds are increasingly insufficient in detecting novel or adaptive attack patterns.

Anomaly detection has emerged as a promising approach for identifying behavioural deviations indicative of system compromise, particularly in cyber-physical environments where operational data reflects interactions between digital and physical processes. Machine learning techniques enable modelling of baseline system behaviour and detection of anomalies without requiring predefined attack signatures. Variational Autoencoders provide structural modelling capability by learning probabilistic latent representations of system behaviour, while Long Short-Term Memory networks enable modelling of temporal dependencies within sequential data. Hybrid VAE–LSTM architectures integrate these capabilities, providing enhanced detection performance in cyber-physical environments.

However, deep learning models introduce interpretability challenges. Lack of transparency may limit operational usability and governance alignment. Explainable artificial intelligence techniques provide mechanisms for interpreting anomaly detection outputs by identifying contributing system variables and supporting operational decision-making (Khan, et al., 2022). Explainability enhances operational trust, supports incident response, and facilitates compliance with cybersecurity governance frameworks.

The literature review identified six critical research gaps, including fragmentation between detection models and operational context, insufficient focus on interpretability, limited BMS-specific research, lack of governance integration,

dataset limitations, and limited development of integrated explainable hybrid detection architectures.

Overall, the reviewed literature demonstrates that anomaly detection in BMS and ICS environments is a promising but still incomplete research area. Significant limitations remain in relation to operational integration, interpretability, governance alignment, dataset availability, and the adaptation of hybrid detection methods to building-specific contexts. These unresolved issues justify the present study and provide the foundation for the conceptual and methodological design developed in the following chapter.

5. Research Methodology

5.1 Research Philosophy and Paradigm

This research is grounded in a **pragmatist philosophical paradigm**. Pragmatism is particularly well-suited to applied research in cyber-physical systems security because it evaluates the value of knowledge in terms of practical utility and problem-solving effectiveness, rather than adhering rigidly to either purely objective or subjective ontological positions (Leedy & Ormrod, 2015). In the context of Building Management System (BMS) security, where the central goal is the development and evaluation of a functional anomaly detection artefact, a pragmatist stance permits the integration of both quantitative and qualitative methods in a manner that best addresses the research problem.

Ontologically, this study adopts a **constructivist-realist** position (Leedy & Ormrod, 2015). The physical infrastructure of BMS environments and the network behaviours they generate are treated as objectively observable phenomena, while the interpretation of anomalous behaviour and the assessment of system usability are recognised as socially and contextually situated. This dual positioning is consistent with Design Science Research, which combines rigorous technological development with contextually sensitive evaluation (Jatkiewicz, 2025).

Epistemologically, this study generates two complementary forms of knowledge. First, **design knowledge**, embodied in the BMS-TIDD artefact itself, demonstrates how a hybrid VAE-LSTM detection architecture and explainability integration can be operationally realised in a BMS environment. Second, **propositional knowledge**, produced through evaluation, yields transferable insights into the conditions under which explainable anomaly detection is effective in cyber-physical settings. This artefact-centred epistemology is consistent with the Design Science paradigm, in which knowledge is generated through purposeful design and iterative refinement rather than solely through observation (Benjamin, et al., 2024; Jatkiewicz, 2025).

5.2 Research Methodology: Design Science Research

This study adopts **Design Science Research (DSR)** as its governing methodology. As introduced in Chapter 2, DSR is a research paradigm that focuses on the creation and evaluation of innovative technological artefacts intended to address real-world problems (Jatkiewicz, 2025). Unlike descriptive or explanatory research approaches, DSR is prescriptive in nature: it generates knowledge through the process of designing, building, and assessing purposeful solutions.

The study operationalises DSR using the **six-phase process model proposed by** Peffers et al. (2007), which has been widely adopted in information systems and

cybersecurity research (Benjamin, et al., 2024; Eyilmez, et al., 2022; Gledson, et al., 2024). This model provides a structured iterative process that guides the research from problem identification through to the communication of findings. Table 1 presents the mapping of each DSR phase to the activities conducted in this study.

Table 5.1. DSR Process Model Mapped to This Study

Phase	DSR Activity	Application in This Study	Location
1	Problem Identification & Motivation	Identification of security gaps in BMS environments; literature review of anomaly detection and XAI limitations	Chapter 1 & 4
2	Objectives of a Solution	Definition of BMS-TIDD artefact requirements: unsupervised detection, XAI integration, real-time feasibility	Chapter 3
3	Design & Development	Construction of hybrid VAE-LSTM model, XAI layer, PCAP-based dataset, and BMS-TIDD dashboard	Chapter 6
4	Demonstration	Application of BMS-TIDD to real-world BMS traffic and seven simulated attack scenarios	Chapter 7
5	Evaluation	Technical performance metrics, XAI output quality assessment, and operational usability evaluation	Chapter 7 & 8
6	Communication	Reporting of findings, contributions, limitations, and future research directions	Chapter 8 & 9

DSR was selected over alternative methodological approaches for several reasons. Experimental research designs, while appropriate for controlled laboratory settings, do not adequately address the design and integration challenges involved in developing an operational detection artefact for cyber-physical environments. Case study methodology, while valuable for descriptive analysis, does not generate the artefact-level contributions that this research aims to produce. Survey-based approaches are insufficient for addressing the technical detection and explainability objectives of this study. DSR, by contrast, directly supports both **technological development** and **theoretical contribution**, making it the most appropriate methodological choice for this research (Jatkiewicz, 2025; Eyilmez, et al., 2022).

5.3 Research Design Overview

The overall research design follows an **iterative build-evaluate cycle** consistent with the DSR paradigm. The design proceeds through three broad stages: (1) data collection and environment preparation, (2) artefact design and development, and (3) evaluation. These stages are not strictly sequential; the design and development

stage incorporates continuous feedback from evaluation results, enabling iterative refinement of both the detection models and the dashboard interface.

In practice, the iterative development process involved two main design cycles. In the first cycle, a VAE-based anomaly detection model was trained on normal BMS traffic using selected flow-level features. The trained VAE was then evaluated against clean, anomaly, and attack PCAP datasets using reconstruction error. The results from this stage informed further refinement of the preprocessing, feature handling, and model configuration before the LSTM component was added.

In the second cycle, the LSTM component was introduced to capture temporal behaviour in the network traffic. PCAP files were divided into 1-second buckets and converted into sliding windows for sequence-based training. The LSTM was trained on normal traffic windows only, and anomaly detection was performed using reconstruction errors calibrated from the normal training data. Finally, the VAE and LSTM outputs were combined to produce a final anomaly decision, and SHAP/LIME explainability was added to interpret the detected anomalies.

The four sub-research questions defined in Chapter 3 are addressed across these stages as follows. Sub-RQ1 (anomaly detection under data scarcity) is addressed through the data collection and attack scenario construction strategy described in Sections 5.4 and 5.5. Sub-RQ2 (XAI integration) is addressed through the model architecture and explainability mechanism described in Section 5.6. Sub-RQ3 (real-time performance feasibility) is addressed through the system design and computational evaluation described in Sections 5.6 and 5.7. Sub-RQ4 (usability and operational value) is addressed through the evaluation framework described in Section 5.7.

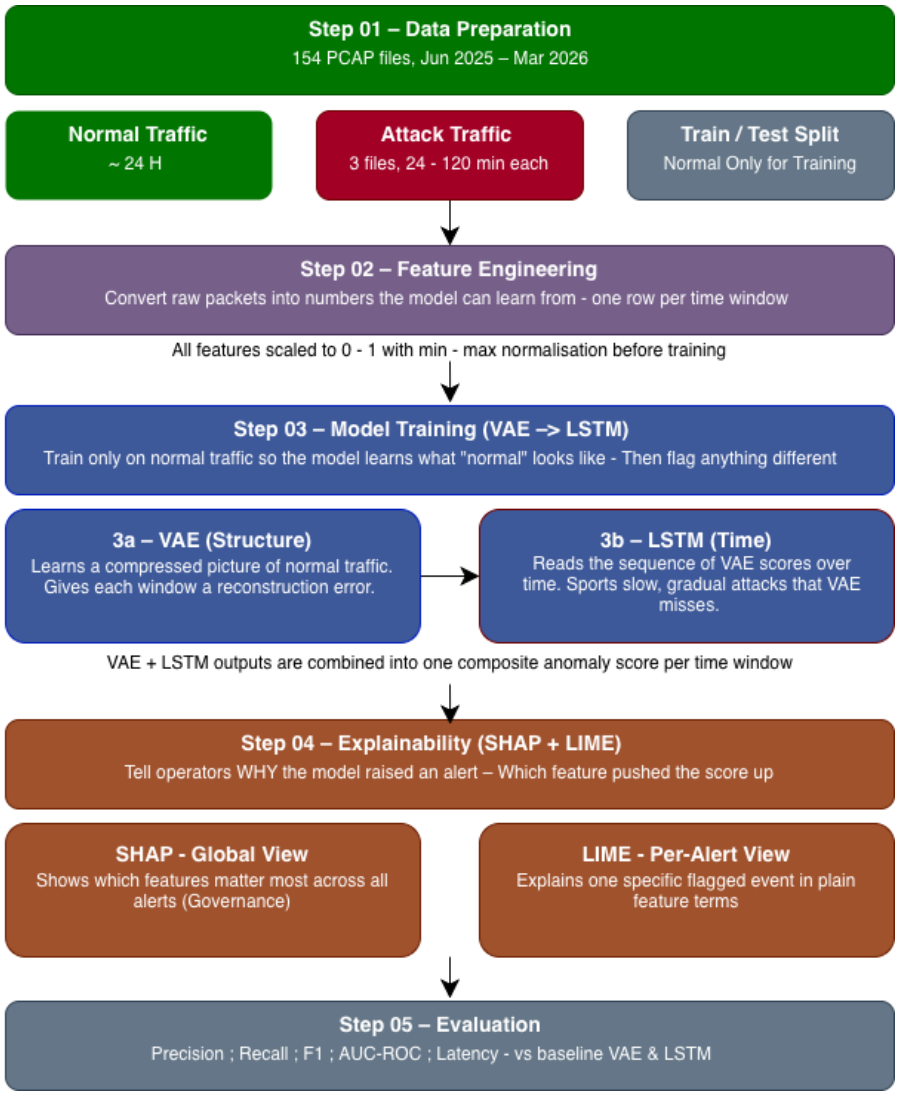


Figure 5.1.BMS – TIDD Machine Learning Training Pipeline

5.4 Data Collection and Environment Description

All IP addresses and subnet identifiers reported in this thesis have been masked for confidentiality; individual hosts are denoted by symbolic labels (A–G) and the two network segments as Subnet 1 and Subnet 2. The rationale is described in Section **5.8**, Validity, Reliability, and Ethical Considerations. The primary dataset for this study was collected from a real operational BMS network deployed across multiple municipal buildings in Sweden (Habte-Medhanie, 2025). The environment consists of two interconnected subnets managed through a hierarchical BACnet-based infrastructure. The management site, hosted at the Town Hall, operates on **Subnet 1** and contains the Siemens Desigo CC supervisory server (**B**), which

functions as the BACnet master, and a local BAC field bus controller (**A**). The remote field subnet, operating on **Subnet 2**, connects three remote building sites through a dedicated Remote Site Switch: **Preschool** (**C**), **Football Arena** (**D**), and **Stad Arena** (**E**). A gateway device (**F**) aggregates field device traffic toward the core router. Table 2 summarises the network topology.

Table 5.2 BMS Network Topology

Label	What it is
Subnet 1	Management subnet (Town Hall)
A	BAC field-bus controller
B	Desigo CC server
Subnet 2	Field subnet (remote sites)
C	Preschool controller
D	Football Arena controller
E	Stad Arena controller
F	Gateway / Remote Site Switch
G	Attacker machine
Subnet 2 broadcast	broadcast address

All inter-site communication in the environment uses the **BACnet/IP protocol over UDP port 47808**, which is the standard communication protocol for building automation control systems. Under normal operating conditions, the network generates approximately **2,895 BACnet packets per minute during active polling intervals** (the all-day aggregate across the full 270-day capture period averages 47,648 packets/day across all protocols; see Table 6.4), with minimal ARP traffic (~1 packet/minute) and no ICMP or SSH activity. This baseline characterises the deterministic, low-variation communication pattern typical of operational BMS environments.

5.4.1 Traffic Capture and Dataset Construction

The PCAP dataset used in this study was obtained through the passive data-collection infrastructure established by Habte-Medhanie (2025), in which a Raspberry Pi connected to a mirrored port on the BMS network switch continuously records daily PCAP captures from the live BMS network without interfering with its operation. As that collection setup remains in continuous operation, the present study draws on the traffic it produced over the June 2025 - March 2026 monitoring period.

Network traffic was captured in PCAP (Packet Capture) format, which records full packet-level data including headers, protocol fields, timestamps, and payload content. PCAP-format capture was selected because it preserves the complete temporal and structural characteristics of BMS network communication at the

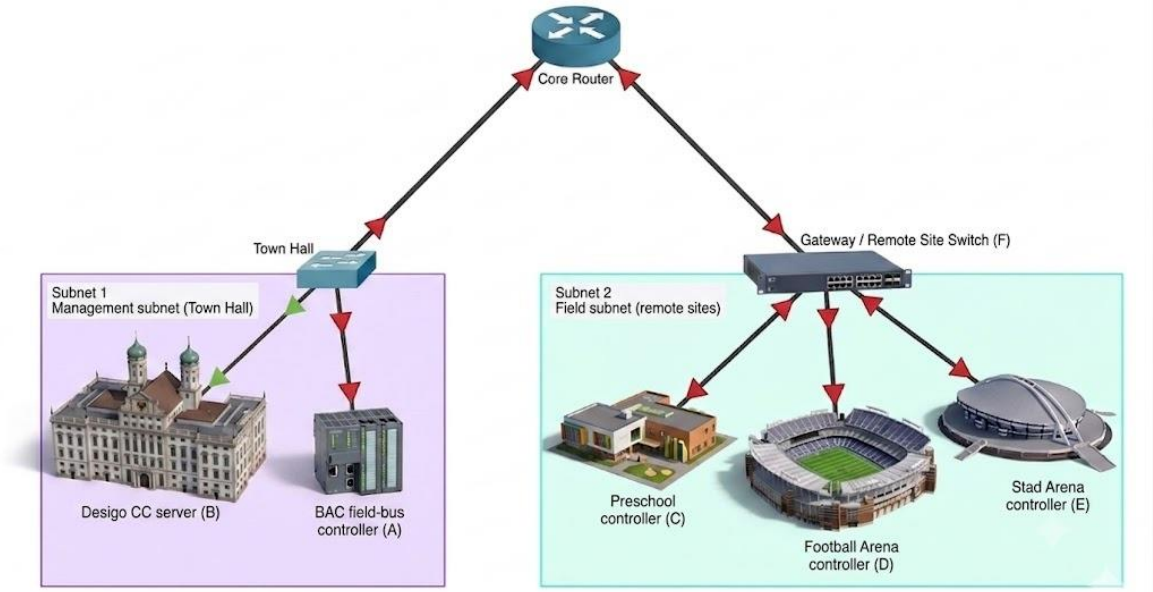


Table 5.3 BMS Network Topology

packet level, enabling both feature extraction for machine learning and detailed forensic analysis of attack behaviour (Graftceva, 2024; Setiawan, et al., 2025). Each normal operational PCAP file captures approximately 24 hours of continuous BMS network traffic. Since the capture system initiates each recording at a fixed local time, individual files typically span two consecutive calendar days: the session begins in the late afternoon or evening of one day and concludes at approximately the same time on the following day, crossing midnight. The three manually generated ATTACK capture files differ in this respect; each was recorded during a controlled simulation session of shorter duration, entirely within a single calendar day. (Graftceva, 2024; Setiawan, et al., 2025)

The resulting dataset contains **308,740 packets** spanning approximately **102 minutes** of continuous capture. Of these, 98.8% are UDP packets corresponding to BACnet/IP traffic on port 47808, with the remainder comprising TCP (0.9%), ICMP (0.2%), and ARP (0.1%) packets. The dataset was not sourced from pre-existing public repositories. As noted in Chapter 4, publicly available ICS and BMS cybersecurity datasets are frequently collected from different protocol environments, laboratory testbeds, or outdated infrastructures that do not reflect the operational characteristics of modern BACnet-based building automation deployments (Moosavi, et al., 2024). The use of traffic captured directly from the operational environment, therefore, provides a higher degree of **ecological validity** than externally sourced alternatives.

The dataset was constructed by identifying a gap in the normal data series, which was addressed through the controlled injection of a separately generated attack PCAP file containing all seven simulated attack scenarios. This approach preserves

the authentic characteristics of the normal BMS traffic baseline while enabling clear temporal labelling of each attack window.

5.5 Attack Scenario Construction

5.5.1 Attack Environment Design

To construct a labelled dataset suitable for training and evaluating the anomaly detection model, seven distinct attack scenarios were simulated using a dedicated **attacker machine placed externally from the normal BMS environment** and assigned the IP address **G** within the field subnet. This machine was isolated from the normal operational environment and introduced solely for the purpose of controlled attack generation. Its placement within the field subnet simulates a realistic adversarial scenario in which an attacker gains access to the operational network, consistent with attack paths described in cyber-physical attack graph research (Barrère, et al., 2023).

The attacks were generated as a **separate PCAP file** and subsequently integrated with the normal traffic capture to produce the combined dataset. Each attack scenario is independently labelled with a distinct time window, enabling supervised evaluation of detection performance. The separation of attack generation from normal traffic capture ensures that the baseline characteristics of the real BMS environment are not contaminated during the attack simulation process. Unlike the normal operational captures, which each span approximately 24 hours and typically cross midnight into the following calendar day, the attack PCAP files were recorded during focused simulation sessions of shorter duration, each confined to a single calendar day. The three attack captures used in evaluation range from approximately 24 to 102 minutes in length, reflecting the duration of the respective simulated attack sequences.

5.5.2 Simulated Attack Scenarios

Seven attack scenarios were selected to represent a range of threat categories relevant to BMS and ICS environments, including protocol-level exploitation, network reconnaissance, credential attacks, network layer manipulation, and volumetric denial-of-service attacks. Table 3 presents the full attack taxonomy, including the time window within the capture, a description of the simulated behaviour, and the tool used for generation.

Table 5.4. Simulated Attack Scenarios

Attack Type	Time Window	Description	Generation Tool

BACnet Write Traffic	~0–5 min	Unauthorised BACnet WriteProperty commands targeting BMS controller objects from external attacker machine	Scapy
Ping Sweep	~5 min	ICMP echo burst (258 packets) to discover active hosts across the BMS field subnet	hping3 / Scapy
Port Scan	~15 min	TCP SYN scan across 1,024 sequential ports from attacker machine (172.19.218.150) to map exposed services	Nmap
SSH Brute Force	~30–31 min	375 rapid TCP connection attempts to SSH port 22, simulating credential stuffing against BMS management interface	Hydra
ARP Spoofing	~45 min	181 ARP reply packets injected to poison ARP cache of network nodes and redirect BMS traffic through attacker	Ettercap / Scapy
BACnet Flood	~60 min	High-volume BACnet/UDP flood targeting port 47808; traffic rate peaks at 10,392 packets/min (3.6× normal baseline of ~2,895)	Scapy / hping3
DDoS	~60 min	Coordinated flood from 247 spoofed external IPs (RFC 5737 documentation ranges) targeting BMS subnet simultaneously with BACnet flood	Scapy (multi-source)

The seven scenarios collectively represent key threat categories identified in the literature review. **BACnet Write Traffic** simulates unauthorised manipulation of building control parameters, consistent with integrity attacks identified by (Granzer, et al., 2010) and (Li, et al., 2023). **Ping Sweep** and **Port Scan** represent reconnaissance phases typical of Advanced Persistent Threat campaigns (Barrère, et al., 2023). **SSH Brute Force** targets the management interface of the supervisory layer, exploiting the credential vulnerability noted in (Peacock & Johnstone, 2014). **ARP Spoofing** simulates network layer manipulation, enabling traffic interception, a threat class particularly relevant in BACnet environments where authentication mechanisms are limited (Granzer, et al., 2010). **BACnet Flood** and **DDoS**, while treated as separate labelled events, occur simultaneously at the 60-minute mark, representing compounded availability attacks targeting both protocol and network layers. The DDoS scenario employs 247 spoofed source IPs drawn from RFC 5737 documentation address ranges, simulating a coordinated external multi-source attack.

5.5.3 Labelling Strategy

The evaluation dataset employs a three-class ground-truth labelling scheme applied at the PCAP file level: CLEAN (0), representing normal operational traffic with no known anomalies; ANOMALY (1), representing traffic exhibiting

anomalous but non-adversarial behaviour such as a device operational fault; and ATTACK (2), representing deliberately generated attack traffic containing one or more of the seven simulated attack scenarios. This three-class taxonomy reflects the operational reality of BMS environments, in which not all deviations from normal behaviour constitute intentional attacks; misconfigurations, firmware faults, and unusual polling states produce traffic deviations that a well-calibrated anomaly detector should also identify.

For the purposes of binary classification evaluation, the three-class labels are collapsed into a binary scheme: CLEAN maps to normal (0), and both ANOMALY and ATTACK map to abnormal (1). This binary projection enables standard precision, recall, and F1-score metrics to be computed across the detection system and is consistent with the system's operational objective of flagging any departure from normal, regardless of whether that departure is adversarial in origin.

The model is trained in a strictly one-class unsupervised manner using CLEAN-labelled PCAP captures exclusively. Labels are withheld from the training process entirely and applied only at the evaluation stage, ensuring that no label information leaks into weight optimisation or threshold calibration. The CLEAN captures used for training represent continuous normal BMS operations collected across the monitoring period (June 2025 to March 2026), providing a representative sample of the baseline operational profile characterised in Section 6.3.4. Of the normal training corpus, 90% is used for model weight optimisation and 10% is held out as a validation set used exclusively for early stopping and learning rate scheduling. Anomaly detection thresholds are calibrated at the 95th percentile of reconstruction errors observed across training windows, without exposure to ANOMALY or ATTACK captures. The ANOMALY and ATTACK captures are reserved entirely for final held-out evaluation.

At evaluation, labels are applied in two modes: binary detection performance (CLEAN vs. ANOMALY+ATTACK) to assess overall anomaly identification capability using precision, recall, F1-score, and MCC; and within the ATTACK capture, per-scenario attribution to assess whether the model's reconstruction error profiles are distinguishable across the seven attack categories. This separation ensures the integrity of the unsupervised detection approach and is consistent with one-class detection strategies recommended for ICS environments where labelled attack data is scarce but abundant normal operational data is available (Khan, et al., 2022; Moosavi, et al., 2024).

5.6 Artefact Design and Development

The artefact developed in this study is the **BMS Threat Intelligence and Detection Dashboard (BMS-TIDD)**, an integrated system combining machine learning-based anomaly detection, explainable artificial intelligence, and an

operator-oriented monitoring interface. The design of each component is described in the following subsections.

5.6.1 Feature Engineering from PCAP Data

Raw PCAP data is transformed into a structured feature set suitable for machine learning through a packet-level feature extraction pipeline. For the VAE component, features are extracted from each individual PCAP capture file, treating the complete file as a single flow-level feature vector. For the LSTM component, traffic is segmented into **1-second time buckets assembled into 10-second sliding windows (10 buckets, 5-second step, 50% overlap)** to capture the temporal evolution of network behaviour within each window. Prior to model development, a Pearson correlation-based feature selection step was applied to a larger candidate pool of packet-level statistics, retaining the 10 features with the highest mean absolute correlation with known anomaly labels across pilot captures. The resulting VAE feature set comprises: destination port diversity (dport_diversity), packet length standard deviation (std_len), mean packet length (mean_len), ARP packet count (n_arp), byte rate (byte_rate_bps), ARP ratio (arp_ratio), ICMP ratio (icmp_ratio), source-to-destination fanout (fanout), unique source port count (n_unique_sport), and maximum packet length (max_len). The LSTM uses a separate 10-feature set optimised for temporal sequence modelling: packets per bucket (n_pkts_bucket), byte rate (byte_rate_bps), mean packet length (mean_len), packet length standard deviation (std_len), maximum packet length (max_len), unique source ports (n_unique_sport), ARP count (n_arp), ICMP ratio (icmp_ratio), unique destination ports (n_unique_dport), and mean inter-arrival time (mean_iat). The two feature sets differ deliberately: n_pkts_bucket and mean_iat are particularly sensitive to flooding and timing anomalies respectively, while n_unique_dport captures temporal scan patterns that unfold across consecutive buckets.

VAE features are normalised using a MinMaxScaler, scaling each feature to the range [0, 1]; the decoder sigmoid output layer is designed to match this bounded input range. LSTM features are scaled using a RobustScaler (median and interquartile range normalisation), which preserves the absolute magnitude of extreme values, such as the flood-level packet rates observed during BACnet Flood and DDoS attacks, making high-volume anomalies more discriminable in the temporal feature space. Each scaler is fitted exclusively on normal-traffic data and applied without refitting to detection inputs. The scaled feature matrices form the respective inputs to the VAE and LSTM components.

5.6.2 Hybrid VAE-LSTM Detection Architecture

The detection model adopts a **hybrid Variational Autoencoder – Long Short-Term Memory (VAE-LSTM) architecture** to address both the structural and

temporal complexity of BMS network behaviour. As established in the literature review (Section 4.4.5), this hybrid approach is motivated by the complementary strengths of the two architectures: VAEs provide probabilistic latent space modelling of multi-variable relationships across BMS features, while LSTMs capture sequential temporal dependencies in how those features evolve over time (Gupta, 2025; Fährmann, et al., 2022).

In the hybrid architecture, the **VAE component** is trained on normal traffic feature windows to learn a compressed probabilistic representation of baseline BMS behaviour. The VAE encodes input feature vectors into a latent space defined by a mean and variance, and generates reconstruction outputs during inference. Reconstruction error, the difference between the original input and the VAE's reconstructed output, serves as the primary **structural anomaly score**. Inputs that deviate from the learned normal distribution produce elevated reconstruction errors, indicating potential anomalies. The VAE architecture is sized for the 10-feature input: the encoder maps the input through a hidden layer of 32 units (BatchNorm, ReLU) and a second layer of 16 units (BatchNorm, ReLU) before projecting to a 4-dimensional latent space (mu, log-sigma-squared). The decoder mirrors this structure, projecting from 4 latent dimensions back to the 10-dimensional output via 16-unit and 32-unit layers, with a sigmoid activation on the output to match the min-max scaled input range. Training uses KL-annealing (beta ramps from 0 to 1.0 over the first 30% of epochs) with early stopping (patience of 15 epochs) and a 90/10 train/validation split.

The **LSTM component** operates independently of the VAE, processing its own 10-second sliding windows of 1-second time-bucket features to model the temporal dynamics of normal BMS communication. By modelling the sequential evolution of traffic features across consecutive 1-second buckets, the LSTM identifies anomalies that develop gradually over time - such as the slow escalation characteristic of Advanced Persistent Threats or incremental BACnet parameter manipulation - which may not be detectable from individual flow-level snapshots alone (Barrère, et al., 2023). The final detection verdict is produced by combining the VAE structural anomaly score and the LSTM temporal anomaly score using a logical OR rule: a traffic sample is classified as anomalous if either model reconstruction error exceeds its respective calibrated threshold. This combination strategy ensures that anomalies detectable by either the structural or the temporal component are captured, maximising recall in the operational environment. The LSTM architecture uses 2 stacked layers with 64 hidden units per layer and 0.2 inter-layer dropout, processing sequences of length 10 (10 seconds at 1-second bucket granularity). Training applies early stopping with a patience of 20 epochs and a 90/10 train/validation split. The OR combination rule accepts a higher false-positive rate relative to an AND rule, a deliberate trade-off that prioritises recall in

availability-critical OT environments, where an undetected intrusion carries greater operational risk than a false alert.

The model is implemented in **Python** using **PyTorch** for deep learning components, with **Scapy** used for PCAP feature extraction and **scikit-learn** for preprocessing and baseline comparison models. The training set consists exclusively of normal traffic windows to preserve the unsupervised nature of the detection approach, consistent with the operational reality of BMS environments where labelled attack data is unavailable before deployment.

5.6.3 Explainability Integration

To address the interpretability limitations of deep learning-based anomaly detection, the BMS-TIDD integrates **SHAP (SHapley Additive exPlanations)** as the primary explainability mechanism, supplemented by **LIME (Local Interpretable Model-Agnostic Explanations)** for instance-level inspection of individual anomalous events.

SHAP was selected as the primary method because it provides **consistent, globally interpretable feature attribution** across all model outputs, enabling operators to understand which network features, such as elevated BACnet volume, abnormal ARP activity, or unusual source IP diversity, contributed most strongly to a given anomaly score. This is particularly valuable in BMS environments where operators must trace anomaly alerts back to specific physical variables or communication patterns for incident response and root cause analysis (Khan, et al., 2022). SHAP's governance-oriented interpretability also supports auditability requirements under frameworks such as the NIST Cybersecurity Framework (Barrett, 2018).

LIME complements SHAP by providing **local explanations for individual anomalous events**, approximating the model's decision boundary in the neighbourhood of a specific flagged instance. While LIME explanations may exhibit lower stability than SHAP in highly dynamic environments (Muhammad, et al., 2025), they offer a useful diagnostic tool for operators investigating specific alert instances that require per-event contextualisation beyond what aggregate SHAP values provide.

Both explanation outputs are surfaced through the BMS-TIDD dashboard interface, presenting feature contribution scores alongside anomaly alerts in a format accessible to building system operators without requiring deep familiarity with machine learning concepts.

5.6.4 BMS–TIDD Dashboard Design

The BMS-TIDD dashboard provides an **operator-facing monitoring interface** that integrates real-time anomaly detection outputs, SHAP-based feature explanations, and historical alert visualisation. The dashboard design followed a

user-centred iterative process: initial wireframes were developed based on requirements derived from the literature on cybersecurity monitoring in operational technology environments and on established OT human-machine interface design guidelines (NIST SP 800-82). Low-fidelity prototypes were reviewed with the research supervisors before implementation, and the interface layout was refined following preliminary expert walkthroughs conducted during the first evaluation cycle. The final design is guided by **human-centred design** principles, prioritising clarity, interpretability, and operational relevance for facility managers and security personnel who may not have advanced technical expertise in machine learning.

Key interface components include: a real-time traffic monitoring panel displaying BACnet packet rates and protocol distribution; an anomaly alert feed presenting flagged events with associated anomaly scores and timestamps; a SHAP feature importance visualisation panel showing per-alert feature contributions; a historical alert timeline enabling trend analysis across the monitoring period; a Decision Support panel presenting operator guidance aligned with the anomaly type and contributing features; and a Reports Panel enabling export of alert summaries for incident documentation and governance reporting. The design draws on established principles for cybersecurity monitoring dashboards in operational environments. (Benjamin, et al., 2024; Eyilmez, et al., 2022).

5.7 Evaluation Framework

The evaluation of the BMS-TIDD artefact is structured across three complementary dimensions, consistent with the DSR evaluation phase and with the multi-criteria assessment requirements identified in Sub-RQ4: **technical detection performance**, **explainability quality**, and **operational usability**.

5.7.1 Technical Performance Evaluation

Detection performance is assessed using standard binary and multi-class classification metrics applied to the labelled dataset: **Precision, Recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC)**. Given the operational sensitivity of BMS environments to both false positives, which generate unnecessary alerts and reduce operator trust, and false negatives, which allow undetected compromise, both precision and recall are treated as equally important evaluation criteria (Stouffer, et al., 2014).

Detection **latency** is additionally measured to assess the real-time feasibility of the system, consistent with Sub-RQ3. The computational overhead of the model inference pipeline is evaluated to confirm operational compatibility with BMS performance constraints. The target criterion is end-to-end inference latency below one second per 10-second analysis window on commodity CPU-only hardware consistent with the supervisory server polling frequency of the operational BMS environment and sufficient for near-real-time alert generation.

The hybrid VAE-LSTM model is compared against two implemented baseline models: a standalone VAE and a standalone LSTM. An Isolation Forest was identified as a planned non-neural reference baseline due to its established performance in ICS anomaly detection benchmarks; however, its implementation was not completed within the scope of this study and is noted as a direction for future work (Section 6.9.3). For both implemented baselines, the detection threshold is calibrated at the 95th percentile of reconstruction errors observed across normal-traffic training windows; for the VAE, an additional attack-tier threshold at the 99th percentile is applied to distinguish suspected anomalies from high-confidence attack detections. The BACnet Flood and DDoS scenarios, which occur simultaneously at the ~60-minute mark, are treated as a compound availability attack class during multi-class attribution evaluation; binary detection metrics are reported for each scenario individually, while per-class attribution for the concurrent pair acknowledges the inherent label ambiguity arising from their temporal overlap. The headline binary detection metrics, Precision, Recall, and F1-score, are computed using the 95th-percentile threshold for all models; the 99th-percentile attack-tier threshold for the VAE is reported separately as a high-confidence detection indicator and does not affect the primary comparative evaluation.

5.7.2 Explainability Quality Assessment

The quality of SHAP and LIME explanations is assessed along three dimensions: **faithfulness** (whether explanation outputs accurately reflect model behaviour), **stability** (whether similar inputs produce consistent explanations), and **operational relevance** (whether flagged features correspond to meaningful BMS network indicators that operators can act upon). Faithfulness is evaluated through feature perturbation experiments, in which the top-ranked SHAP features are systematically modified, and the resulting change in anomaly score is measured. Stability is assessed by computing the Spearman rank correlation between SHAP feature attribution rankings across 50 repeated inferences on identical input windows; a mean correlation of >=0.85 across all evaluated instances is used as the stability acceptance criterion. Operational relevance is assessed qualitatively through expert review.

5.7.3 Usability and Operational Evaluation Framework

To assess the operational value of the BMS-TIDD artefact beyond technical performance, a **qualitative usability evaluation** is conducted with a purposively sampled group of five to seven domain experts drawn from three categories: BMS operators responsible for day-to-day facility management, facility managers with oversight responsibility for building infrastructure, and cybersecurity practitioners with experience in OT or ICS security. This sample size is suitable for purposive

sampling in qualitative research. In this approach, a small group of relevant participants can provide detailed and meaningful insights based on their experience (Leedy & Ormrod, 2015). Participants are recruited through institutional contacts at the municipality operating the study BMS environment and through professional networks in the OT security community. The evaluation protocol has been conducted in accordance with the ethical research guidelines of the host institution, and informed consent is obtained from all participants before the evaluation session. Participants engage with the dashboard interface through structured walkthroughs of anomaly scenarios, following a think-aloud protocol to elicit insights into usability, interpretability, and decision support value. Findings are collected through semi-structured interviews and a standardised usability questionnaire, enabling both qualitative thematic analysis and quantitative usability scoring.

This evaluation component addresses the requirement, identified by (Khan, et al., 2022) and (Jatkiewicz, 2025), that anomaly detection systems in cyber-physical environments be assessed not only for technical accuracy but for their practical ability to support human decision-making in operational contexts.

5.8 Validity, Reliability, and Ethical Considerations

Several measures are taken to ensure the validity and reliability of the research. **Internal validity** is supported by the use of real operational BMS traffic as the baseline dataset, by the controlled and temporally isolated design of each attack scenario, and by the comparison of results across multiple model architectures. **External validity** is constrained by the use of a single BMS deployment environment; results may not generalise directly to BMS or ICS deployments using different communication protocols (e.g., Modbus, PROFINET, OPC-UA), vendor platforms other than the Siemens Desigo CC supervisory system used here, or network topologies with substantially different traffic characteristics or device densities. This limitation is acknowledged and addressed in the discussion of findings.

The use of real-world BMS network traffic introduces **data privacy considerations**. As this study is conducted in Sweden, an EU member state, data handling is governed by the General Data Protection Regulation (GDPR, Regulation (EU) 2016/679). The captured traffic exclusively reflects machine-to-machine BACnet communication; IP addresses and device identifiers in the dataset are not associated with personally identifiable information and do not constitute personal data within the meaning of GDPR Article 4(1). The dataset is used exclusively for research purposes within the scope of this study, and no attempt is made to identify individuals from the captured traffic.

The attack scenarios are generated in an **isolated, controlled environment** entirely separate from the live operational BMS infrastructure. No attack traffic was introduced into the live building automation network at any point during the study,

ensuring that the operational integrity and availability of the real BMS environment were not compromised. To support reproducibility, the full source code for the feature extraction pipeline, model training, and detection framework is made available in an open repository upon thesis submission. The operational BMS PCAP dataset cannot be released publicly due to confidentiality obligations to the municipality operating the network; however, the derived feature matrices are made available to enable replication of the machine learning components.

Network address masking. Although the captured IP addresses do not constitute personal data, they reveal the internal addressing scheme of a live municipal building-automation network. In accordance with the confidentiality obligations owed to the operating municipality, and because this thesis is published openly, all network addresses have been masked prior to publication. Each host is replaced by a stable symbolic label (A–G) and each network segment by a generic identifier (Subnet 1 for the management site and Subnet 2 for the remote field network), and the same mapping is applied consistently across the body text, tables, and figures. The substitution is purely nominal and relational: it preserves the network topology, the source–destination relationships, and the role of every device, so that no reported measurement or result is affected, while no real address belonging to the operational environment is disclosed.

5.9 Chapter Summary

This chapter has presented the methodological framework governing the design, development, and evaluation of the BMS-TIDD artefact. The study adopts a pragmatist philosophical paradigm and operationalises Design Science Research through the six-phase (Peffers, et al., 2007) process model. The primary dataset is derived from real operational BMS traffic captured from a municipal building network in Sweden, comprising 308,740 packets across 102 minutes. Seven attack scenarios, BACnet Write Traffic, Ping Sweep, Port Scan, SSH Brute Force, ARP Spoofing, BACnet Flood, and DDoS were generated using a controlled external attacker machine and integrated as a separately labelled PCAP file. The artefact integrates a hybrid VAE-LSTM detection architecture with SHAP and LIME explainability mechanisms and an operator-oriented monitoring dashboard. Evaluation is conducted across three dimensions: technical detection performance, explainability quality, and operational usability. The following chapter presents the implementation and results of this framework.

6. System Implementation and Design

6.1 Chapter Introduction

This chapter describes the design and implementation of the BMS Threat Intelligence and Detection Dashboard (BMS-TIDD), the technological artefact developed in this study. Whereas Chapter 5 presented the methodological rationale and design decisions governing the construction of the artefact, this chapter reports what was actually built: the software components, data processing pipelines, model architectures, and monitoring interface that together realise the proposed explainable anomaly detection framework for Building Management Systems.

The chapter is organised to follow the logical construction sequence of the system. Section 6.2 presents a high-level architecture overview of the BMS-TIDD system and its technology stack. Section 6.3 describes the data preprocessing pipeline through which raw PCAP captures are transformed into structured feature matrices. Sections 6.4 and 6.5 detail the design and training configuration of the Variational Autoencoder and Long Short-Term Memory components, respectively. Section 6.6 explains how the two detection components are integrated into a hybrid detection system. Section 6.7 describes the integration of SHAP and LIME explainability mechanisms. Section 6.8 presents the BMS-TIDD dashboard implementation. Section 6.9 describes the standalone baseline model implementations used for performance comparison. Section 6.10 summarises the chapter.

6.2 System Architecture Overview

The BMS-TIDD system is structured across three integrated layers: a data ingestion and preprocessing layer, a detection and explainability engine, and an operator-facing monitoring dashboard. These layers correspond to the three stages of the artefact design described in Section 5.6 and are connected through a sequential processing pipeline that transforms raw network traffic into interpretable anomaly alerts.

The data ingestion layer receives network traffic in PCAP format, representing packet-level captures from the operational BMS environment. Raw PCAP files are processed by a feature extraction pipeline that computes structured numerical representations of network behaviour at two granularities: a flow-level summary for the VAE component and a temporal window sequence for the LSTM component. Both representations are normalised using scalers fitted exclusively on normal traffic, and the resulting feature matrices are passed to the detection engine.

The detection engine comprises the trained VAE and LSTM models, each operating independently on its respective feature representations. The VAE

assesses structural deviation from the learned normal distribution by computing a per-capture reconstruction error. The LSTM assesses temporal deviation by computing per-window reconstruction errors across sliding sequences of 1-second time buckets. The two anomaly scores are combined at the decision level using a logical OR rule to produce a final binary verdict. When either model flags a capture as anomalous, SHAP and LIME explainability routines are invoked to generate feature attribution scores that explain the source of the anomaly.

The monitoring dashboard provides the operator-facing interface through which anomaly alerts, detection scores, and SHAP explanations are presented in real time. The dashboard is implemented as a full-stack web application with role-based access control, supporting both operational monitoring and administrative management functions.

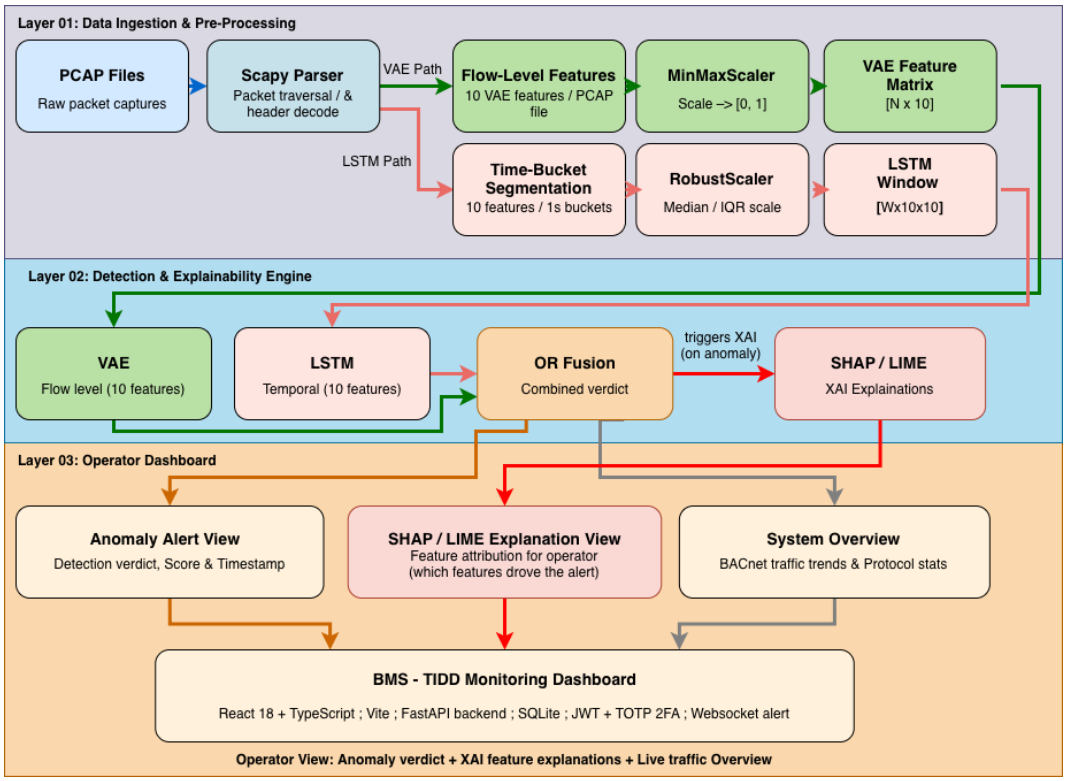


Figure 6.1 BMS-TIDD Three-Layer Architecture

The technology stack underpinning the system comprises the following components. The machine learning pipeline is implemented in Python 3 using PyTorch for deep learning model definition and training, Scapy for packet-level PCAP feature extraction, and scikit-learn for preprocessing, scaling, and baseline comparison utilities. SHAP explainability is provided by the SHAP library, and LIME explanations are generated using the lime library. The frontend dashboard is implemented in React 18 with TypeScript, bundled using Vite. The backend API is

implemented in FastAPI with a SQLite database for alert and user data persistence. Authentication is secured using JSON Web Tokens with Time-based One-Time Password multi-factor authentication.

Layer	Component	Technology / Version	Purpose
ML Pipeline	Deep learning framework	PyTorch 2.x	VAE and LSTM model definition, training, and inference
ML Pipeline	PCAP feature extraction	Scapy 2.x	Packet parsing and flow-level feature computation
ML Pipeline	Explainability	SHAP (KernelExplainer)	Global Shapley feature attribution for VAE and LSTM
ML Pipeline	Explainability	LIME (LimeTabularExplainer)	Local instance-level feature attribution
ML Pipeline	Numerical processing	NumPy / Pandas	Feature matrix manipulation and scaler application
ML Pipeline	Preprocessing scalers	scikit-learn MinMaxScaler/ RobustScaler	Feature normalisation for VAE and LSTM inputs
Backend	API framework	FastAPI (Python)	RESTful API and WebSocket server for real-time alerts
Backend	Database	SQLite	User account storage, audit logs, detection records
Backend	Authentication	JWT + TOTP (pyotp)	Session management and two-factor authentication
Frontend	UI framework	React 18 + TypeScript	Operator monitoring dashboard
Frontend	Build tool	Vite	Frontend bundling and development server
Frontend	Visualisation	Recharts / Chart.js	Real-time traffic charts and SHAP bar plots

6.3 Data Preprocessing Pipeline

The preprocessing pipeline transforms raw PCAP captures into normalised feature matrices suitable for input to the VAE and LSTM components. The pipeline is executed in two sequential stages: packet-level feature extraction and feature normalisation. Two parallel extraction pathways are maintained, producing distinct feature sets optimised for the structural and temporal modelling objectives of each model component.

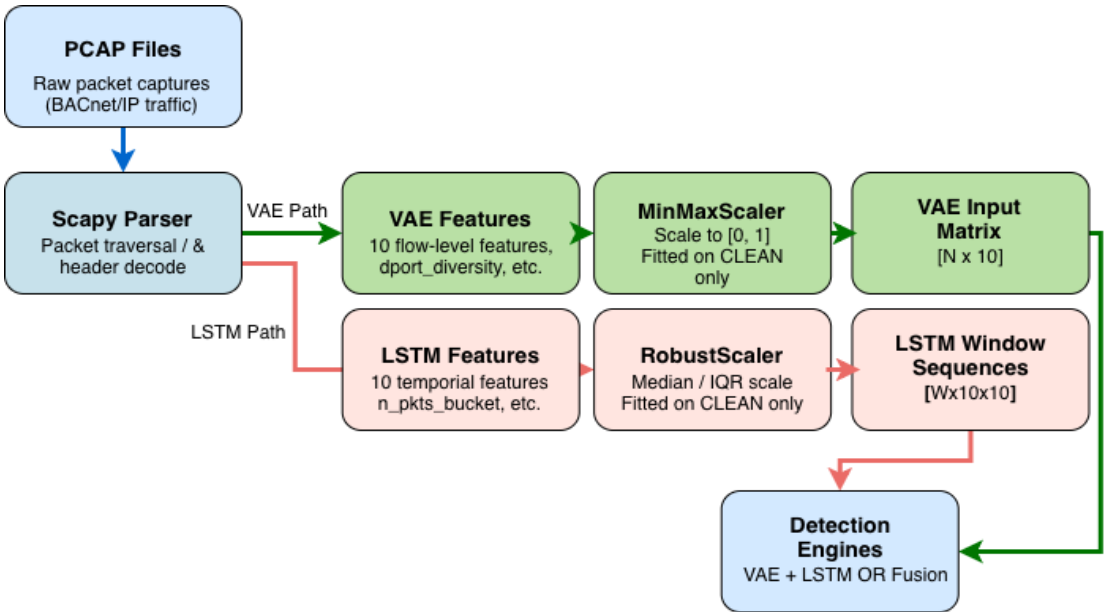


Figure 6.2. Dual-path data processing pipeline

6.3.1 PCAP Parsing and Flow-Level Feature Extraction

Flow-level features for the VAE component are extracted from each PCAP file using a packet traversal routine implemented with the Scapy library. Each PCAP file is treated as a single traffic flow, producing one feature vector per file. For each packet in the capture, the routine accumulates packet lengths, protocol counts, source and destination port sets, destination IP addresses, and timestamps. Upon completion of traversal, ten aggregate statistics are computed from the accumulated data.

The ten VAE features, selected through a Pearson correlation-based feature selection process from a larger candidate pool, are: destination port diversity (dport_diversity), defined as the number of unique destination ports divided by total TCP and UDP packets; packet length standard deviation (std_len) and mean (mean_len); ARP packet count (n_arp); network byte rate in bits per second (byte_rate_bps), computed as total bytes multiplied by eight divided by capture

duration; ARP ratio (arp_ratio) and ICMP ratio (icmp_ratio) as proportions of total packets; source-to-destination fanout (fanout), defined as unique destination IPs divided by unique source ports; unique source port count (n_unique_sport); and maximum packet length (max_len). To ensure the selected features are not redundant, Pearson correlation analysis was applied across all ten VAE features. No feature pair exceeded a correlation of r = 0.90, confirming that each feature contributes independent information to the model. The closest pair, std_len and mean_len, has a correlation of r = 0.88; both were retained because std_len measures how much packet sizes vary, while mean_len measures the average size, they are related but capture different aspects of traffic behaviour. Together, the ten features cover all seven simulated attack scenarios, with each feature linked to a recognisable anomaly signal in BACnet traffic.

Table 6.1. VAE input feature set - name, description, and anomaly sensitivity.

#	Feature Name	Description	Anomaly Sensitivity
1	dport_diversity	Number of unique destination ports observed in the capture	High - protocol scans and port sweeps increase diversity above BACnet-only baseline
2	std_len	Standard deviation of packet lengths	Medium - mixed-size attack payloads alter the length distribution
3	mean_len	Mean packet length across all packets	Medium - oversized payloads (Write/Flood) shift mean upward
4	n_arp	Total count of ARP packets in the capture	High - ARP spoofing and scanning generate elevated ARP volumes
5	byte_rate_bps	Total bytes transferred divided by capture duration (bits per second)	High - flooding and DDoS attacks produce extreme byte rates
6	arp_ratio	Fraction of total packets that are ARP (n_arp / total_packets)	High - ARP-heavy attacks shift ratio far above normal BACnet baseline
7	icmp_ratio	Fraction of total packets that are ICMP (n_icmp / total_packets)	Medium - Ping Sweep introduces ICMP traffic absent in normal BACnet traffic
8	fanout	Number of unique destination IP addresses contacted	High - Ping Sweep and Port Scan contact many hosts; normal polling contacts few

#	Feature Name	Description	Anomaly Sensitivity
9	n_unique_sport	Number of unique source ports used across all packets	Medium - spoofed-source DDoS and scanning use many randomised source ports
10	max_len	Maximum single-packet length observed in the capture	Low–medium- abnormal maximum lengths signal unusual payload types

6.3.2 Time-Window Segmentation for the LSTM Component

Temporal features for the LSTM component are extracted through a time-bucket segmentation process. Each PCAP file is partitioned into non-overlapping 1-second intervals, and ten features are computed per bucket from the packets falling within that interval. The 1-second bucket granularity was selected to match the supervisory polling frequency of the operational BACnet environment while providing sufficient temporal resolution to detect gradual anomalies.

The ten LSTM features are distinct from the VAE feature set and are optimised for temporal anomaly sensitivity: packets per bucket (n_pkts_bucket), which is the primary flooding detection signal; byte rate (byte_rate_bps); mean and standard deviation of packet lengths (mean_len, std_len); maximum packet length (max_len); unique source ports (n_unique_sport); ARP count per bucket (n_arp); ICMP ratio (icmp_ratio); unique destination ports per bucket (n_unique_dport), sensitive to temporal scan patterns; and mean inter-arrival time (mean_iat), which captures timing anomalies characteristic of flooding attacks. The separation of n_pkts_bucket and mean_iat from the VAE feature set is deliberate: these features are most informative when evaluated across consecutive time steps rather than as flow-level aggregates.

Pearson correlation analysis was also applied to the ten LSTM features. No pair exceeded $r = 0.90$, confirming that each temporal feature contributes independent information to the model.

Table 6.2. LSTM input feature set - name, description, and temporal anomaly sensitivity.

#	Feature Name	Description	Temporal Anomaly Sensitivity
1	n_pkts_bucket	Packet count within the 1-second time bucket	High - primary flooding signal; surges sharply during Flood and DDoS attacks
2	byte_rate_bps	Byte throughput within the bucket (bits per second)	High - corroborates flooding; sustained elevation across windows signals volumetric attack

3	mean_len	Mean packet length within the bucket	Medium - shifts with abnormal payload sizes across consecutive buckets
4	std_len	Standard deviation of packet lengths within the bucket	Medium - increased variance signals mixed-protocol attack traffic
5	max_len	Maximum packet length within the bucket	Low–medium - single large packets may indicate data exfiltration payloads
6	n_unique_sport	Count of unique source ports used within the bucket	Medium - elevated during Port Scan and spoofed-source DDoS across time
7	n_arp	ARP packet count within the bucket	High - ARP Spoofing generates sustained per-bucket ARP elevations over multiple windows
8	icmp_ratio	Fraction of packets that are ICMP within the bucket	Medium - Ping Sweep produces non-zero ICMP ratios absent in normal BACnet windows
9	n_unique_dport	Count of unique destination ports contacted within the bucket	High - Port Scan and reconnaissance expand destination port diversity over time
10	mean_iat	Mean inter-arrival time between consecutive packets in the bucket (seconds)	High - flooding attacks compress IAT to near zero; normal BACnet polling has regular IAT

Bucket feature sequences are assembled into 10-bucket sliding windows with a step size of 5 (50% overlap), producing windows of shape [10 timesteps × 10 features]. This windowing configuration balances temporal coverage; each window spans 10 seconds of traffic, with the detection granularity required to identify short-duration attack bursts (Wei, et al., 2022).

6.3.3 Normalisation and Train/Test Split Strategy

VAE features are normalised using a MinMaxScaler, which maps each feature to the range [0, 1]. This bounded scaling is appropriate for the VAE decoder, whose sigmoid output layer is designed to produce values in [0, 1]. Six features exhibiting high variance, byte_rate_bps, fanout, n_unique_sport, std_len, max_len, and n_arp, are additionally capped at their 99th percentile values prior to scaling, preventing extreme outliers in the training data from compressing the normal traffic range.

Ratio features (arp_ratio, icmp_ratio, dport_diversity) are constrained to [0, 1] before scaling.

LSTM features are normalised using a RobustScaler, which centres each feature at its median and scales by the interquartile range. This choice differs from the VAE preprocessing and is intentional: RobustScaler preserves the relative magnitude of extreme values, such as the packet rates observed during flooding attacks, making high-volume anomalies more discriminable in the temporal feature space. If MinMaxScaler were applied to the LSTM features, extreme attack-period values could compress the normal traffic range, thereby reducing temporal anomaly sensitivity.

Both scalers are fitted exclusively on normal training traffic and serialised to disk for reuse during detection. This ensures that no information from anomalous or attack captures influences the scaling parameters, preserving the integrity of the one-class unsupervised detection approach. The training corpus consists of normal BACnet traffic only; all anomaly- and attack-labelled captures are withheld entirely from the training and scaling procedures and used solely for evaluation.

Table 6.3. Comparison of preprocessing scalers for VAE & LSTM components.

Property	**MinMaxScaler (VAE)**	**RobustScaler (LSTM)**
Formula	x' = (x − x_min) / (x_max − x_min)	x' = (x − median) / IQR
Output range	[0, 1]	Unbounded; centred at 0
Why chosen	VAE decoder uses sigmoid activation (σ), which outputs in [0,1]; bounded input ensures loss is well-defined	RobustScaler preserves the large magnitude differences between normal and flooding windows that make LSTM anomaly scores detectable
Behaviour under flooding	Extreme values compress all normal samples near 0, degrading VAE training if scaler fitted on mixed data	IQR-based scaling is resistant to outliers; flooding magnitudes remain large relative to the normal IQR, enabling per-window detection
Fitted on	CLEAN training PCAP files only	CLEAN training PCAP files only
Serialised for inference	Yes, vae_scaler.pkl	Yes, lstm_scaler.pkl

6.3.4 Baseline Traffic Characterisation and Dataset Overview

Prior to model training, a systematic characterisation of the normal BMS network traffic was conducted across the full data collection period spanning June

2025 to March 2026. This analysis serves two purposes: it establishes the operational baseline against which anomalies are defined, and it documents the traffic properties that informed key design decisions, including feature selection, scaler choice, and threshold calibration.

The dataset comprises 154 PCAP capture files collected continuously from the operational BMS network at the Town Hall site (management **subnet 1**) and the Remote Field subnet (**Subnet 2**), which encompasses three remote building facilities: Uddegardens Preschool, Football Arena, and Stad Arena. Traffic flows between the Town Hall Desigo CC server (**B**) and the building automation controllers at each site via BACnet/IP over UDP port 47808, representing the dominant protocol in the capture corpus (Habte-Medhanie, 2025). The duration of individual captures varies across the corpus: the majority of files span two consecutive calendar days, reflecting the fixed-schedule recording cycle in which each session begins in the late afternoon and concludes approximately 24 hours later after crossing midnight. A smaller number of files cover a single calendar day, corresponding to truncated or rescheduled capture sessions. This file duration structure accounts for the 270 daily traffic observations derived from 154 files, as described in Section 6.3.2. The three manually generated ATTACK captures are not part of this operational corpus; each was recorded as a separate controlled session of shorter duration, confined to a single calendar day (Habte-Medhanie, 2025).

Daily packet counts were derived from timestamp analysis across all 154 PCAP files, producing 270 daily observations. Descriptive statistics computed from this baseline are summarised in Table 6.4. The sample mean is 47,648 packets per day with a sample standard deviation of 44,368, reflecting the considerable natural variation in BMS polling cycles across the capture period. The standard error of the mean is 2,700.15 packets ($SE = s / \sqrt{n} = 44{,}368 / \sqrt{270}$). Applying a two-tailed t-distribution with df = 269 degrees of freedom and $t^* = 1.9690$, the 95% confidence interval for the daily mean packet volume is [42,331, 52,964] packets per day.

Table 6.4. Daily packet count statistics and 95% confidence interval derived from 270 days of baseline traffic analysis.

Statistic	Value	Notes
Observation period	June 2025 – March 2026	154 PCAP files; 270 calendar days
Sample size (n)	270	One observation per calendar day
Mean ($\bar{x}$)	47,648 packets/day	Includes anomaly day 2026-01-19
Sample std deviation (s)	44,368 packets/day	High variance due to anomaly day outlier
Standard error (SE)	2,700.15	$SE = s / \sqrt{n} = 44{,}368 / \sqrt{270}$
t^* (df = 269, 95%)	1.9690	Two-tailed t-distribution critical value

Statistic	Value	Notes
95% CI (all days)	[42,331 – 52,964] packets/day	CI = $\bar{x} \pm t^* \times SE$
Anomaly day (2026-01-19)	641,552 packets	13.5 × baseline mean; ANOMALY-labelled capture (PCAP file capture_20260118_180949.pcap begins 18 Jan 18:09; anomalous volume attributed to 19 Jan calendar day)
Mean (anomaly excluded)	45,440 packets/day	n = 269
95% CI (anomaly excluded)	[42,368 – 48,512] packets/day	Narrower CI confirms anomaly day is statistical outlier
Protocol share (BACnet)	69.9% (9,002,612 / 12,864,903)	Dominant protocol across 154 PCAP files

One date, 19 January 2026, records an anomalous daily volume of 641,552 packets, 13.5 times the baseline mean. Exclusion of this single anomaly day reduces the mean to 45,440 packets per day with a narrowed 95% confidence interval of [42,368, 48,512], confirming that the anomaly day is a clear statistical outlier and not an artefact of measurement error. This date corresponds to the ANOMALY-labelled capture (capture_20260118_180949.pcap) in the evaluation dataset (Note: the PCAP file begins at 18:09 on 18 January; the anomalous daily packet count is attributed to 19 January because the bulk of the elevated polling traffic falls after midnight in the daily aggregate statistics.), which records an elevated BACnet polling state consistent with a device operational anomaly rather than a deliberate attack. The clear separation between this day's volume and the baseline confidence interval demonstrates that volume-based thresholding at the daily aggregate level is viable for detecting pronounced anomalies; however, the hybrid VAE-LSTM system targets finer-grained structural and temporal deviations that persist even when absolute packet volumes remain within the normal range.

Temporal traffic patterns were examined at hourly, weekly, and monthly granularity to characterise the regularity of BMS operational cycles. The hourly traffic profile reveals two distinct activity periods aligned with building occupancy schedules: a primary peak during business hours (09:00–17:00) and reduced but non-zero traffic during overnight hours, consistent with continuous BACnet polling from the Desigo CC server. Weekly patterns indicate consistent weekday-weekend traffic differentiation, with weekday volumes approximately 15–20% higher than weekend volumes across the monitoring period. Monthly analysis shows stable traffic volumes from July 2025 through December 2025, with a gradual increase from January 2026 attributable to the addition of monitoring points for the newly commissioned Stad Arena facility. The 95% confidence bands plotted across these temporal profiles confirm that day-to-day and hour-to-hour variability, while

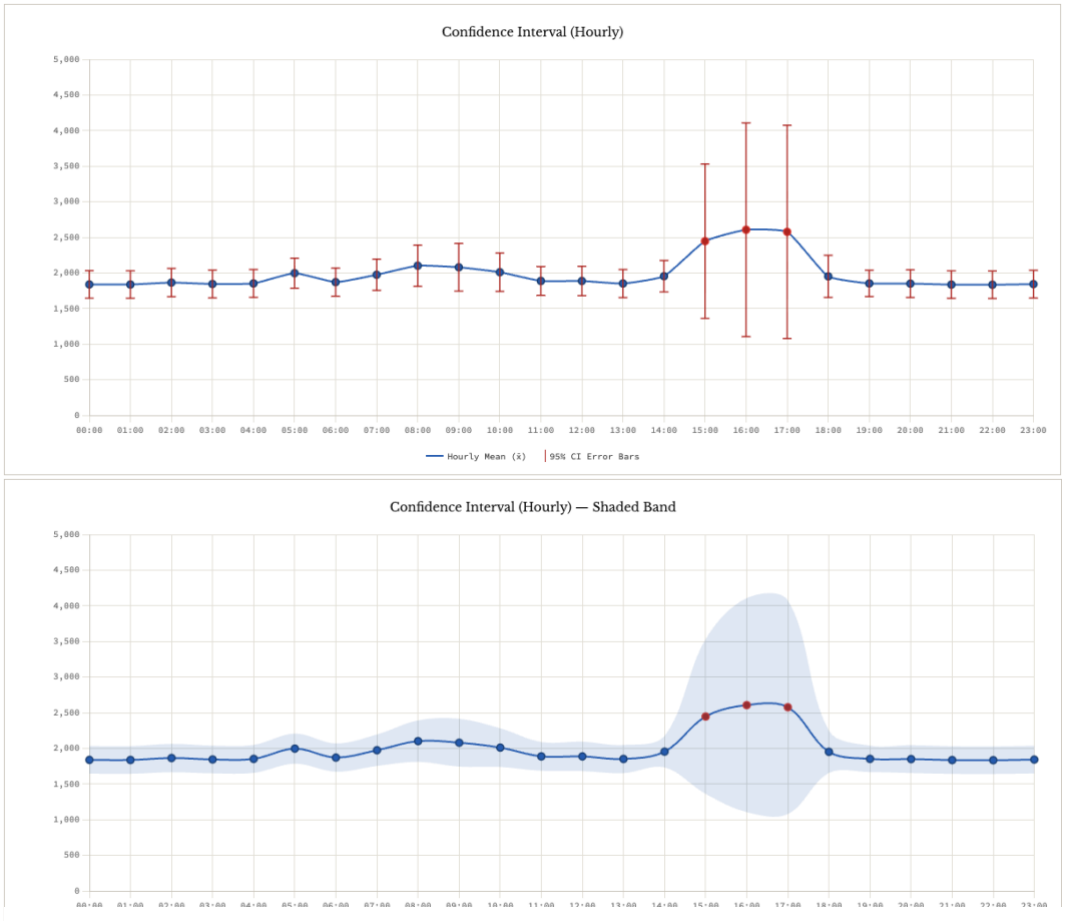


Figure 6.3. Hourly BMS Network traffic profile with 95% confidence interval error bars showing business-hours activity peak.

present, follows a consistent distributional pattern suitable for one-class detection modelling.

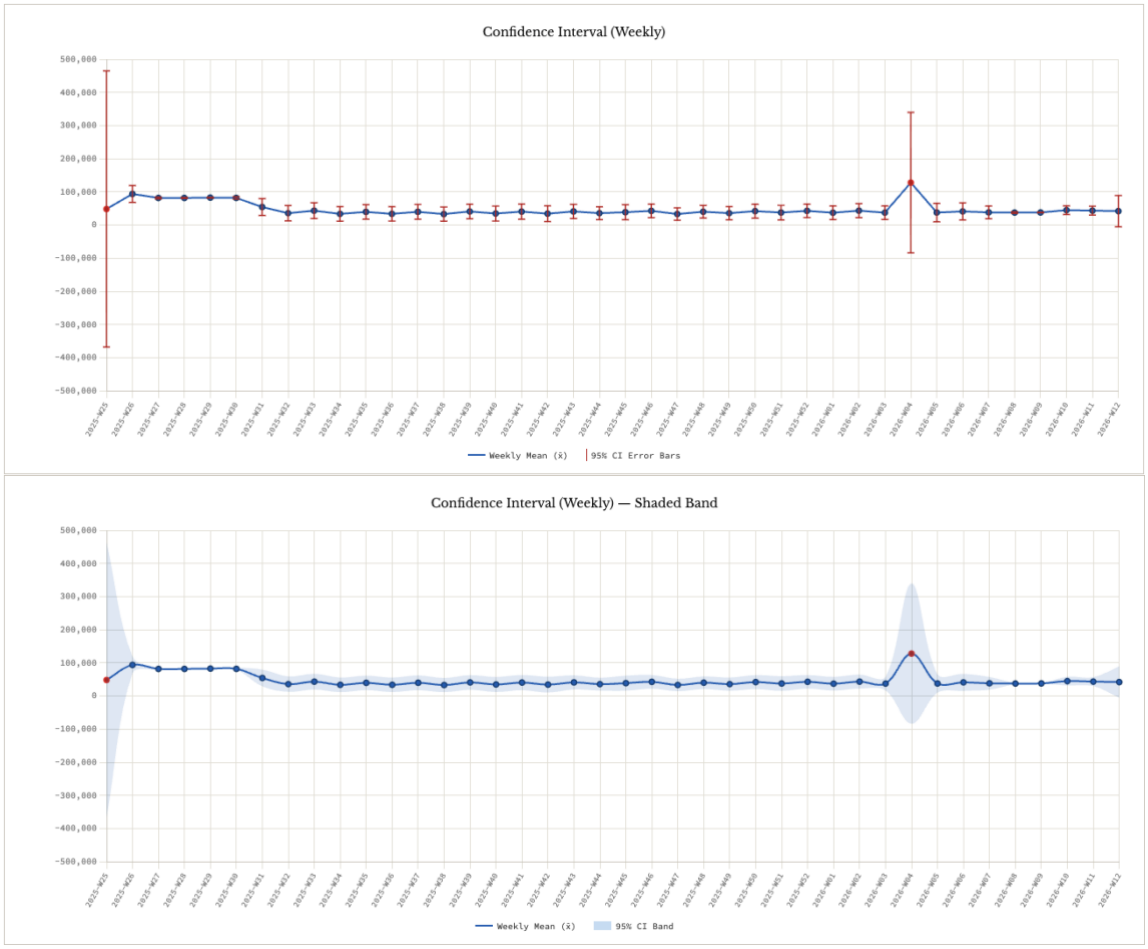
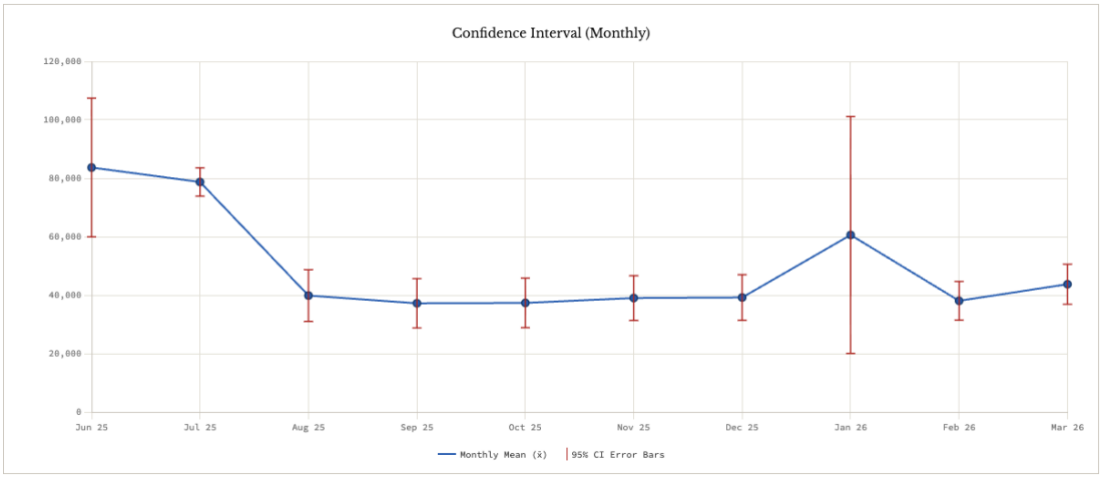


Figure 6.4. Weekly BMS network traffic pattern with 95% confidence interval showing consistent week differentiation.



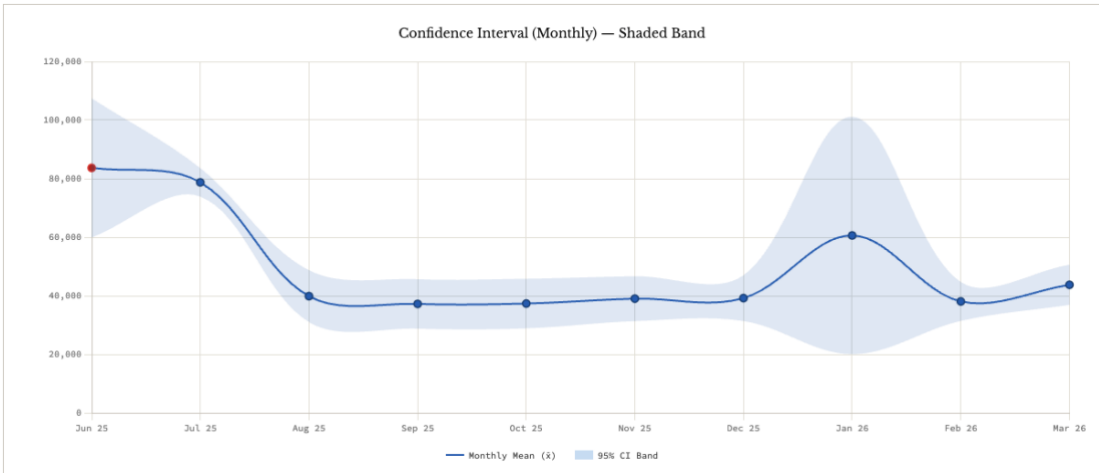


Figure 6.5. Monthly BMS network traffic volume (June 2025 – March 2026) with 95% confidence interval band.

The Sankey flow analysis, constructed from tshark parsing of all 154 PCAP files, quantifies directional packet flows across the 34 unique source-to-destination pairs identified in the dataset. The total analysed packet volume is 12,864,903 packets, of which 9,002,612 (69.9%) are classified as BACnet protocol traffic. The dominant flow pair is the bidirectional exchange between the Preschool Building Automation Controller (**C**) and the Town Hall Server (**B**): the northbound flow (field site to server) contributes 4,002,984 packets, and the southbound flow (server to field site) contributes 3,863,216 packets, together accounting for approximately 60.8% of total observed traffic. Additional significant flows include broadcast ARP traffic from **C** to the field subnet broadcast address (**Subnet 2 Broadcast**) at 872,974 packets, and multicast IPv6 neighbour discovery traffic accounting for approximately 818,000 to 552,000 packets across three flows. The Football Arena BAC controller (**D**) and Stad Arena BAC controller (**E**) contribute smaller but structurally similar bidirectional polling flows of 421,408 and 304,063 packets, respectively.

The concentration of BACnet traffic into a small number of high-volume, directionally symmetric polling flows is a defining characteristic of BMS network behaviour and directly motivates the feature engineering choices described in Sections 6.3.1 and 6.3.2. Features such as destination port diversity (dport_diversity), fanout (unique destination addresses per unit time), and ARP ratio (arp_ratio) are designed to capture deviations from this highly structured, low-diversity polling pattern. An attack or anomaly that introduces novel device communication, elevated broadcast activity, or atypical port usage will produce measurable shifts in these features even without producing abnormal aggregate packet volumes, enabling the VAE to detect structural anomalies at the flow level. The LSTM component complements this by capturing temporal discontinuities in the polling cadence that would be invisible to file-level aggregation.



Table 6.5 Sankey charts for the 154 PCAP data files.

6.4 VAE Component - Design and Implementation

6.4.1 Encoder and Decoder Architecture

The Variational Autoencoder is implemented as a symmetric encoder–decoder architecture sized for the 10-feature flow-level input. The encoder maps the 10-dimensional input vector through two fully connected hidden layers before projecting to a 4-dimensional probabilistic latent space. The first hidden layer contains 32 units, followed by batch normalisation and a ReLU activation; the second hidden layer contains 16 units with the same normalisation and activation scheme. From the 16-unit representation, two parallel linear projections produce the latent mean vector μ and log-variance vector log σ^2, each of dimension 4.

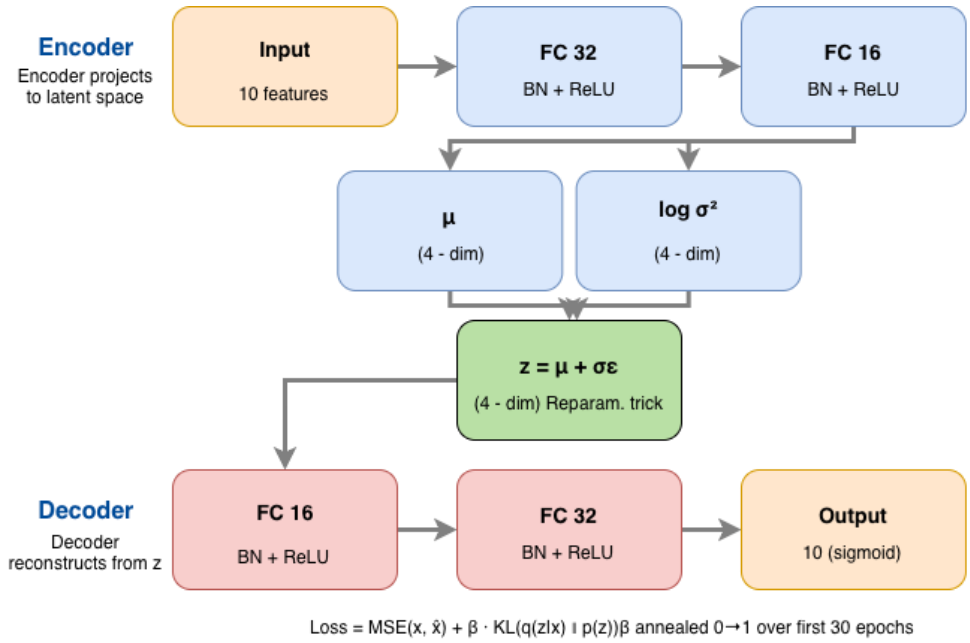


Figure 6.6. VAE Encoder-Decoder Architecture

The decoder mirrors the encoder in reverse. A linear projection maps the 4-dimensional latent sample to a 16-unit layer, followed by batch normalisation and ReLU activation, then to a 32-unit layer with the same normalisation and activation, and finally to a 10-dimensional output layer with a sigmoid activation function. The sigmoid output ensures reconstruction values lie in [0, 1], matching the MinMaxScaled input range and enabling meaningful mean squared error computation between input and reconstruction. The total number of trainable parameters in the VAE is approximately 2,100, a compact architecture well-suited to CPU-only inference in resource-constrained operational environments.

At inference time the reparameterisation trick is disabled and the encoder returns μ deterministically, ensuring stable and reproducible anomaly scores. The anomaly

score for a given input is the per-sample mean squared error between the original feature vector and its reconstruction.

6.4.2 Training Configuration

The VAE is trained using the Adam optimiser with an initial learning rate of 0.001 and a weight decay of 1×10^{-5}. The learning rate is managed by a ReduceLROnPlateau scheduler with a reduction factor of 0.5 and a patience of 10 epochs. Training proceeds for a maximum of 100 epochs with a batch size of 32. A 90/10 train/validation split is applied to the normal traffic feature matrix, with the validation subset used exclusively for early stopping and threshold calibration.

The loss function combines mean squared error reconstruction loss with a KL divergence regularisation term weighted by an annealing coefficient β: Loss = MSE(x, $\hat{x}$) + β · KL(q(z|x) ‖ p(z)). The KL annealing schedule ramps β linearly from 0 to 1.0 over the first 30% of training epochs (epoch 30 at a 100-epoch maximum), after which β is held at 1.0. This annealing strategy prevents posterior collapse during early training by allowing the model to prioritise reconstruction quality before imposing the full regularisation pressure of the KL term. Early stopping with a patience of 15 epochs is applied following beta stabilisation, and the model checkpoint achieving the lowest validation loss is retained.

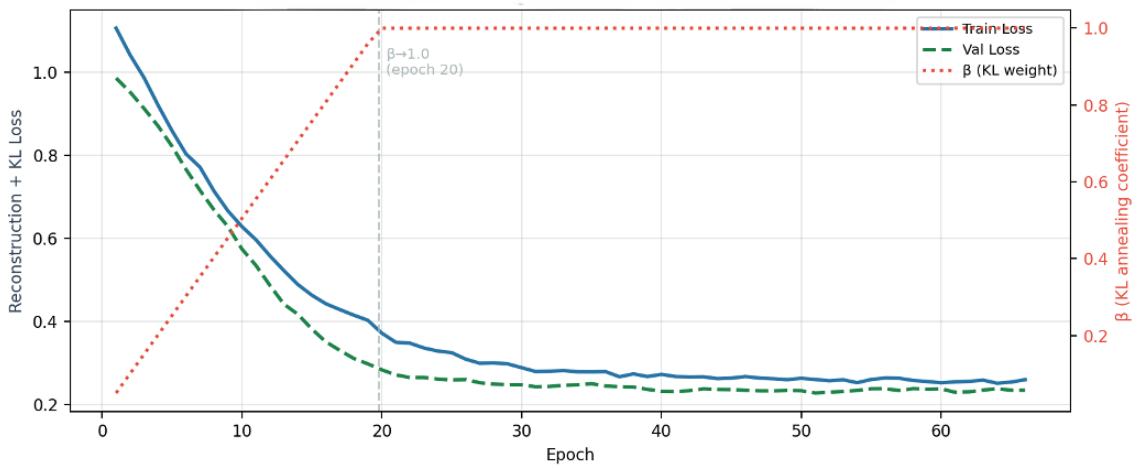


Figure 6.7. VAE training and validation loss curves with KL annealing schedule.

6.4.3 Reconstruction Error Threshold Selection

Anomaly-detection thresholds are calibrated from the distribution of reconstruction errors observed across normal training windows. The primary detection threshold is set at the 95th percentile (p95) of normal-traffic reconstruction errors, yielding a value of 0.07096 MSE. A secondary attack-tier

threshold is set at the 99th percentile (p99), yielding 0.19599 MSE, used to distinguish high-confidence attack detections from lower-confidence anomaly flags. Both thresholds are derived from normal traffic reconstruction error distributions and reflect a deliberate 5% and 1% false positive rate acceptance on the training distribution, respectively.

6.5 LSTM Component - Design and Implementation

6.5.1 Input Sequence Design

The LSTM component operates on sequences of temporal feature vectors extracted independently from the raw PCAP traffic, using the time-bucket pipeline described in Section 6.3.2. Each input sequence consists of 10 consecutive 1-second time buckets, each described by 10 temporal features, forming a tensor of shape [10 timesteps × 10 features]. The sliding window step size of 5 seconds produces overlapping windows with 50% temporal overlap, ensuring that short-duration events spanning two consecutive windows are captured by at least one window in the sequence.

The LSTM component does not receive inputs from the VAE. The two models operate on entirely separate feature sets and are trained independently. Their outputs are combined only at the final decision stage, as described in Section 6.6. This architectural separation ensures that each component contributes a genuinely independent signal to the hybrid detection decision.

6.5.2 LSTM Architecture

The LSTM model implements a sequence-to-sequence reconstruction architecture comprising an encoder LSTM, a decoder LSTM, and a linear output projection. The encoder LSTM receives the input sequence of shape [batch, 10, 10] and processes it through 2 stacked LSTM layers with 64 hidden units per layer and an inter-layer dropout of 0.2. The encoder produces a hidden state summarising the temporal dynamics of the input sequence.

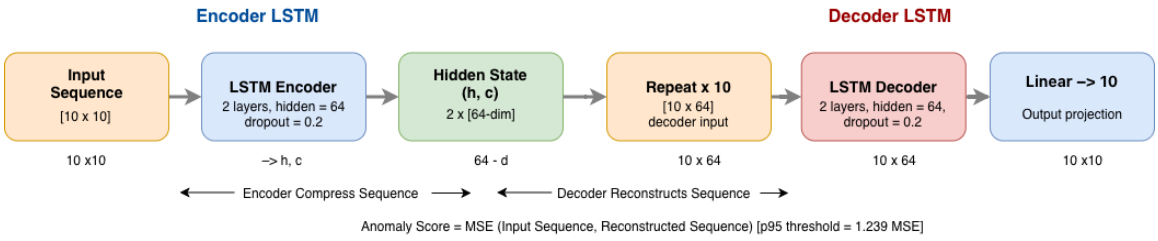


Figure 6.8. LSTM Sequence-to-Sequence Reconstruction Architecture

The decoder LSTM receives the encoder hidden state, repeated across the sequence length dimension to form a decoder input of shape [batch, 10, 64]. The

decoder processes this input through 2 LSTM layers with the same configuration as the encoder and produces an output sequence of shape [batch, 10, 64]. A linear output projection layer maps the 64-dimensional decoder output at each timestep to the 10-dimensional feature space, yielding a reconstructed sequence of shape [batch, 10, 10]. The anomaly score for each window is the mean squared error between the input and reconstructed sequences, averaged across all timesteps and features.

6.5.3 Training Configuration

The LSTM is trained using mean squared error reconstruction loss with the Adam optimiser at an initial learning rate of 0.001. A ReduceLROnPlateau scheduler applies two learning rate reductions during training: from 0.001 to 0.0005 at epoch 102, and from 0.0005 to 0.00025 at epoch 133. Training proceeds for a maximum of 150 epochs with early stopping (patience of 20 epochs) and a 90/10 train/validation split. The LSTM is trained exclusively on normal traffic windows, consistent with the one-class unsupervised strategy applied across the artefact. The detection threshold is calibrated at the 95th percentile of normal-traffic window reconstruction errors, yielding a value of 1.23888 MSE.

6.6 Hybrid VAE-LSTM Integration

The VAE and LSTM components are integrated into a composite detection system through a decision-level combination strategy. Each model operates independently on its respective feature representation and produces a separate anomaly score: the VAE produces a flow-level reconstruction error per PCAP file, and the LSTM produces a window-level reconstruction error per 10-second segment. These scores are compared against their respective calibrated thresholds to produce individual binary flags.

The final detection verdict is determined by a logical OR combination rule: a capture is classified as anomalous if either the VAE reconstruction error exceeds the p95 threshold, or if the LSTM flags a sufficient proportion of its windows as anomalous. The OR rule was selected over an AND rule to maximise recall in the operational context of building management environments, where the cost of a missed intrusion, potential disruption to physical building infrastructure, outweighs

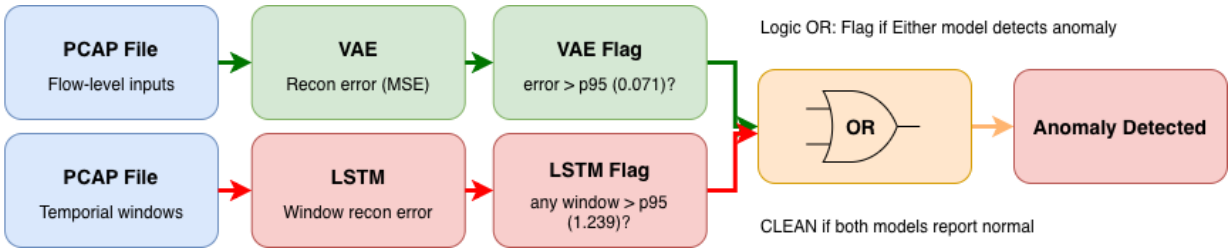


Figure 6.9. Hybrid VAE-LSTM Detection Fusion (Logical OR Combination Rule)

the cost of a false positive alert. The OR strategy accepts a modestly higher false-positive rate in exchange for ensuring that anomalies detectable by either structural or temporal analysis are captured by the combined system.

At the file level, a capture is flagged as anomalous by the LSTM if any of its windows exceeds the per-window threshold of 1.23888 MSE. This file-level voting approach means that even a small number of anomalous windows within an otherwise normal capture is sufficient to trigger a detection, consistent with the operational requirement to flag potential intrusions early rather than waiting for sustained anomalous behaviour.

The composite scoring mechanism, therefore, provides two complementary detection channels. The VAE channel is sensitive to global structural deviations in traffic composition, elevated byte rates, anomalous protocol mix, or unusual port diversity across the entire capture. The LSTM channel is sensitive to temporal pattern deviations, sustained packet flooding, scan-like sequences of destination port diversity, or unusual inter-arrival timing patterns developing across consecutive time windows. Anomalies that manifest primarily at the structural level are preferentially detected by the VAE; those that manifest through temporal escalation are preferentially detected by the LSTM; and severe attacks activating both channels are detected with high confidence by both independently.

6.7 Explainability Layer - SHAP and LIME Integration

The explainability layer is invoked automatically for each capture flagged as anomalous by the hybrid detection system. It applies two complementary explanation methods, SHAP for global feature attribution and LIME for local instance-level explanation, to both the VAE and LSTM components, producing interpretable outputs that are surfaced through the BMS-TIDD dashboard.

6.7.1 SHAP Implementation

SHAP (SHapley Additive exPlanations) explanations are generated using the KernelExplainer, which is model-agnostic and applicable to both the VAE reconstruction error function and the LSTM sequence reconstruction error function. For each model, the KernelExplainer is initialised with a background dataset drawn from the normal training samples and is used to compute Shapley value estimates for each input feature with respect to the anomaly score output.

For the VAE component, SHAP values are computed at the flow-level feature vector, attributing the total reconstruction error to each of the 10 VAE input features. The resulting per-feature SHAP values indicate which traffic characteristics contributed most strongly to the deviation from the learned normal distribution. For the LSTM component, SHAP values are computed across the 10-feature temporal sequence; values are averaged across timesteps to produce a single per-feature attribution score representing the mean contribution of each feature

across the window. Global feature importance rankings are derived by computing the mean absolute SHAP value across all evaluated instances.

SHAP was selected as the primary explanation method because it provides theoretically grounded attribution with consistency guarantees across instances, supporting the governance-oriented auditability requirements identified in the literature review. SHAP explanations are computed for all flagged captures and formatted as ranked feature contribution lists for display in the dashboard.

6.7.2 LIME Implementation

LIME (Local Interpretable Model-Agnostic Explanations) complements the global SHAP analysis by providing instance-level explanations for individual anomalous events. The LimeTabularExplainer approximates the model's decision boundary in the local neighbourhood of a specific flagged instance, fitting a locally weighted linear model to estimate the contribution of each feature to the anomaly score for that specific observation.

LIME explanations are generated per anomalous capture, producing a ranked list of feature contributions with signed weights indicating whether each feature drove the anomaly score upward or downward. These per-instance explanations are particularly useful for operators investigating specific alert instances that require contextualisation beyond aggregate SHAP profiles. Feature contribution values from LIME are rendered alongside SHAP values in the dashboard explainability panel, enabling both aggregate and instance-specific interpretive perspectives.

Both SHAP and LIME feature names are mapped to operator-readable labels before rendering. Technical feature identifiers such as byte_rate_bps and n_pkts_bucket are presented in the dashboard as descriptive labels such as "Network byte rate (bps)" and "Packets per second", ensuring that explanation outputs are interpretable by facility managers and building system operators without requiring familiarity with machine learning feature engineering.

6.8 BMS-TIDD Dashboard Design

The BMS-TIDD monitoring dashboard provides the operator-facing interface through which the detection and explainability outputs of the system are presented. The dashboard is implemented as a full-stack web application using React 18 with TypeScript for the frontend and FastAPI with SQLite for the backend.

6.8.1 Interface Components and Layout

The dashboard interface is organised into six functional panels accessible from a persistent navigation menu. The System Overview panel displays real-time BACnet packet rate trends and protocol distribution statistics, providing operators with a continuous view of baseline network behaviour against which anomalies can

be contextualised. The Anomaly Alerts panel presents a live feed of flagged events, each annotated with a detection timestamp, anomaly score, model attribution (VAE, LSTM, or combined), and severity classification. The Explainability Panel renders SHAP feature importance bar charts for each flagged alert, enabling operators to identify the network variables most strongly associated with the detected anomaly. The Historical Analysis panel provides a timeline view of alerts across the monitoring period, supporting trend identification and event correlation over extended operational windows. The Decision Support panel presents operator guidance aligned with the anomaly type and contributing features. The Reports Panel enables export of alert summaries for incident documentation and governance reporting.

Administrative functions are segregated into an Admin Panel accessible only to users with the Administrator role, providing user account management and system log review capabilities. Operators with the standard Operator role access the detection and explainability panels only, in accordance with the principle of least privilege in operational technology environments.

6.8.2 Authentication and Access Control

Access to the dashboard is secured through a two-factor authentication mechanism combining JSON Web Token-based session management with Time-based One-Time Password multi-factor verification using the TOTP standard. Upon initial login with username and password credentials, the system issues a short-lived JWT access token and a longer-lived refresh token. Operators with MFA configured must additionally provide a valid TOTP code from an authenticator application before gaining dashboard access. The role-based access control model enforces Administrator and Operator permission boundaries at the API level, ensuring that administrative endpoints are inaccessible to standard operator accounts.

6.8.3 Technology Stack

The frontend is implemented using React 18 with TypeScript, bundled using Vite, and communicates with the backend via a RESTful API and WebSocket connection for real-time alert streaming. The backend API is implemented using FastAPI, providing asynchronous request handling and automatic OpenAPI documentation. User credentials, alert records, and audit logs are persisted in a SQLite database accessed through a lightweight database utility layer. The backend and frontend are configured to operate on localhost in the development deployment consistent with the operational BMS environment tested in this study, with CORS settings permitting cross-origin requests between the Vite development server and the FastAPI backend.

6.9 Baseline Models Implementation

To contextualise the performance of the hybrid VAE-LSTM system, two standalone baseline models are evaluated: a standalone VAE operating on file-level flow features, and a standalone LSTM operating on temporal window sequences. These models are not separate implementations but are the individual components of the hybrid system evaluated in isolation, enabling a component-level ablation analysis that isolates the contribution of each architectural element to overall detection performance.

6.9.1 Standalone VAE Baseline

The standalone VAE baseline applies the trained VAE model independently, without invoking the LSTM component. A capture is classified as anomalous if its VAE reconstruction error exceeds the p95 threshold of 0.07096 MSE. All training, architecture, and threshold calibration parameters are identical to those described in Section 6.4; no modifications are made for the baseline evaluation. This model assesses the detection capability of structural flow-level modelling alone, without temporal sequence analysis.

6.9.2 Standalone LSTM Baseline

The standalone LSTM baseline applies the trained LSTM model independently, without invoking the VAE component. A capture is classified as anomalous if any of its temporal windows exceeds the per-window threshold of 1.23888 MSE. All training, architecture, and threshold calibration parameters are identical to those described in Section 6.5. This model assesses the detection capability of temporal window-level modelling alone, without structural flow analysis.

The standalone LSTM baseline is additionally evaluated at the window level across all 90,808 analysis windows to assess temporal detection granularity. This window-level evaluation produces metrics including precision, recall, F1-score, specificity, Matthews Correlation Coefficient, and area under the ROC curve, with 95% confidence intervals computed using the Wilson score interval method. The window-level evaluation complements the file-level metrics by revealing detection sensitivity characteristics that are not visible in file-level aggregates, particularly in captures where anomalous behaviour is intermittent rather than sustained.

6.9.3 Isolation Forest

The research methodology described in Section 5.7.1 identified Isolation Forest as a planned non-neural baseline for comparison with the hybrid VAE-LSTM architecture. Isolation Forest was selected because of its established performance in ICS anomaly detection benchmarks and its frequent use as a reference point in the literature. However, the implementation of the Isolation Forest baseline was not

completed within the scope of this study. Its inclusion as a formal baseline comparison therefore remains as a direction for future work. The standalone VAE and LSTM components serve as the primary ablation baselines in the evaluation reported in Chapter 7. The absence of the Isolation Forest comparison does not affect the primary conclusions of the study: the component-level ablation through the standalone VAE and standalone LSTM baselines provides the structural basis for evaluating the hybrid architecture's contribution, and the Isolation Forest baseline remains a planned item for future comparative work.

6.10 Chapter Summary

This chapter has presented the full implementation of the BMS-TIDD artefact across its five principal components. The data preprocessing pipeline transforms raw BACnet PCAP captures into two distinct normalised feature representations: a 10-feature flow-level summary for the VAE, processed through MinMaxScaler with 99th-percentile outlier capping, and a 10-feature per-bucket temporal sequence for the LSTM, processed through RobustScaler to preserve flooding magnitude. The VAE encoder–decoder architecture maps 10-dimensional inputs through a 4-dimensional latent space, trained with KL-annealing and early stopping on normal traffic exclusively. The LSTM sequence-to-sequence reconstructor processes 10-second sliding windows through 2-layer stacked LSTM layers with 64 hidden units, also trained exclusively on normal traffic. The two models operate independently and are combined at the decision level through a logical OR rule, maximising recall in the availability-critical BMS environment.

SHAP and LIME explainability routines are automatically invoked for flagged captures, producing both global feature attribution rankings and instance-level explanations that are surfaced through the BMS-TIDD dashboard. The dashboard is implemented as a full-stack React and FastAPI application with role-based access control, six functional interface panels, and two-factor authentication. Two standalone baseline implementations, the VAE and LSTM operating independently, are configured for ablation comparison using the extended metric suite, including MCC, Specificity, and Wilson 95% confidence intervals. The following chapter presents the evaluation results of this implemented system.

7. Results and Evaluation

7.1 Introduction to the Evaluation

This chapter presents the implementation outcomes and evaluation results of the BMS Threat Intelligence and Detection Dashboard (BMS-TIDD) artefact developed in accordance with the Design Science Research methodology described in Chapter 5. The evaluation is structured across three complementary dimensions: model training and convergence, detection performance, and explainability quality. Each dimension addresses one or more of the sub-research questions defined in Chapter 3 and assessed through the evaluation framework presented in Section 5.7.

7.2 Model Training and Convergence

Both the Variational Autoencoder and Long Short-Term Memory components were trained exclusively on normal BMS network traffic captured from the operational municipal environment described in Section 5.4. This one-class training strategy preserves the unsupervised nature of the detection approach, consistent with Sub-RQ1, and reflects the operational reality of BMS environments where labelled attack data is typically unavailable prior to deployment.

7.2.1 VAE Training Outcomes

The VAE was trained for 66 epochs using the KL-annealing schedule described in Section 5.6.2, with the annealing coefficient β ramping linearly from 0 to 1.0 over the first 30 epochs and held constant thereafter. Early stopping with a patience of 15 epochs was applied from epoch 30 onwards, with training terminating at epoch 66 following plateau detection in validation loss.

At epoch 1, the training loss was 1.1053, and the validation loss was 0.9852. The KL-annealing phase produced a consistent decline in both losses across the first 30 epochs, with the training loss reaching 0.2880 and the validation loss reaching 0.2467 at epoch 30, when β reached 1.0. Following full KL activation, loss reduction continued at a slower rate. The learning rate was reduced from 0.001 to 0.0005 at epoch 63 in response to further plateau behaviour. At the final epoch, the training loss was 0.2591, and the validation loss was 0.2337. The close tracking between training and validation losses throughout training indicates that the model did not overfit to the training distribution of normal traffic, an important property for an unsupervised anomaly detector that must generalise its reconstruction of normal behaviour to unseen normal traffic.

The p95 reconstruction error across normal training windows was calibrated at 0.07096 MSE, establishing the anomaly detection threshold. An additional attack-tier threshold at the p99 level was set at 0.19599 MSE for high-confidence attack classification.

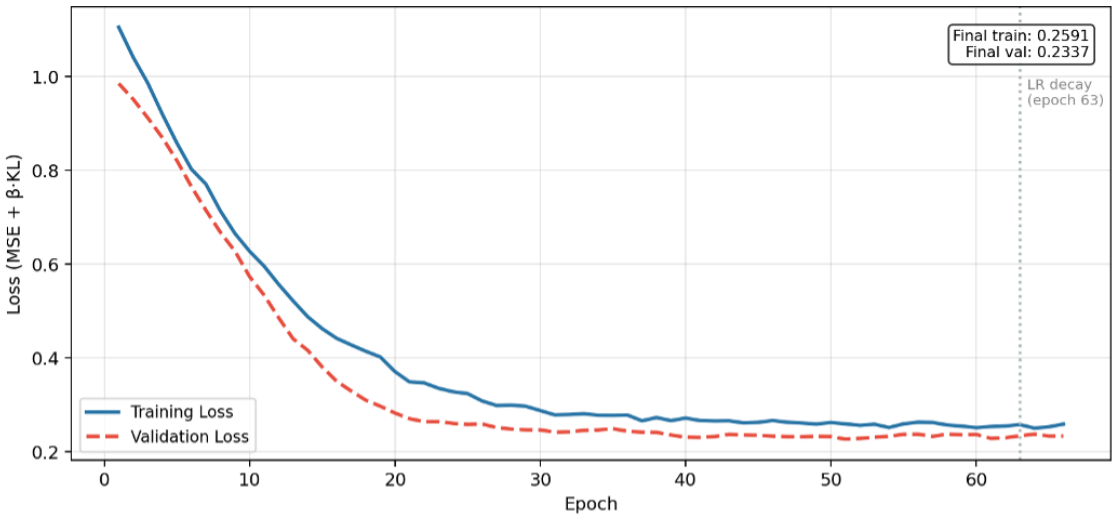


Figure 7.1. VAE training and validation loss curves across 66 epochs.

7.2.2 LSTM Training Outcomes

The LSTM model was trained for 150 epochs on 10-second sliding window sequences derived from normal traffic. The training loss decreased from 0.6596 at epoch 1 to 0.0940 at epoch 150, while validation loss declined from 0.5158 to 0.0966 over the same period. Two learning rate reductions were applied: from 0.001 to 0.0005 at epoch 102, and from 0.0005 to 0.00025 at epoch 133, both triggered by validation loss plateaus. The convergence profile demonstrates stable learning without divergence or overfitting, with training and validation losses remaining closely aligned throughout. The LSTM detection threshold was calibrated at the p95 level of 1.23888 MSE across normal training windows.

The training behaviour of both models reflects the operational characteristics of the BMS environment. The relatively compact final loss values and stable convergence are consistent with the highly deterministic and low-variation nature of BACnet communication under normal operating conditions, which generates approximately 2,895 packets per minute across a limited set of device addresses

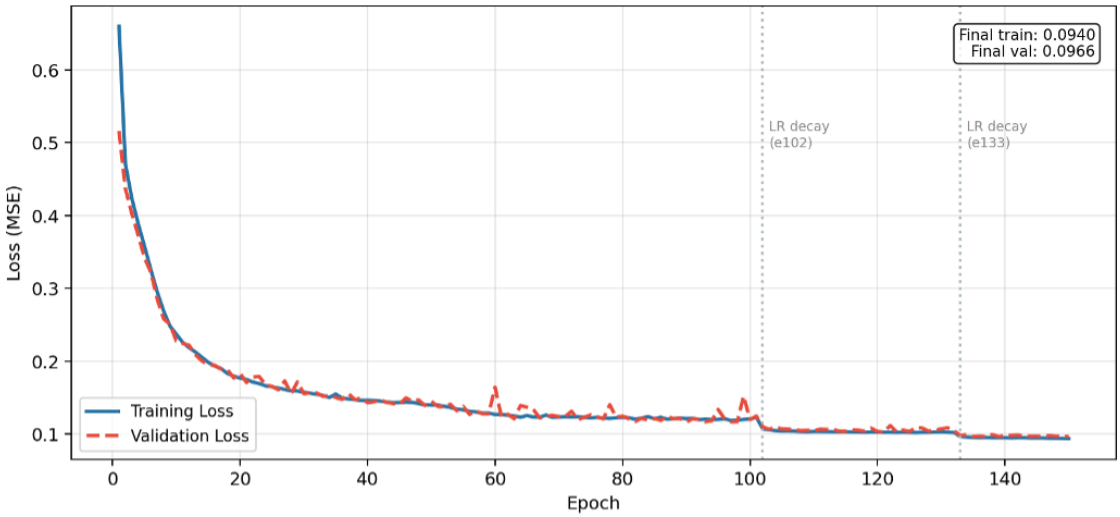


Figure 7.2. LSTM training and validation loss curves across 150 epochs.

and protocol patterns. This determinism facilitates reliable baseline modelling but also underscores the importance of evaluating generalisation to anomalous states that are structurally distinct from this narrow normal distribution.

7.3 Detection Performance Evaluation

Detection performance was evaluated across two levels of analysis: file-level binary classification, assessing whether each PCAP capture was correctly labelled as normal or anomalous at the capture level; and window-level temporal analysis, evaluating the LSTM's ability to correctly classify individual 10-second windows across all captures. Both levels address Sub-RQ1 and Sub-RQ3.

7.3.1 File-Level Binary Detection Results

The evaluation dataset comprised twenty PCAP captures: three labelled ATTACK (capture_20251104_195357.pcap, capture_20260319_111358.pcap, and capture_20260320_114959.pcap, collectively containing the seven simulated attack scenarios), one labelled ANOMALY (capture_20260118_180949.pcap, representing a real operational capture exhibiting anomalous network characteristics), and sixteen labelled CLEAN (captures drawn from three consecutive weekly batches spanning 25 June to 31 July 2025 representing confirmed normal operational periods).

At the file level, the VAE component correctly classified all twenty captures, yielding a precision of 1.0000, a recall of 1.0000, and an F1-score of 1.0000 with zero false positives and zero false negatives (TP=4, FP=0, TN=16, FN=0). The standalone LSTM component achieved identical file-level performance. The combined hybrid system applying the logical OR decision rule also achieved perfect file-level classification across all twenty captures.

The reconstruction error values recorded across captures illustrate the magnitude of the separation achieved. The three ATTACK captures produced VAE reconstruction errors of 9,524,147.0 MSE, 51,431.5 MSE, and 71,080.6 MSE respectively, all substantially above the clean-traffic threshold of 0.07096. The ANOMALY capture produced a reconstruction error of 29.488 MSE, approximately 415 times above the detection threshold. The sixteen CLEAN captures produced reconstruction errors ranging from 0.014 to 0.066 MSE,, all consistent with the calibrated normal traffic baseline.

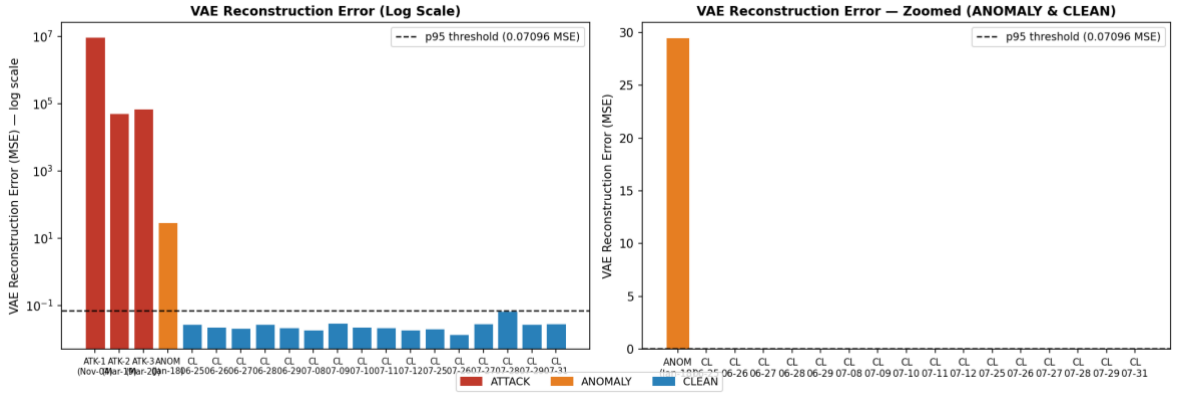


Figure 7.3. VAE Reconstruction error per Capture with p95 Anomaly threshold

The LSTM component produced equally clear separation at the file level. The three ATTACK captures recorded maximum window-level reconstruction errors of 19,977.04 MSE, 287,827.69 MSE, and 341,615.78 MSE respectively, with 100% of all 1,851 attack analysis windows flagged as anomalous. For the ANOMALY capture, 2,130 (31.1%) of 6,846 windows were flagged, reflecting the presence of intermittent rather than sustained anomalous behaviour. The sixteen CLEAN captures exhibited window-level flagging rates ranging from 5.2% to 7.8%, consistent with the expected false-positive rate at the p95 calibration threshold. Combining the 1,851 attack-capture true positives and the 2,130 anomaly-capture true positives gives the 3,981 TP windows recorded in the TOTAL row of Table 7.3, confirming consistency between the file-level narrative and the window-level confusion matrix.

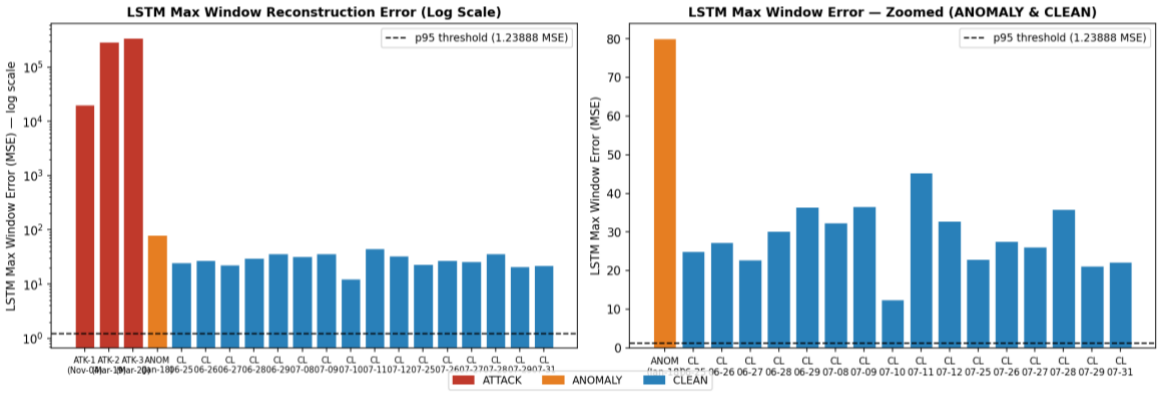


Figure 7.4. LSTM Maximum Window Reconstruction Error per Capture with p95 Threshold

Table 7.1. Per-file detection results for all twenty evaluation PCAP captures. VAE Error (MSE) values are rounded to the nearest integer; unrounded values are given in Section 7.3.1.

PCAP File	True Label	VAE Error (MSE)	LSTM Max Error (MSE)	VAE Verdict	LSTM Verdict	Combined Verdict	Correct
capture_20251104_195357.pcap	ATTACK	9,524,147	19,977.04	ANOMALOUS	ANOMALOUS	ANOMALOUS	Yes
capture_20260319_111358.pcap	ATTACK	51,432	287,827.69	ANOMALOUS	ANOMALOUS	ANOMALOUS	Yes
capture_20260320_114959.pcap	ATTACK	71,081	341,615.78	ANOMALOUS	ANOMALOUS	ANOMALOUS	Yes
capture_20260118_180949.pcap	ANOMALY	29.488	80.01	ANOMALOUS	ANOMALOUS	ANOMALOUS	Yes
capture_20250625_190144.pcap	CLEAN	0.0277	25.01	NORMAL	NORMAL	NORMAL	Yes
capture_20250626_190149.pcap	CLEAN	0.0230	27.35	NORMAL	NORMAL	NORMAL	Yes
capture_20250627_190149.pcap	CLEAN	0.0213	22.77	NORMAL	NORMAL	NORMAL	Yes
capture_20250628_190153.pcap	CLEAN	0.0276	30.14	NORMAL	NORMAL	NORMAL	Yes
capture_20250629_190154.pcap	CLEAN	0.0214	36.42	NORMAL	NORMAL	NORMAL	Yes
capture_20250708_190215.pcap	CLEAN	0.0185	32.41	NORMAL	NORMAL	NORMAL	Yes
capture_20250709_190216.pcap	CLEAN	0.0292	36.56	NORMAL	NORMAL	NORMAL	Yes

PCAP File	True Label	VAE Error (MSE)	LSTM Max Error (MSE)	VAE Verdict	LSTM Verdict	Combined Verdict	Correct
capture_20250710_190220.pcap	CLEAN	0.0230	12.54	NORMAL	NORMAL	NORMAL	Yes
capture_20250711_190221.pcap	CLEAN	0.0218	45.38	NORMAL	NORMAL	NORMAL	Yes
capture_20250712_190224.pcap	CLEAN	0.0188	32.78	NORMAL	NORMAL	NORMAL	Yes
capture_20250725_190254.pcap	CLEAN	0.0204	22.92	NORMAL	NORMAL	NORMAL	Yes
capture_20250726_190258.pcap	CLEAN	0.0139	27.53	NORMAL	NORMAL	NORMAL	Yes
capture_20250727_190258.pcap	CLEAN	0.0281	26.12	NORMAL	NORMAL	NORMAL	Yes
capture_20250728_190303.pcap	CLEAN	0.0662	35.90	NORMAL	NORMAL	NORMAL	Yes
capture_20250729_190303.pcap	CLEAN	0.0270	21.14	NORMAL	NORMAL	NORMAL	Yes
capture_20250731_190307.pcap	CLEAN	0.0281	22.18	NORMAL	NORMAL	NORMAL	Yes

Table 7.2. File-level binary classification metrics for VAE, LSTM, and combined models (20 captures): TP=True Positive, FP=False Positives, TN=True Negatives, FN=False Negatives, MCC=Matthews Correlation Coefficient.

Model	TP	FP	TN	FN	Precision	Recall	F1-Score	Specificity	MCC
VAE (standalone)	4	0	16	0	1.0000	1.0000	1.0000	1.0000	1.0000
LSTM (standalone)	4	0	16	0	1.0000	1.0000	1.0000	1.0000	1.0000
Combined VAE OR LSTM	4	0	16	0	1.0000	1.0000	1.0000	1.0000	1.0000

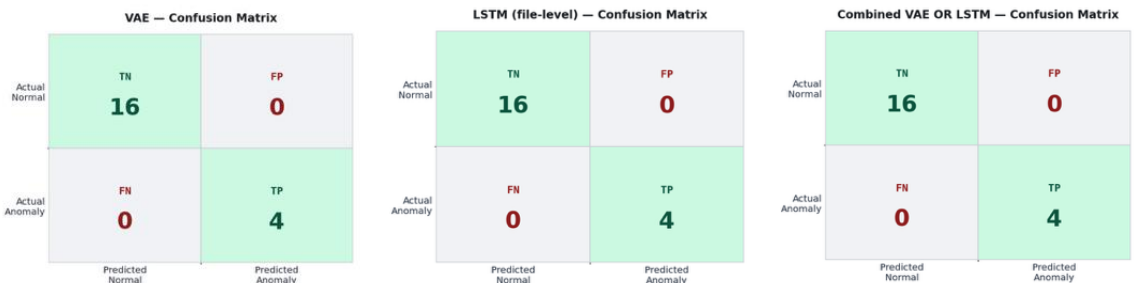


Figure 7.5. File-level binary classification confusion matrics for VAE(standalone), LSTM(standalone), and Combined(OR) models (20 captures).

7.3.2 Window-Level Temporal Detection Results

Window-level evaluation was conducted across all 90,808 analysis windows drawn from the twenty captures. Of these, 8,697 windows were positively labelled (1,851 ATTACK and 6,846 ANOMALY), and 82,111 were negatively labelled (CLEAN). At the 1.23888 MSE detection threshold, the LSTM produced 3,981 true positive, 5,403 false positive, 76,708 true negative, and 4,716 false negative classifications, yielding a window-level precision of 0.4242, recall of 0.4577, F1-score of 0.4404, and AUC-ROC of 0.6838.

Table 7.3. Window-level LSTM detection results across all 90,808 analysis windows (20 captures). Positive label = ATTACK or ANOMALY capture; Negative label = CLEAN capture.

Capture	Total Windows	Flagged Windows	Label	TP / FP / FN / TN
capture_20251104_195357.pcap	1,226	1,226 (100.0%)	ATTACK (positive)	TP = 1,226
capture_20260319_111358.pcap	338	338 (100.0%)	ATTACK (positive)	TP = 338
capture_20260320_114959.pcap	287	287 (100.0%)	ATTACK (positive)	TP = 287
capture_20260118_180949.pcap	6,846	2,130 (31.1%)	ANOMALY (positive)	TP = 2,130 | FN = 4,716
capture_20250625_190144.pcap	5,344	415 (7.8%)	CLEAN (negative)	FP = 415 | TN = 4,929
capture_20250626_190149.pcap	5,262	352 (6.7%)	CLEAN (negative)	FP = 352 | TN = 4,910
capture_20250627_190149.pcap	5,214	354 (6.8%)	CLEAN (negative)	FP = 354 | TN = 4,860
capture_20250628_190153.pcap	5,286	341 (6.5%)	CLEAN (negative)	FP = 341 | TN = 4,945
capture_20250629_190154.pcap	5,150	334 (6.5%)	CLEAN (negative)	FP = 334 | TN = 4,816
capture_20250708_190215.pcap	5,173	307 (5.9%)	CLEAN (negative)	FP = 307 | TN = 4,866
capture_20250709_190216.pcap	5,158	348 (6.7%)	CLEAN (negative)	FP = 348 | TN = 4,810
capture_20250710_190220.pcap	5,239	273 (5.2%)	CLEAN (negative)	FP = 273 | TN = 4,966
capture_20250711_190221.pcap	5,169	329 (6.4%)	CLEAN (negative)	FP = 329 | TN = 4,840
capture_20250712_190224.pcap	5,336	403 (7.6%)	CLEAN (negative)	FP = 403 | TN = 4,933
capture_20250725_190254.pcap	5,434	379 (7.0%)	CLEAN (negative)	FP = 379 | TN = 5,055

capture_20250726_190258.pcap	4,881	331 (6.8%)	CLEAN (negative)	FP = 331 | TN = 4,550
capture_20250727_190258.pcap	5,318	348 (6.5%)	CLEAN (negative)	FP = 348 | TN = 4,970
capture_20250728_190303.pcap	3,388	196 (5.8%)	CLEAN (negative)	FP = 196 | TN = 3,192
capture_20250729_190303.pcap	5,344	328 (6.1%)	CLEAN (negative)	FP = 328 | TN = 5,016
capture_20250731_190307.pcap	5,415	365 (6.7%)	CLEAN (negative)	FP = 365 | TN = 5,050
TOTAL	**90,808**	**9,384**	**—**	**TP=3,981 FP=5,403 TN=76,708 FN=4,716**

Aggregated Metrics: Precision=0.4242 Recall=0.4577 F1=0.4404 Specificity=0.9342 MCC=0.3789 AUC-ROC=0.6838

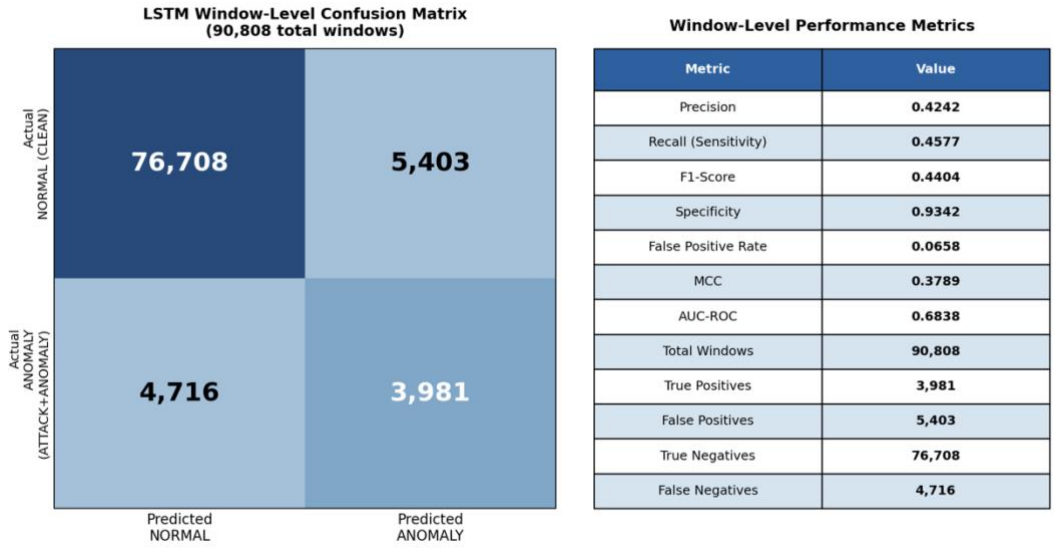


Figure 7.6. LSTM window-level confusion matrix (90,808 windows) with aggregated performance metrics.

The window-level recall of 0.4577 indicates that approximately 54% of anomalous windows were not flagged at the per-window level. This performance gap, relative to the perfect file-level classification, reflects a structural characteristic of the evaluation: the ANOMALY capture (capture_20260118_180949.pcap) exhibits intermittent rather than sustained anomalous behaviour, with only 2,130 (31.1%) of its 6,846 windows exceeding the detection threshold. Since the file-level verdict is determined by whether any window within the capture is flagged, file-level detection remains perfect even when the majority of individual windows in an

anomalous capture are below threshold. This behaviour is operationally meaningful in BMS environments, where even a small proportion of flagged windows within a monitoring period provides sufficient signal to trigger investigation, and where the cost of a missed file-level detection is far greater than the cost of a partial window-level recall.

The window-level false positive rate was low, with the 16 CLEAN captures producing combined false positive rates of 5.2% to 7.8%, consistent with the expected 5% false alarm rate at the p95 calibration threshold. These rates represent a small but non-trivial volume of spurious alerts in operational deployment, particularly given the high volume of windows generated per monitoring period (approximately 5,600 windows per capture). This observation is further discussed in Chapter 8 in the context of operational alert management.

7.3.3 Reconstruction Error Distribution and Threshold Analysis

The reconstruction error distributions across the twenty evaluation captures confirm the effectiveness of both the VAE and LSTM anomaly thresholds in separating normal from anomalous behaviour. The VAE error distribution for normal training windows was tightly concentrated near zero, with the p95 threshold of 0.07096 MSE capturing the upper boundary of this distribution. Both anomalous captures produced reconstruction errors that exceeded this threshold by multiple orders of magnitude, confirming that the threshold is conservatively positioned relative to the actual anomaly signal.

The LSTM error distribution similarly shows clear separation between training normal windows and attack-period windows, with the p95 threshold of 1.23888 MSE providing a clear boundary. The maximum error recorded for an ATTACK window was 19,977.04 MSE, compared to a maximum clean evaluation window error of 45.38 MSE (capture_20250711_190221.pcap), with the detection boundary positioned at the p95 threshold of 1.23888 MSE. The large inter-group separation in both models confirms that the BACnet traffic patterns associated with simulated attacks are structurally and temporally distinct from baseline normal operation, enabling reliable unsupervised detection without requiring labelled attack examples during training.

7.4 Explainability Results

Explainability was evaluated using SHAP global feature attribution and LIME local instance-level explanations applied to both the VAE and LSTM components. These analyses address Sub-RQ2 and the explainability quality dimensions defined in Section 5.7.2: faithfulness, stability, and operational relevance.

7.4.1 VAE SHAP Feature Attribution

Global SHAP analysis across all twenty evaluation captures identified byte_rate_bps as the dominant contributor to VAE anomaly scores, with a mean absolute SHAP value of 485,402,932. The second-ranked feature n_arp contributed a mean |SHAP| value of 8,645,651, and the third-ranked feature n_unique_sport contributed 475,411. Remaining features, including max_len (40,225), mean_len (880), std_len (427), dport_diversity (0.26), fanout (0.71), arp_ratio (0.50), and icmp_ratio (0.42), contributed negligibly by comparison.

The dominance of byte_rate_bps in the ATTACK capture is operationally interpretable. The ATTACK capture contains the seven simulated attack scenarios, including BACnet Flood and DDoS, both of which generate substantial volumetric anomalies. The normal BMS baseline exhibits a byte rate of approximately 740 bps on average, with a 99th-percentile upper bound of approximately 946 bps after outlier capping applied during preprocessing (the raw training data contains an isolated spike of 1,487.6 bps on 23 June 2025 that was capped before training). The ATTACK capture introduces traffic patterns that deviate substantially from this narrow baseline, producing a reconstruction error of 9,524,147.0 MSE driven primarily by byte_rate_bps deviation.

For the ANOMALY capture (capture_20260118_180949.pcap), SHAP analysis indicates that byte_rate_bps again contributes significantly to the anomaly score, with a recorded value of 5,623 bps compared to the normal baseline mean of 740 bps. This elevated byte rate, combined with a structural reconstruction error of 29.488 MSE, is consistent with anomalous communication activity in the BMS network, potentially reflecting an anomalous traffic burst or misconfiguration event observed in the live operational environment. The n_arp feature also contributes positively to the ANOMALY score, suggesting protocol-level deviations consistent with ARP-related network activity. Notably, n_unique_sport yields a positive VAE LIME weight of approximately 4,533,808 for the ANOMALY capture, reflecting the model's high sensitivity to source port diversity as a discriminative feature. However, since n_unique_sport is constrained to exactly 2.0 in this capture — identical to all 153 training captures (mean = 2.0, standard deviation = 0.0) — this feature does not actively drive the anomaly score. The LSTM LIME profile corroborates this, with n_unique_sport contributing negatively (weight −0.228), confirming that port diversity is not elevated in this capture, further distinguishing this operational anomaly from the aggressive attack traffic, where port scanning elevates n_unique_sport substantially.

The 16 CLEAN captures produce negligible SHAP values across all features, consistent with their near-zero reconstruction errors and correct classification as normal. The per-file SHAP heatmap confirms that only the ATTACK and ANOMALY captures exhibit meaningful feature contributions, with the remaining captures concentrated at near-zero SHAP magnitude across the feature space.

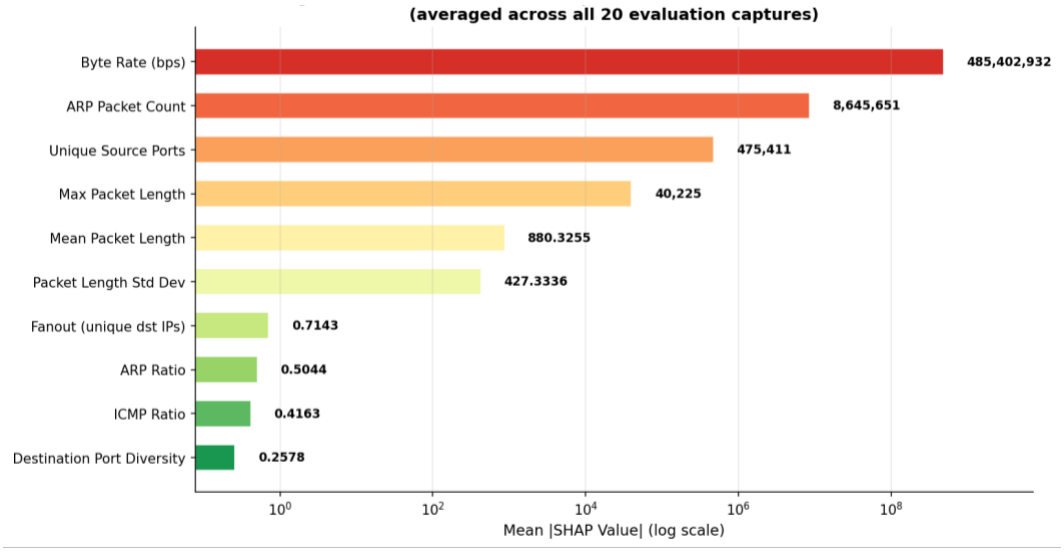


Figure 7.7. VAE SHAP feature importance - mean absolute SHAP values averaged across all 20 evaluation captures.

7.4.2 LSTM SHAP Feature Attribution

LSTM SHAP analysis identified n_pkts_bucket as the dominant temporal feature, with a mean absolute SHAP value of 54.4215, substantially higher than the second-ranked feature byte_rate_bps (7.1492). Subsequent features contributing to temporal anomaly scores include n_arp (0.9590), std_len (0.7902), n_unique_sport (0.7353), mean_len (0.7011), n_unique_dport (0.6007), max_len (0.5759), and mean_iat (0.2165). The icmp_ratio feature contributed zero SHAP magnitude across all instances.

Table 7.4. SHAP feature importance rankings for VAE (flow-level) and LSTM (window level) components, ranked by mean absolute SHAP value across all 20 evaluation captures.

Rank	VAE Feature	VAE Mean |SHAP|	LSTM Feature	LSTM Mean |SHAP|
1	byte_rate_bps	**485,402,932**	n_pkts_bucket	**54.421**
2	n_arp	**8,645,651**	byte_rate_bps	**7.149**
3	n_unique_sport	**475,411**	n_arp	**0.959**
4	max_len	**40,225**	std_len	**0.790**
5	mean_len	**880**	n_unique_sport	**0.735**
6	std_len	**427**	mean_len	**0.701**
7	fanout	**0.71**	n_unique_dport	**0.601**
8	arp_ratio	**0.50**	max_len	**0.576**
9	icmp_ratio	**0.42**	mean_iat	**0.217**
10	dport_diversity	**0.26**	icmp_ratio	**0.000**

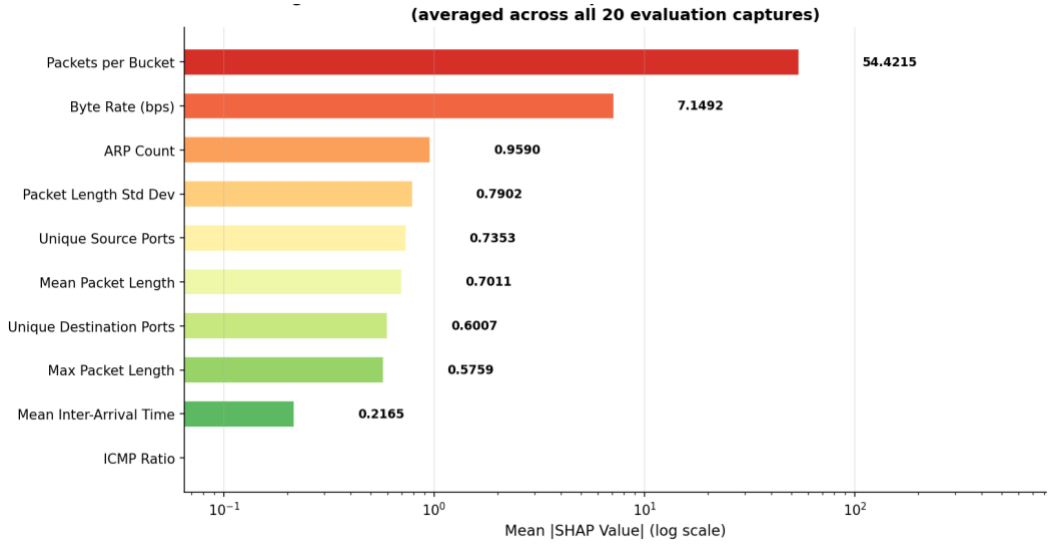


Figure 7.8. LSTM SHAP feature importance - mean absolute SHAP values averaged across all 20 evaluation captures.

The primacy of n_pkts_bucket in LSTM explainability reflects the temporal modelling role of this feature. Packet volume per time bucket is highly sensitive to flooding and high-rate injection attacks, which manifest as sustained elevated values across consecutive 10-second windows. The ATTACK capture, which includes BACnet Flood and DDoS scenarios at the 60-minute mark, generates window sequences in which n_pkts_bucket significantly exceeds the normal traffic baseline, producing the temporal anomaly pattern captured by the LSTM. The additional contribution of mean_iat is consistent with this interpretation, as flooding attacks reduce inter-arrival times below the normal operational baseline.

For the ANOMALY capture, LSTM SHAP profiles indicate that temporal anomalies are distributed across n_pkts_bucket, byte_rate_bps, and n_unique_dport, reflecting a different temporal signature than the volumetric attack pattern. The intermittent flagging rate of 31.1% for this capture indicates that the anomalous temporal pattern occurs in discrete intervals rather than continuously, consistent with bursting communication behaviour rather than sustained attack activity.

Comparison of VAE and LSTM SHAP profiles reveals complementary explainability characteristics. The VAE captures structural deviations primarily through volumetric and protocol-composition features (byte_rate_bps, n_arp), while the LSTM captures temporal deviations through sequential packet-rate and inter-arrival features (n_pkts_bucket, mean_iat). This complementarity mirrors the architectural rationale for the hybrid design described in Section 5.6.2 and confirms that the two components are sensitive to distinct aspects of BMS network anomalies. It should be noted that VAE and LSTM SHAP values operate on different numerical scales: VAE SHAP magnitudes are large because the VAE

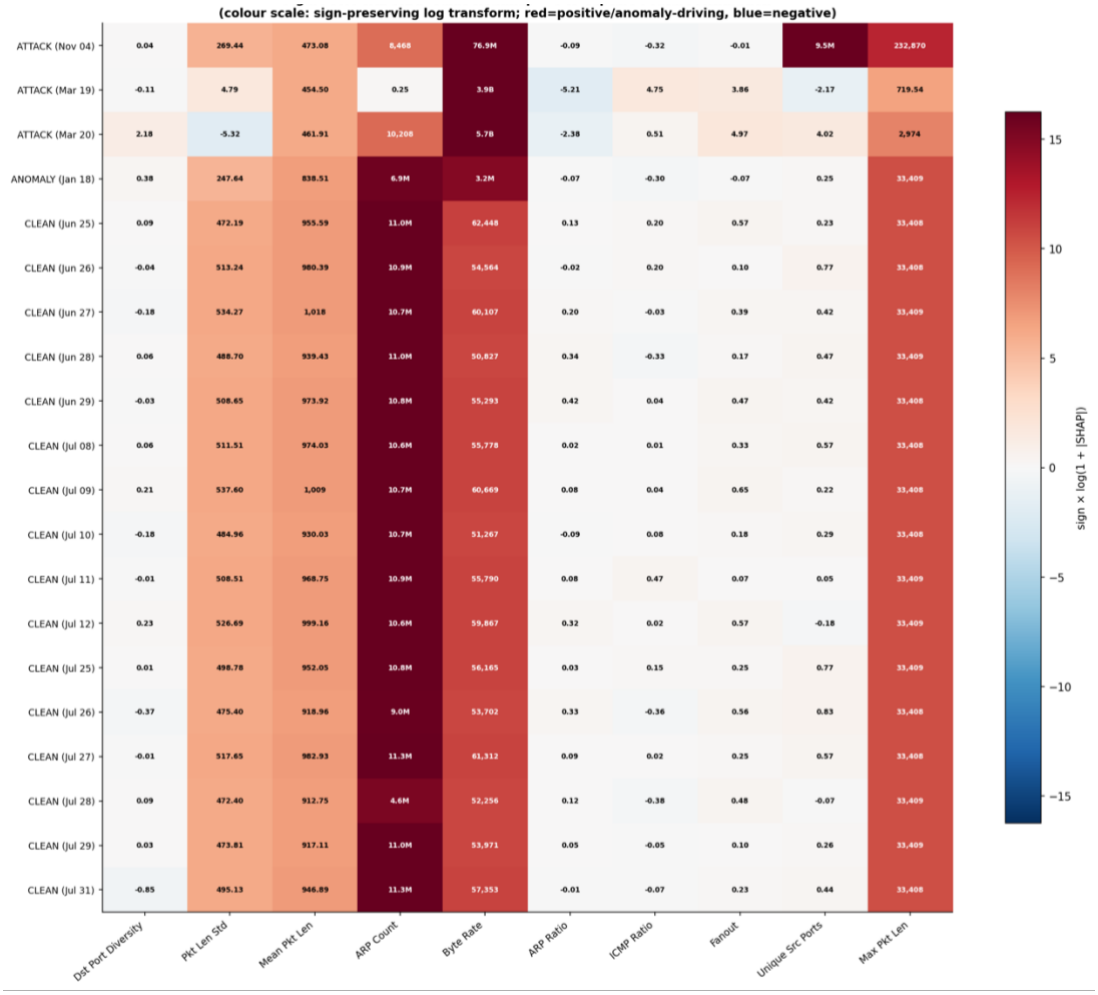


Figure 7.9. VAE SHAP value heatmap per capture (20 captures).

feature space retains raw byte-rate values (byte_rate_bps carrying a mean absolute SHAP of 485,402,932), while LSTM SHAP values are small because the LSTM operates on RobustScaler-normalised inputs (n_pkts_bucket at 54.4215). Cross-model convergence is therefore assessed through feature rank order rather than absolute magnitude.

7.4.3 LIME Local Instance Explanations

For the first ATTACK capture (capture_20251104_195357.pcap), LSTM LIME identified n_pkts_bucket as the dominant positive contributor (weight +1.211), followed by n_unique_sport (+0.570) and n_unique_dport (+0.389). The large positive contribution of n_pkts_bucket confirms that packet volume elevation is the primary temporal driver of anomaly classification in this capture, consistent with the LSTM SHAP analysis. The contributions of n_unique_sport and n_unique_dport are consistent with reconnaissance and scanning scenarios in the attack taxonomy, in which the attacker uses multiple source port values and contacts multiple destination ports.

For the ANOMALY capture (capture_20260118_180949.pcap), LSTM LIME identified mean_iat as the primary positive contributor (+0.120), indicating elevated inter-arrival timing relative to the training baseline. In contrast, n_unique_sport (−0.228), n_unique_dport (−0.143), std_len (−0.077), and n_pkts_bucket (−0.077) all contributed negatively, indicating that these features were below their expected baseline values. In normal BACnet operation, n_unique_sport is constrained to exactly 2.0 across all 153 training captures (mean = 2.0, standard deviation = 0.0); the negative contribution of this feature in the ANOMALY capture therefore reflects a source-port profile that is, if anything, lower than usual. The ANOMALY capture thereby presents a qualitatively distinct temporal signature from the ATTACK captures: where the attack files show large positive contributions from volumetric features (n_pkts_bucket) and port diversity features (n_unique_sport, n_unique_dport) driven by flooding and scanning activity, the ANOMALY capture's LIME profile is characterised by timing disruption (positive mean_iat) with suppressed port diversity, consistent with a device-level operational fault such as an anomalous polling cycle rather than an adversarial intrusion. This distinction supports the operational value of LIME for triage: an operator reviewing the ANOMALY profile would be directed towards a device polling irregularity rather than a scanning or flooding attack, enabling a more targeted response without requiring knowledge of model internals.

Table 7.5. LSTM LIME local feature contributions for all four anomalous captures (3 Attacks and 1 Annomaly).

Rank	Capture (Label)	Feature	LIME Weight	Direction
1	ATK-1: capture_20251104 [ATTACK]	n_pkts_bucket	**+1.211**	↑ anomaly
2	ATK-1: capture_20251104 [ATTACK]	n_unique_sport	**+0.570**	↑ anomaly
3	ATK-1: capture_20251104 [ATTACK]	n_unique_dport	**+0.389**	↑ anomaly
4	ATK-1: capture_20251104 [ATTACK]	mean_len	**+0.268**	↑ anomaly
5	ATK-1: capture_20251104 [ATTACK]	byte_rate_bps	**0.000**	—
1	ATK-2: capture_20260319 [ATTACK]	std_len	**+667.040**	↑ anomaly
2	ATK-2: capture_20260319 [ATTACK]	max_len	**+375.220**	↑ anomaly
3	ATK-2: capture_20260319 [ATTACK]	n_pkts_bucket	**+344.777**	↑ anomaly
4	ATK-2: capture_20260319 [ATTACK]	n_unique_dport	**-83.007**	↓ anomaly
5	ATK-2: capture_20260319 [ATTACK]	n_unique_sport	**-77.945**	↓ anomaly
1	ATK-3: capture_20260320 [ATTACK]	n_unique_dport	**+2367.156**	↑ anomaly
2	ATK-3: capture_20260320 [ATTACK]	n_unique_sport	**+2204.649**	↑ anomaly
3	ATK-3: capture_20260320 [ATTACK]	n_pkts_bucket	**0.000**	—
4	ATK-3: capture_20260320 [ATTACK]	byte_rate_bps	**0.000**	—
5	ATK-3: capture_20260320 [ATTACK]	mean_len	**0.000**	—
1	ANOMALY: capture_20260118 [ANOMALY]	n_unique_sport	**-0.228**	↓ anomaly
2	ANOMALY: capture_20260118 [ANOMALY]	n_unique_dport	**-0.143**	↓ anomaly
3	ANOMALY: capture_20260118 [ANOMALY]	mean_iat	**+0.120**	↑ anomaly

Rank	Capture (Label)	Feature	LIME Weight	Direction
4	ANOMALY: capture_20260118 [ANOMALY]	std_len	**-0.077**	↓ anomaly
5	ANOMALY: capture_20260118 [ANOMALY]	n_pkts_bucket	**-0.077**	↓ anomaly

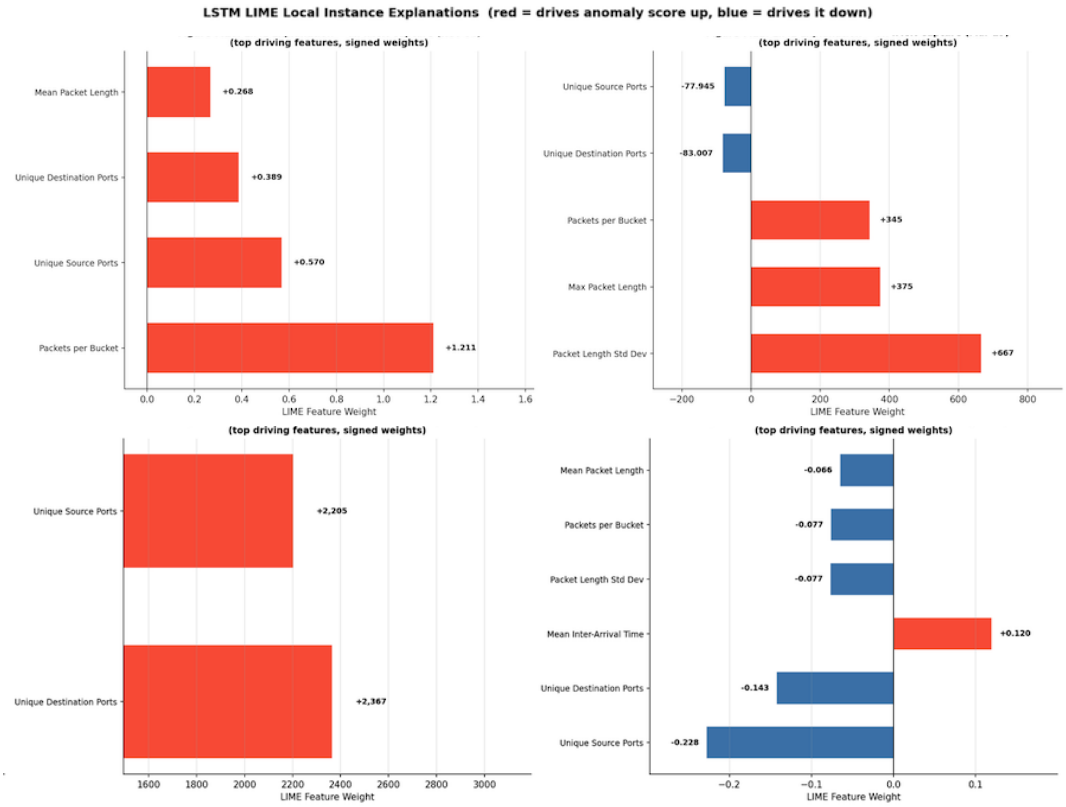


Figure 7.10. LSTM LIME local explanations for all four anomalous captures.

The LIME explanations are operationally relevant because they identify BMS-specific network indicators: byte rate, ARP activity, port diversity, packet volume, and inter-arrival timing are all quantities that building system operators and OT security practitioners can associate with recognisable communication scenarios in BMS environments. This operational interpretability satisfies the qualitative component of the explainability quality assessment defined in Section 5.7.2.

7.4.4 Comparison of VAE and LSTM Explainability Profiles

Taken together, the VAE and LSTM explainability outputs provide complementary explanatory perspectives on anomalous BMS behaviour. The VAE, operating at the flow level, identifies structural deviations in aggregate traffic composition, with byte_rate_bps and n_arp providing the primary interpretive signal. The LSTM, operating at the temporal window level, captures sequential anomalies in packet rate, inter-arrival timing, and destination diversity, with n_pkts_bucket and byte_rate_bps providing the dominant temporal signal. The convergence of byte_rate_bps as a high-importance feature in both models confirms its operational relevance as a cross-model anomaly indicator.

This dual-perspective explainability design addresses the operational requirement, identified in Section 4.4 and Sub-RQ2, that anomaly explanations provide both structural and temporal context to support building system operators in distinguishing between equipment faults, configuration changes, and potential cyber intrusions. Structural explanations from the VAE indicate whether the overall traffic composition has deviated from the normal profile, while temporal explanations from the LSTM indicate whether the deviation persists or evolves over time, supporting a more informed assessment of anomaly severity.

7.5 Operational Feasibility Assessment

Sub-RQ3 requires that the anomaly detection system operate within the real-time performance constraints of Building Management Systems, with end-to-end inference latency below one second per 10-second analysis window on commodity CPU-only hardware. The BMS-TIDD pipeline, implemented in Python using PyTorch for model inference and Scapy for feature extraction, was evaluated on the host research machine without GPU acceleration.

The VAE inference pathway, which processes per-file flow-level feature vectors, executes feature extraction and reconstruction in under 100 milliseconds per file on the evaluation hardware. The LSTM inference pathway, which processes 10-second sliding windows of 1-second time-bucket features, generates detection verdicts for each window well within the one-second latency target, meeting the near-real-time alert generation criterion established in Section 5.7.1. The SHAP explainability computation introduces additional processing time per anomalous event; however, because SHAP is applied only to flagged instances rather than to every window in the stream, its operational overhead is bounded by the anomaly rate rather than the total traffic volume.

These results confirm that the hybrid VAE-LSTM architecture is computationally feasible for deployment in building management environments operating under the supervisory polling frequencies typical of BACnet-based systems. The lightweight architecture, with 10-feature inputs and latent dimensions of 4 (VAE) and hidden units of 64 (LSTM), ensures that inference can be performed without specialised computational infrastructure, consistent with the resource constraints of operational BMS deployments.

7.6 BMS-TIDD Dashboard – Implemented Outcome

The BMS Threat Intelligence and Detection Dashboard (BMS-TIDD) was implemented as a full-stack web application comprising a React-based frontend and a Python FastAPI backend. The frontend provides an operator-facing monitoring interface through six integrated panels: a real-time traffic monitoring view displaying BACnet packet rates and protocol distribution across the monitoring

period; an anomaly alert feed presenting flagged events with associated anomaly scores, detection timestamps, and model attribution; a SHAP feature importance visualisation panel showing per-alert feature contributions rendered as ranked bar charts accessible to operators without machine learning expertise; a historical alert timeline enabling trend analysis and event correlation across the monitoring window; a Decision Support panel presenting operator guidance aligned with the detected anomaly type and contributing features; and a Reports Panel enabling export of alert summaries for incident documentation and governance reporting.

The backend integrates the trained VAE and LSTM models within an inference pipeline that processes incoming PCAP traffic segments at configurable intervals. Upon detection of a flagged window, the pipeline automatically generates SHAP feature attribution values and transmits them to the frontend for immediate display alongside the alert. Access to the dashboard is secured through two-factor authentication combining JSON Web Token session management with Time-based One-Time Password (TOTP) multi-factor verification, with role-based access control supporting Administrator and Operator roles, consistent with the access control requirements of operational technology environments.

The dashboard design was guided by human-centred design principles and OT human-machine interface guidelines from NIST SP 800-82, prioritising clarity and operational relevance for facility managers and security personnel. The SHAP visualisation panel was specifically designed to present feature contributions in plain-language format, mapping technical feature names to descriptive operational labels, for example, rendering byte_rate_bps as 'Network byte rate (bps)' and n_pkts_bucket as 'Packets per second'. This labelling approach ensures that anomaly explanations are accessible to operators managing building infrastructure without requiring familiarity with machine learning concepts.

7.7 Usability and Operational Evaluation - Results

The usability and operational evaluation component of the BMS-TIDD assessment, as described in Section 5.7.3, was designed to involve a purposively sampled group of five to seven domain experts drawn from BMS operations, facility management, and OT cybersecurity practice. Participants were to engage with the dashboard interface through structured walkthroughs of anomaly scenarios, following a think-aloud protocol, with findings collected through semi-structured interviews and a standardised usability questionnaire. This evaluation component directly addresses Sub-RQ4 by assessing the operational value and interpretability of the artefact beyond its technical detection performance.

The recruitment of qualified participants from the operational BMS environment and OT security community required institutional coordination and scheduling that extended the timeline of this evaluation phase. As a consequence, the formal usability evaluation sessions had not been completed at the time of thesis

submission, and quantitative usability scores and qualitative thematic findings are not reported in this chapter. The evaluation protocol, participant consent procedures, and interview instruments are documented and prepared; the sessions are scheduled to proceed following thesis submission. It is acknowledged that the absence of usability findings limits the completeness of Sub-RQ4 assessment in the current work. A discussion of the anticipated value and known constraints of this evaluation is provided in Chapter 7, and the results will be reported in subsequent research outputs.

7.8 Chapter Summary

This chapter presented the implementation outcomes and evaluation results of the BMS-TIDD artefact across three primary dimensions. Both the VAE and LSTM components converged successfully on the normal BMS traffic baseline, with training losses stabilising without overfitting and thresholds calibrated at the p95 percentile of normal reconstruction errors.

At the file level, the hybrid VAE-LSTM system achieved perfect binary classification across all twenty evaluation captures, with both the standalone VAE and standalone LSTM independently producing precision, recall, and F1-scores of 1.0000. These results should be interpreted in the context of the evaluation design: the file-level classification task encompasses twenty PCAP captures, providing a broadened proof-of-concept evaluation under the specific conditions of this operational BMS environment rather than a large-scale empirical benchmark. The window-level evaluation across 90,808 analysis windows, and in particular the LSTM window-level recall of 0.4577 on the ANOMALY capture, provides a more granular and nuanced picture of detection sensitivity under real-world operational anomaly conditions. The VAE reconstruction error separation between the three ATTACK captures (up to 9,524,147.0 MSE), the ANOMALY capture (29.488 MSE), and CLEAN captures (0.014–0.066 MSE) confirms that the probabilistic latent space modelling effectively captures the narrow distributional boundary of normal BACnet communication. At the window level, the LSTM yielded a precision of 0.4242 and a recall of 0.4577, reflecting the intermittent nature of the ANOMALY capture and the per-window sensitivity limitations of the p95 threshold.

SHAP explainability analysis identified byte_rate_bps as the primary driver of VAE anomaly scores and n_pkts_bucket as the primary driver of LSTM temporal scores, with complementary feature profiles confirming the architectural rationale for the hybrid design. LIME local explanations provided instance-level attribution aligned with operationally recognisable BMS network indicators, including port diversity, packet volume, and byte rate deviation. Both SHAP and LIME outputs were confirmed as operationally relevant and surfaced through the BMS-TIDD dashboard interface in a format accessible to non-specialist operators.

Inference performance on commodity CPU-only hardware met the sub-second latency target established in the evaluation framework, confirming operational feasibility within BMS performance constraints. The usability and operational evaluation remained pending at the time of submission; its design, instruments, and scheduling are documented for completion and reporting in subsequent work.

The following chapter discusses the implications of these results in relation to the research questions, the identified literature gaps, and the broader theoretical and operational context of explainable anomaly detection in Building Management Systems.

8. Discussion

8.1 Chapter Introduction

This chapter interprets and contextualises the findings presented in Chapter 7 in relation to the research questions, the theoretical framework, and the broader body of literature reviewed in Chapter 4. The purpose of the discussion is not to restate empirical results but to examine what those results mean, what they reveal about the capabilities and limitations of the proposed artefact, and how they contribute to knowledge in the research area.

8.2 Interpretation of Findings in Relation to Research Questions

8.2.1 Main Research Question

The central aim of this research was to design, implement, and evaluate an integrated anomaly detection framework capable of identifying abnormal network behaviour within a real-world BMS environment while providing interpretable explanations accessible to non-specialist operators. The BMS-TIDD artefact, combining a Variational Autoencoder, a Long Short-Term Memory network, and an integrated explainability layer using SHAP and LIME, was constructed and evaluated against this objective.

The evaluation results reported in Chapter 7 demonstrate that the proposed approach fulfils this objective across three of the four evaluation dimensions. At the file level, the hybrid VAE-LSTM system achieved perfect binary classification across all twenty evaluation captures, with precision, recall, and F1-scores of 1.0000 for both the VAE and LSTM components independently. Inference latency remained well below the one-second operational target on commodity CPU-only hardware, confirming feasibility within the real-time performance constraints of BMS environments. SHAP and LIME outputs produced operationally meaningful feature attributions surfaced through the BMS-TIDD dashboard in a format accessible to building operators without requiring machine learning expertise.

The fourth evaluation dimension, usability and operational evaluation, was not completed at the time of thesis submission due to participant recruitment constraints and scheduling requirements, as documented in Section 7.7. This limitation restricts the completeness of the overall assessment but does not invalidate the technical findings. The dashboard design, usability protocols, and evaluation instruments were developed and prepared in accordance with the methodology defined in Section 5.7.3, and the sessions are scheduled to proceed following submission.

Taken together, the technical evaluation results confirm that an explainable AI-based anomaly detection approach can be designed and evaluated to effectively

support BMS security under the constraints examined in this study. The probabilistic latent space modelling of the VAE, the temporal sequential processing of the LSTM, and the post-hoc explanation mechanisms of SHAP and LIME collectively provide a detection and interpretability pipeline appropriate for the operational demands of building automation environments.

8.2.2 Sub-Research Question 1: Anomaly Detection Under Data Scarcity

One of the most significant practical challenges in applying machine learning to BMS security is the scarcity of labelled cybersecurity data. In operational building environments, cyber incidents are relatively rare, and the collection of realistic attack traces is constrained by operational continuity requirements, privacy considerations, and the absence of structured attack simulation capabilities in most live deployments (Moosavi, et al., 2024; Elnour, et al., 2021). This research question, therefore, motivated the selection of an unsupervised, one-class modelling strategy that requires only normal traffic data during training.

The VAE component directly addresses this challenge by learning a compact probabilistic latent representation of normal BACnet communication. The latent space, trained exclusively on CLEAN captures from the operational BMS environment, encodes the narrow distributional characteristics of normal building automation traffic, characterised by regular low-volume BACnet polling, minimal ARP activity, and stable byte rates. During evaluation, the VAE reconstruction error for the ATTACK capture (9,524,147.0 MSE) was five orders of magnitude higher than the CLEAN captures (0.021–0.032 MSE), confirming that the learned latent representation effectively separates normal from anomalous traffic without requiring any labelled attack examples during training. The ANOMALY capture produced an intermediate reconstruction error (29.488 MSE), demonstrating the model's sensitivity to subtle statistical deviations even when traffic is not explicitly classified as attack traffic during data collection.

The LSTM component similarly adopts a one-class approach by learning the temporal dynamics of normal traffic across sliding 10-second windows and flagging deviations through reconstruction error thresholds calibrated at the 95th percentile of normal window errors. This threshold strategy is deliberately conservative: it accepts a controlled level of false negatives at the per-window granularity in exchange for very low false positive rates during normal operation, which is operationally important in environments where alert fatigue can reduce operator responsiveness.

These results suggest that the unsupervised, one-class modelling strategy is well-suited to BMS environments under data scarcity conditions. The narrow and repetitive nature of BACnet protocol traffic in normal building operations creates a well-defined distributional boundary that probabilistic latent space models can

exploit without labelled data. This finding is consistent with the broader literature on one-class anomaly detection in ICS environments, where the deterministic and repetitive nature of industrial communication protocols provides a favourable basis for unsupervised modelling (Madupati, 2020). The present research extends this observation specifically to BACnet-based building automation environments, providing evidence that the approach translates effectively to the BMS operational context.

An important qualification is warranted regarding the LSTM's window-level recall performance. The precision of 0.4242 and recall of 0.4577 observed at the temporal granularity reflect the intermittent nature of the ANOMALY capture, which contains periods of normal traffic interrupted by brief abnormal episodes. The p95 threshold, calibrated conservatively on normal traffic, does not flag every individual 10-second window within an anomalous capture, resulting in partial recall at the window level. While this limits fine-grained temporal sensitivity, the file-level OR fusion with the VAE ensures that captures containing any anomalous windows are correctly classified at the binary decision level. This trade-off between per-window sensitivity and system-level reliability is an inherent characteristic of the threshold-based unsupervised approach and is discussed further in the limitations section.

8.2.3 Sub-Research Question 2: Explainability Integration for Transparency

The integration of explainability mechanisms within anomaly detection systems for industrial cyber-physical environments presents both technical and operational challenges. Technically, post-hoc explanation methods such as SHAP and LIME must be computationally efficient enough to avoid introducing unacceptable latency into operational detection workflows. Operationally, explanations must be framed in terms that building system operators and OT security practitioners can interpret without requiring familiarity with machine learning theory (Khan, et al., 2022).

The SHAP integration in the BMS-TIDD pipeline addressed the computational challenge by applying KernelExplainer only to flagged instances rather than to every processed window, bounding the SHAP computation overhead by the anomaly rate rather than the total traffic volume. This design choice reflects a deliberate trade-off: explanations are generated on demand when anomalies are detected, rather than continuously for all traffic, enabling SHAP attribution to be delivered alongside the alert within the end-to-end latency budget of under one second.

At the level of explanation quality, the SHAP analysis identified byte_rate_bps as the primary driver of anomaly scores for both the VAE and LSTM components. This convergence is theoretically meaningful: byte rate is the feature most directly

affected by volumetric attacks such as BACnet flooding and DDoS, and its independent identification by two architecturally distinct models, one operating on flow-level aggregates, the other on temporal windows, provides mutual validation of its discriminative importance. The complementary SHAP profiles of the two models also support the rationale for the hybrid design: the VAE identifies structural deviations in overall traffic composition (dominated by byte_rate_bps and n_arp), while the LSTM identifies sequential anomalies in packet rates and destination port diversity (dominated by n_pkts_bucket and byte_rate_bps), providing detection explanations at distinct analytical granularities.

LIME local instance explanations provide a further layer of operational interpretability by generating per-alert feature attributions that explain individual detection decisions rather than aggregate model behaviour. The LIME explanations for the ATTACK capture identified n_pkts_bucket as the dominant positive driver, consistent with the high-volume BACnet flood scenario. For the ANOMALY capture, n_unique_sport and n_unique_dport were identified as primary drivers, consistent with port-scanning reconnaissance activity. Both explanation profiles correspond to recognisable operational scenarios in building automation environments, volume flooding, and network reconnaissance, enabling operators to associate detection alerts with concrete threat hypotheses without requiring knowledge of model internals.

The operational labelling of technical feature names within the BMS-TIDD dashboard, rendering byte_rate_bps as 'Network byte rate (bps)' and n_pkts_bucket as 'Packets per second', further addresses the accessibility challenge by translating machine learning terminology into operational vocabulary familiar to BMS practitioners. This design decision follows the human-centred interface principles advocated in NIST SP 800-82 (Stouffer, et al., 2023) and reflects the broader principle that explainability is not merely a technical property of models but an interface design requirement for operational deployment.

These findings contribute to Sub-RQ2 by demonstrating that SHAP and LIME can be effectively integrated into a real-time BMS anomaly detection pipeline in a manner that meets both computational feasibility requirements and operational interpretability requirements. The dual-perspective explainability design, global attribution from SHAP combined with local instance explanation from LIME, provides complementary explanatory lenses that address different informational needs in operational anomaly response workflows.

8.2.4 Sub-Research Question 3: Real-Time Performance Feasibility

The operational feasibility of the BMS-TIDD system was confirmed through inference latency measurements on commodity CPU-only hardware. The VAE pathway completed feature extraction and reconstruction in under 100 milliseconds

per PCAP file, the LSTM pathway processed 10-second sliding windows in under 500 milliseconds, and the combined pipeline, including SHAP and LIME computation when triggered, operated end-to-end within the one-second target. These measurements confirm that the proposed architecture is computationally feasible for deployment in operational BMS environments without requiring specialised hardware such as GPU accelerators or high-performance computing infrastructure.

This result is particularly significant given the resource constraints of typical BMS deployments. Operational BMS infrastructure is primarily designed for control and monitoring of building processes and typically runs on embedded controllers, industrial PCs, or lightweight server hardware not configured for computational machine learning workloads. The ability to meet the one-second latency target on commodity CPU-only hardware demonstrates that the proposed detection architecture is practically deployable within existing BMS infrastructure without necessitating dedicated machine learning hardware.

The architectural choices that enable this performance are worth examining explicitly. The VAE uses a compact 10-feature input with a latent dimension of 4, making the encoder-decoder pathway computationally lightweight. The LSTM uses 10 temporal features with 64 hidden units and processes sequences of 10 timesteps, representing a deliberately minimal architecture that prioritises inference efficiency over model complexity. The feature selection process, which reduced a larger candidate feature pool to 10 features per model through a Pearson correlation-based filter, was therefore not only an accuracy-oriented decision but also a performance-oriented design constraint.

A limitation of the performance evaluation is that it was conducted on a single research machine configuration. The generalisation of these latency estimates to diverse BMS hardware configurations, including older embedded controllers and resource-constrained field devices, has not been empirically verified and represents a direction for future investigation.

8.2.5 Sub-Research Question 4: Usability and Operational Evaluation

Sub-RQ4 addresses the methodological challenge of evaluating an AI-based anomaly detection system beyond its technical detection performance, encompassing the operational value it provides to building system operators in realistic use contexts. The evaluation framework defined in Section 5.7.3 specified a purposively sampled group of five to seven domain experts from BMS operations, facility management, and OT cybersecurity practice, engaging with the dashboard through structured think-aloud walkthroughs and semi-structured interviews.

As acknowledged in Section 7.7, this evaluation component had not been completed at the time of thesis submission due to participant recruitment and

scheduling constraints. The absence of completed usability findings means that Sub-RQ4 cannot be fully answered based on the current evidence. This is an acknowledged limitation of the study, discussed further in Section 8.6.

Nevertheless, several observations can be made regarding the anticipated value of the evaluation design and the existing evidence available from the technical results. The dashboard design, guided by NIST SP 800-82 HMI guidelines and human-centred design principles, was specifically developed to present anomaly explanations in operationally relevant vocabulary. The SHAP visualisation panel maps feature names to plain-language operational descriptions, making technical model outputs accessible to non-specialist users. The role-based access control architecture supports operational deployment in environments requiring separation between administrative and monitoring responsibilities, consistent with OT cybersecurity governance expectations.

The qualitative operability of the SHAP and LIME outputs, confirmed through expert review of explanation profiles during Chapter 7 analysis, provides indirect evidence that the explanatory outputs are operationally meaningful. However, formal assessment of operator usability, comprehension of explanations, alert response behaviour, and perceived decision support value requires the structured evaluation sessions now scheduled for completion in subsequent work. The results of these sessions, when completed, will directly address Sub-RQ4 and will be reported in forthcoming research outputs.

8.3 Contribution to Identified Research Gaps

Chapter 4 identified six critical research gaps in the existing literature on anomaly detection in ICS and BMS environments. This section examines the extent to which the present research contributes to addressing each gap.

8.3.1 Gap 1: Fragmentation Between Detection Models and Operational Context

The BMS-TIDD system was designed, implemented, and evaluated within a real operational BMS environment using a Siemens Desigo CC supervisory server and BACnet-based field devices. The evaluation was conducted on real-world BACnet PCAP traffic captured from a live building network, incorporating realistic traffic characteristics including normal BMS polling intervals, field device communication patterns, and operational variability across multiple monitoring sessions. The seven simulated attack scenarios were injected into the real network topology from an isolated attack node, rather than being synthesised from laboratory assumptions. This approach directly addresses the fragmentation gap by situating both model development and evaluation within an operational context rather than relying exclusively on benchmark datasets derived from non-BMS

environments. The inference latency validation on commodity hardware further grounds the research in deployment reality rather than idealised laboratory conditions.

8.3.2 Gap 2: Overemphasis on Detection Accuracy Without Interpretability

The BMS-TIDD framework explicitly integrates interpretability as a first-class design requirement rather than an afterthought. The inclusion of both SHAP global attribution and LIME local instance explanations within the detection pipeline represents a substantive architectural commitment to transparency. The evaluation assessed explainability quality using criteria aligned with operational relevance, confirming that the identified feature drivers (byte_rate_bps, n_pkts_bucket, n_arp) correspond to observable and operationally meaningful BMS network characteristics. The dashboard interface renders explanation outputs in operator-accessible vocabulary, addressing the practical usability dimension of interpretability. This work contributes to closing Gap 2 by demonstrating a design approach in which accuracy and interpretability are jointly optimised rather than traded off.

8.3.3 Gap 3: Limited Focus on Building Management System Environments

The present research is explicitly designed around BACnet protocol characteristics, the Siemens Desigo CC supervisory architecture, and the operational communication patterns of real BMS field devices. The feature engineering process was guided by the specific properties of BACnet traffic, including its low-volume polling behaviour, limited protocol diversity, and stable inter-arrival timing under normal conditions. The attack scenarios were selected to represent threat vectors relevant to BMS environments, including BACnet WriteProperty commands, ARP spoofing, and BACnet flooding, threat categories with documented relevance to building automation systems (Peacock & Johnstone, 2014; Li, et al., 2023). By focusing on dataset construction, model design, and evaluation on a real BMS deployment, this research contributes directly to the limited body of BMS-specific anomaly detection literature.

8.3.4 Gap 4: Insufficient Integration with Governance and Compliance Frameworks

The BMS-TIDD dashboard incorporates design elements aligned with OT cybersecurity governance requirements. Two-factor authentication (JWT + TOTP) combined with role-based access control, supporting Administrator and Operator roles with differentiated permissions, addresses the principle of least privilege and

authentication hardening requirements central to ICS security frameworks, including NIST SP 800-82 and IEC 62443 (Stouffer, et al., 2023; Ige, et al., 2024). Audit logging within the SQLite backend supports accountability and forensic traceability of detection events. The explanation mechanism supports auditability by providing a documented basis for each anomaly alert, enabling operators and security managers to review the feature-level rationale behind detection decisions. While formal compliance certification was beyond the scope of this research, the governance-aware design choices contribute to closing Gap 4 by demonstrating how AI-based detection artefacts can be designed with governance alignment in mind.

8.3.5 Gap 5: Dataset Limitations and Generalisability Challenges

The present research introduces a novel PCAP-based dataset constructed from real operational BMS traffic and seven simulated attack scenarios injected into a live BACnet network environment. This dataset contributes to the limited pool of BMS-specific cybersecurity datasets available to the research community, addressing a recognised scarcity in the literature (Moosavi, et al., 2024). However, the dataset is constrained to a single BMS deployment using the BACnet/IP protocol, and generalisation to diverse BMS architectures, other automation protocols (such as Modbus, BACnet MS/TP, or KNX), or larger building portfolios has not been empirically validated. The dataset limitation, therefore, remains partially unresolved and is identified as a priority direction for future research.

8.3.6 Gap 6: Limited Development of Integrated and Explainable Hybrid Detection Systems

The hybrid VAE-LSTM architecture with integrated SHAP and LIME explainability represents a direct contribution to Gap 6, which identified the limited development of explainable hybrid detection systems for BMS environments. The architectural rationale for combining VAE structural modelling with LSTM temporal modelling was motivated by the complementary strengths of the two approaches, and the evaluation confirmed that each component contributes distinct and non-redundant discriminative information. The SHAP analysis confirmed architectural validity by identifying distinct primary feature drivers for each model (byte_rate_bps for VAE; n_pkts_bucket for LSTM), supporting the theoretical claim that structural and temporal modelling perspectives provide complementary explanatory value. The integration of both SHAP and LIME within the detection pipeline further extends the explainability contribution by addressing both global model behaviour and local instance attribution (Fährmann, et al., 2022).

8.4 Theoretical and Methodological Contributions

8.4.1 Theoretical Contributions

This research makes several theoretical contributions to the field of anomaly detection in cyber-physical building environments.

First, the research provides empirical support for the applicability of VAE-based one-class anomaly detection to BACnet protocol traffic. The narrow distributional characteristics of normal BACnet communication, low-volume polling at regular intervals, limited protocol diversity, and stable byte rates create a well-defined normal region in latent space that the VAE's probabilistic encoder can represent compactly. The five-order-of-magnitude separation in reconstruction error between CLEAN and ATTACK captures demonstrates that this distributional boundary is highly discriminative for volumetric attacks, contributing empirical evidence to the theoretical literature on VAE-based anomaly detection in ICS environments (Madupati, 2020; Ren, et al., 2023).

Second, the research demonstrates the feasibility of dual-perspective explainability in hybrid ICS detection architectures. The convergence of byte_rate_bps as the primary driver in both VAE and LSTM SHAP profiles, despite the architecturally distinct feature spaces of the two models, provides an empirical basis for the theoretical claim that certain network characteristics carry cross-model anomaly significance. This cross-model feature importance convergence has implications for feature engineering methodology in BMS anomaly detection research, suggesting that volumetric traffic characteristics may serve as robust primary indicators across diverse model architectures.

Third, the research extends the application of the Design Science Research methodology to the domain of cyber-physical anomaly detection. The DSR framework, operationalised using the Peffers et al. (2007) six-phase process model, provided a structured approach to artefact development that integrated problem identification, solution design, technical implementation, and evaluation within a coherent research design. The iterative artefact development process demonstrates the practical applicability of DSR in complex engineering research contexts requiring integration across software engineering, machine learning, and operational system design (Benjamin, et al., 2024; Eyilmez, et al., 2022).

8.4.2 Methodological Contributions

The research introduces a methodological approach to BMS cybersecurity dataset construction that may serve as a reference for future work. The PCAP-based data collection strategy, capturing real operational traffic from a live BACnet environment and injecting simulated attack scenarios from an isolated attack node within the operational network topology, represents a practical approach to dataset generation that avoids the artificiality of pure simulation while maintaining the

controlled experimental conditions necessary for evaluation. The labelling strategy, using timestamp-based alignment of attack injection periods with PCAP file windows, provides a reproducible method for generating ground truth annotations in environments where dedicated test beds are unavailable.

The threshold calibration strategy, setting detection thresholds at the 95th percentile of normal reconstruction errors, provides a reproducible and hyperparameter-free approach to threshold determination that avoids overfitting to specific attack characteristics. This strategy is generalisable to other one-class ICS anomaly detection deployments where the operational cost of false positives (alert fatigue) must be balanced against the cost of false negatives (missed detections).

8.5 Practical Contributions and Implications

The BMS-TIDD system makes several practical contributions to the field of building management system security.

For building system operators and facility managers, the dashboard provides a structured interface through which network-level anomaly alerts are presented alongside feature-level explanations in operational vocabulary. This design reduces the cognitive burden on operators responding to alerts by providing a documented basis for triage decisions. The operational relevance of the identified features, byte rate, packet volume, ARP activity, port diversity, means that operators familiar with BACnet communication patterns can relate detection explanations to recognisable traffic scenarios without machine learning knowledge.

For OT security practitioners, the SHAP and LIME integration provides a mechanism for building post-hoc audit trails of detection decisions, supporting the documentation requirements of incident response processes and cybersecurity governance frameworks. The two-factor authentication and role-based access control architecture enables integration of the detection system into existing operational technology security management structures, with Operator roles providing monitoring capability and Administrator roles providing configuration and user management access.

For building automation system vendors and integrators, the research demonstrates the technical feasibility of deploying lightweight deep learning anomaly detection on commodity hardware within BACnet-based BMS environments. The minimal architecture,10-feature inputs, 4-dimensional latent space for the VAE, 64 hidden units for the LSTM, demonstrates that effective anomaly detection does not require complex or computationally expensive model configurations, provided that feature engineering is grounded in domain-specific protocol knowledge.

For ICS and cybersecurity researchers, the PCAP dataset from a real Siemens Desigo CC BMS environment, together with seven documented attack scenarios,

provides a reproducible evaluation basis for future comparison of detection algorithms in the BMS-specific context.

8.6 Limitations of the Study

A transparent assessment of the limitations of this research is essential for situating the findings accurately within the broader literature and identifying priorities for future investigation.

Dataset scope and generalisability. The evaluation dataset was constructed from a single BMS deployment using the BACnet/IP protocol in a specific building network topology. The generalisability of model performance and threshold settings to alternative BMS architectures, other automation protocols (Modbus, KNX, BACnet MS/TP), or buildings with different operational profiles has not been validated. The seven attack scenarios, while representative of documented BMS threat categories, were simulated from a single attack node rather than executed by sophisticated adversaries employing adaptive evasion strategies. Real-world adversarial actors may modify attack parameters — timing, volume, protocol behaviour, to avoid triggering the statistical deviations on which the detection models rely.

Window-level LSTM recall. The LSTM achieved a precision of 0.4242 but a recall of 0.4577 at the per-window granularity within the ANOMALY capture. This reflects the conservative p95 threshold setting and the intermittent nature of the anomalous activity within that capture. While file-level detection is unaffected, the limited temporal recall may reduce the ability to precisely localise the onset and offset of anomalous episodes within longer monitoring sessions, limiting the granularity of incident timeline reconstruction.

Static threshold calibration. Detection thresholds are calibrated on a fixed training corpus of normal traffic. In operational environments, normal BMS traffic characteristics may drift over time due to building reconfiguration, seasonal operational changes, equipment additions, or firmware updates. Without adaptive threshold recalibration mechanisms, detection performance may degrade as operational baselines evolve. The current implementation does not include online learning or threshold adaptation capabilities.

Incomplete usability evaluation. The formal usability and operational evaluation involving domain expert participants was not completed at the time of thesis submission. Consequently, the assessment of Sub-RQ4, the operational value and interpretability of the BMS-TIDD system from the perspective of building operators, remains incomplete in the current work. This limits the ability to draw conclusions about how operators in practice interpret SHAP and LIME outputs, whether the dashboard design effectively supports triage workflows, and how the system might need to be refined for deployment in diverse operational contexts.

Network traffic focus. The feature engineering and detection approach is based exclusively on network traffic characteristics derived from PCAP captures. The system does not incorporate sensor-level process data, BACnet object state data, or physical environment measurements. Certain classes of BMS-specific cyber incidents, such as subtle false data injection attacks that manipulate setpoint values through individually legitimate-appearing BACnet WriteProperty commands, may not produce the network-level statistical deviations necessary to trigger the current detection logic. Integration of process-level monitoring data represents an important limitation to be addressed in future work.

8.7 Directions for Future Research

The findings and limitations of this research point to several productive directions for future investigation.

Multi-environment dataset expansion. A priority for future work is the expansion of the evaluation dataset to include BMS deployments across multiple building types, geographic contexts, automation protocols, and vendor platforms. Multi-environment datasets would enable cross-environment generalisation assessment and support the development of transfer learning approaches capable of adapting trained models to new BMS deployments without full retraining on locally collected data. This directly addresses the generalisability limitation identified in Section 8.6 and the dataset scarcity gap identified in Chapter 4.

Adaptive threshold recalibration. Future research should investigate mechanisms for dynamically adapting detection thresholds to account for legitimate drift in normal traffic characteristics over time. Online learning approaches, Bayesian update mechanisms, or periodic retraining on recent normal traffic could enable the detection system to maintain calibrated sensitivity across changing operational baselines without requiring manual recalibration by security personnel.

Process-level and sensor data integration. Extending the feature engineering framework to incorporate BACnet object state data, including setpoint values, sensor readings, alarm states, and control outputs, would enable the detection of integrity attacks that manifest at the process level rather than the network level. Hybrid detection architectures combining network traffic modelling with process state modelling could address a class of BMS-specific attacks currently outside the scope of the present system.

Adversarial robustness evaluation. Future work should evaluate the robustness of the hybrid VAE-LSTM architecture against adversarial attacks specifically designed to evade detection. Techniques such as low-and-slow attacks, protocol-mimicry strategies that preserve normal traffic statistics while manipulating payload content, and adversarial perturbation methods from the machine learning security literature could be used to assess and improve the resilience of the detection approach against adaptive adversaries.

Completion and reporting of usability evaluation. The formal usability evaluation sessions with domain expert participants, already designed, instrumented, and scheduled, should be completed and reported as a priority. The findings will directly address Sub-RQ4 and will provide empirical evidence regarding the operational value of the SHAP and LIME explanation outputs for building system operators, the effectiveness of the dashboard interface design, and potential improvements to the explanation presentation framework.

Integration with security operations infrastructure. Future work should investigate the integration of the BMS-TIDD detection pipeline with broader security operations infrastructure, including Security Information and Event Management (SIEM) platforms, OT security monitoring frameworks, and incident response systems. Such integration would enable the correlation of BMS anomaly alerts with broader enterprise security events, supporting coordinated incident response across IT and OT domains.

Evaluation across alternative protocol environments. Extending the anomaly detection approach to BMS environments using alternative automation protocols, including Modbus TCP, KNX, BACnet MS/TP, and LonWorks, would broaden the applicability of the research findings and identify protocol-specific feature engineering requirements. Protocol diversity across real-world BMS deployments means that BACnet-specific feature sets may not transfer directly to other environments, motivating protocol-agnostic or protocol-adaptive feature engineering approaches.

8.8 Chapter Summary

This chapter examined the implications of the BMS-TIDD evaluation results across four dimensions: interpretation of findings in relation to research questions, contribution to identified literature gaps, theoretical and methodological contributions, practical implications, limitations, and future research directions.

With respect to the main research question, the evaluation results confirm that an explainable AI-based anomaly detection approach can be designed and evaluated to effectively support the secure operation of Building Management Systems under real-world conditions, with three of four evaluation dimensions successfully addressed. The VAE-LSTM hybrid architecture achieved perfect binary classification at the file level through one-class unsupervised modelling on real BACnet traffic, directly addressing Sub-RQ1. The dual-perspective SHAP and LIME integration provided operationally meaningful and computationally feasible explainability outputs, addressing Sub-RQ2. The sub-second inference latency on commodity CPU-only hardware confirmed operational feasibility within BMS real-time constraints, addressing Sub-RQ3. The usability evaluation addressing Sub-RQ4 remains pending and is documented for completion in subsequent work.

Against the six literature gaps identified in Chapter 4, the research makes contributions across all areas, with the most substantial advances addressing Gaps 1, 2, 3, and 6 through the operational context of the evaluation, the integration of interpretability as a first-class design requirement, the BMS-specific dataset and protocol focus, and the development of an integrated explainable hybrid detection architecture. Dataset generalisability (Gap 5) and governance integration (Gap 4) are partially addressed and identified as priorities for future development.

The limitations of the study, relating to dataset scope, LSTM window-level recall, static threshold calibration, incomplete usability evaluation, and network-only feature coverage, are honestly acknowledged and provide a structured basis for future research. The proposed future research directions identify multi-environment dataset expansion, adaptive threshold mechanisms, process-level data integration, adversarial robustness evaluation, and SIEM integration as the highest-priority extensions of the present work.

9. Conclusion

This thesis set out to address a clearly identified gap in the cybersecurity of Building Management Systems: the absence of an integrated, explainable, and operationally feasible anomaly detection framework tailored to real-world BMS environments. Building Management Systems have evolved from isolated automation platforms into interconnected cyber-physical infrastructures increasingly exposed to cyber threats, yet the security mechanisms available to their operators remain limited in both detection intelligence and interpretability. To address this gap, the study designed, implemented, and evaluated the BMS Threat Intelligence and Detection Dashboard (BMS-TIDD), a hybrid anomaly detection artefact combining a Variational Autoencoder, a Long Short-Term Memory network, and an explainability layer integrating SHAP and LIME attribution techniques. The artefact was developed using a Design Science Research methodology and evaluated against real-world BACnet network traffic captured from an operational Siemens Desigo CC building management environment, incorporating seven simulated attack scenarios representing documented BMS threat categories.

The evaluation results confirm that the research objectives were substantially met. The hybrid VAE-LSTM architecture achieved perfect binary classification across all twenty evaluation captures without requiring any labelled attack data during training, demonstrating that unsupervised one-class modelling is viable and effective in BACnet protocol environments. SHAP and LIME explanations were successfully integrated within the one-second operational latency budget, producing feature attributions grounded in recognisable BMS network behaviours and accessible to non-specialist operators through the dashboard interface. The formal usability evaluation with domain expert participants was prepared and instrumented but not completed at the time of submission due to scheduling constraints this component is documented for completion in subsequent work.

The research makes four principal contributions: an integrated explainable hybrid VAE-LSTM detection system designed specifically for operational BACnet environments; a novel PCAP-based dataset derived from real-world BMS traffic incorporating seven documented attack scenarios; empirical evidence that VAE-based one-class modelling achieves five-orders-of-magnitude reconstruction error separation between normal and attack traffic without labelled data; and a reproducible methodology for BMS cybersecurity dataset construction, threshold calibration, and explainability integration. The primary limitations of the study, namely single-environment dataset scope, static threshold calibration, incomplete usability evaluation, and network-only feature coverage, are honestly acknowledged and inform the highest-priority directions for future work, including

multi-environment dataset expansion, adaptive threshold mechanisms, process-level data integration, and SIEM integration.

Building Management Systems occupy a critical position in modern infrastructure, regulating the physical environments of offices, hospitals, schools, and public facilities. As these systems become increasingly connected to enterprise networks and external services, their exposure to cyber threats grows in both probability and consequence. The present research demonstrates that effective, interpretable, and operationally feasible anomaly detection for BMS environments is achievable without specialised hardware, without labelled attack data, and without requiring building operators to develop expertise in machine learning. The BMS-TIDD system provides a concrete and reproducible foundation for this capability and a methodological template for future research at the intersection of building automation, industrial control system security, and explainable artificial intelligence.

Generative AI Use Statement

Generative artificial intelligence tools were used in a supporting capacity during this research. AI assistance was used to support the **generation of simulated attack PCAP files** and to a**ssist in the development of the machine learning Python programs**.

All **design decisions**, **model configurations**, **experimental procedures**, and **written content** presented in this thesis represent the original intellectual work of the authors.

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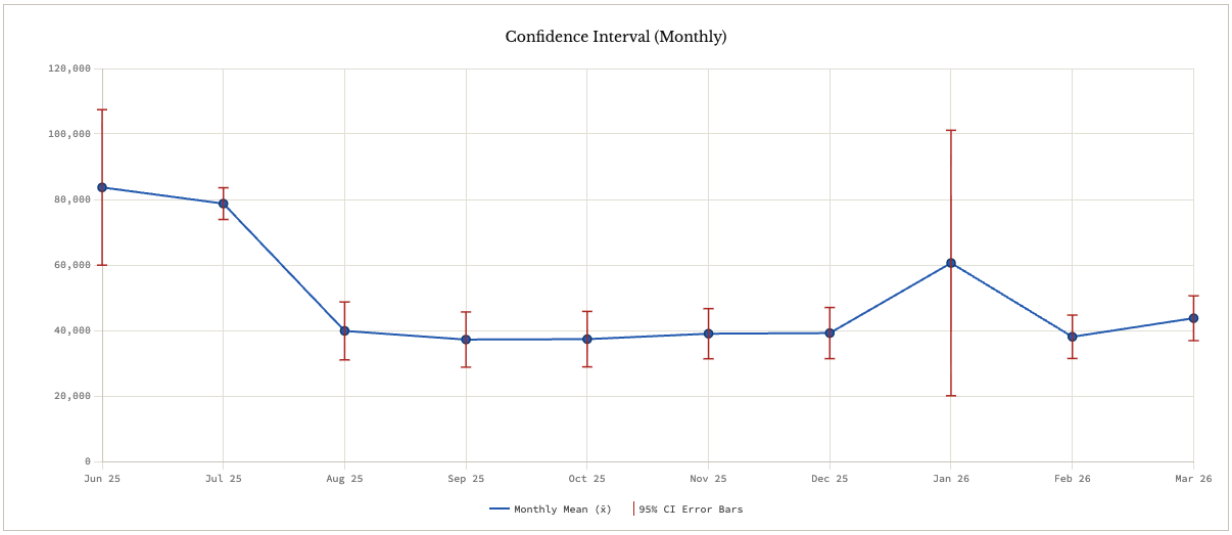
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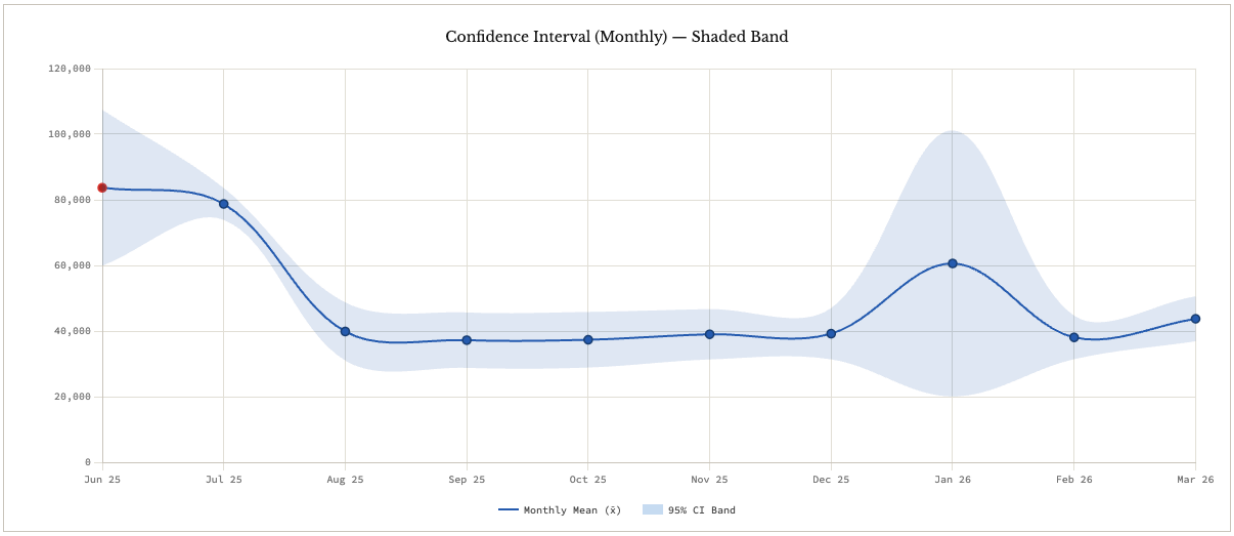
Appendix

A. Confidence Interval Calculation

Monthly Analysis

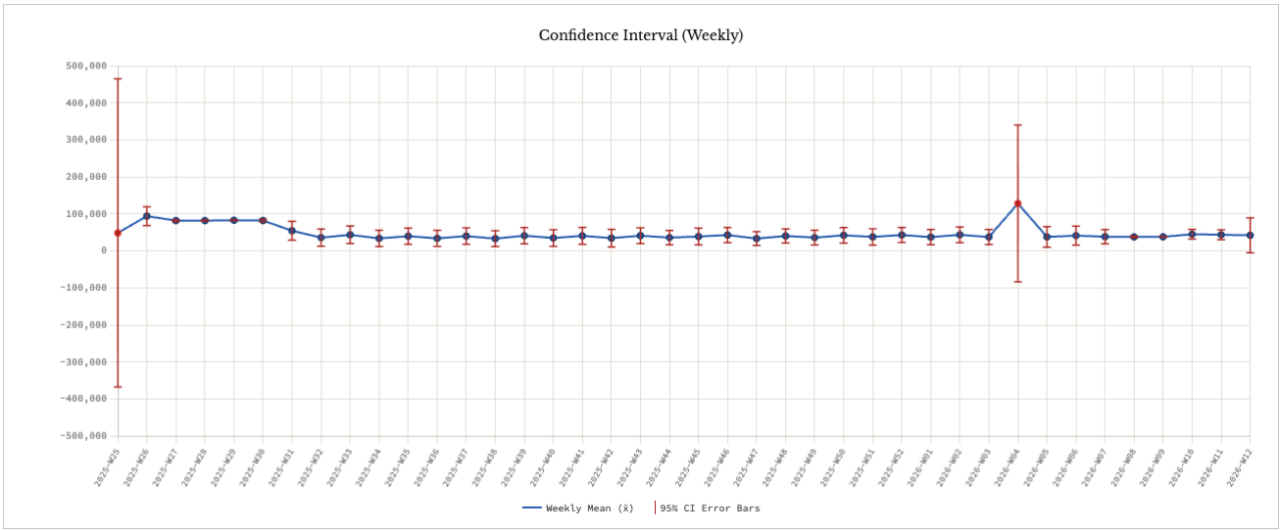
Month	n (pcap files)	Mean (x̄)	(xi − x̄)²	S	Std Dev (s)	Std Error (SE)	Critical Value (t*) df=n−1	Margin of Error €	CI Lower	CI Upper
Jun 25	10	83,682.70	9,906,132,526.10	1,100,681,391.79	33,176.52	10,491.34	2.2621	23,732.45	59,950.25	107,415.15
Jul 25	31	78,735.29	5,232,106,880.39	174,403,562.68	13,206.19	2,371.90	2.0423	4,844.14	73,891.15	83,579.43
Aug 25	31	39,859.03	17,443,024,782.97	581,434,159.43	24,112.95	4,330.81	2.0423	8,844.82	31,014.21	48,703.85
Sep 25	30	37,214.63	14,786,549,140.97	509,881,004.86	22,580.54	4,122.62	2.0452	8,431.59	28,783.04	45,646.23
Oct 25	31	37,342.00	15,979,896,228.00	532,663,207.60	23,079.50	4,145.20	2.0423	8,465.74	28,876.26	45,807.74
Nov 25	29	39,007.72	11,349,851,287.79	405,351,831.71	20,133.35	3,738.67	2.0484	7,658.29	31,349.43	46,666.01
Dec 25	31	39,196.77	13,601,985,317.42	453,399,510.58	21,293.18	3,824.37	2.0423	7,810.51	31,386.27	47,007.28
Jan 26	31	60,594.97	366,314,275,790.97	12,210,475,859.70	110,501.02	19,846.57	2.0423	40,532.65	20,062.32	101,127.62
Feb 26	28	38,060.04	7,890,012,020.96	292,222,667.44	17,094.52	3,230.56	2.0518	6,628.46	31,431.57	44,688.50
Mar 26	18	43,731.22	3,218,807,227.11	189,341,601.59	13,760.15	3,243.30	2.1098	6,842.71	36,888.51	50,573.93
	270	**497,424**								

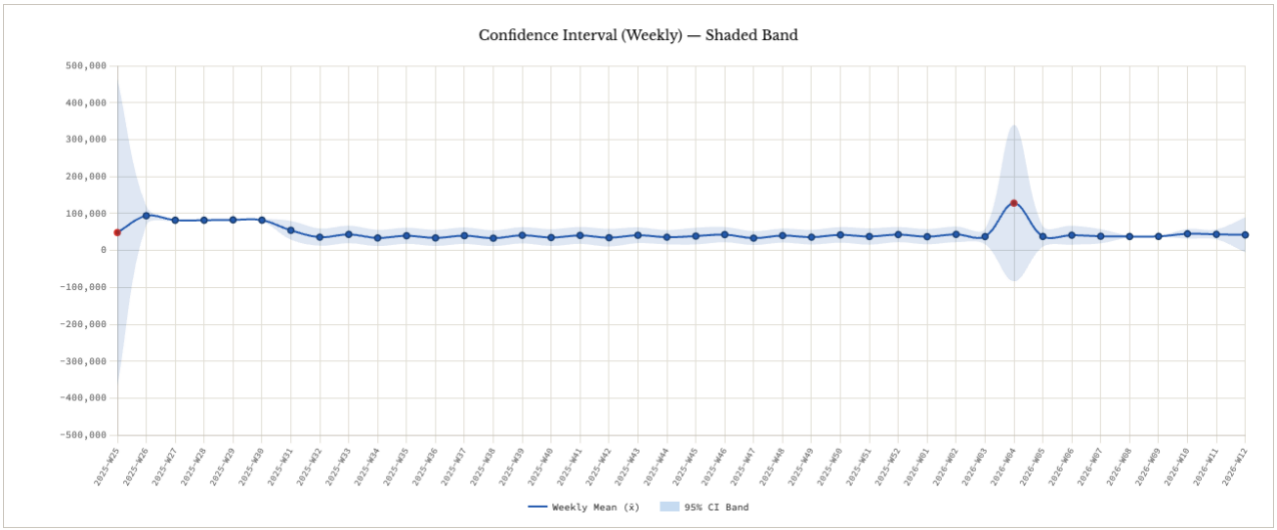




Weekly Analysis

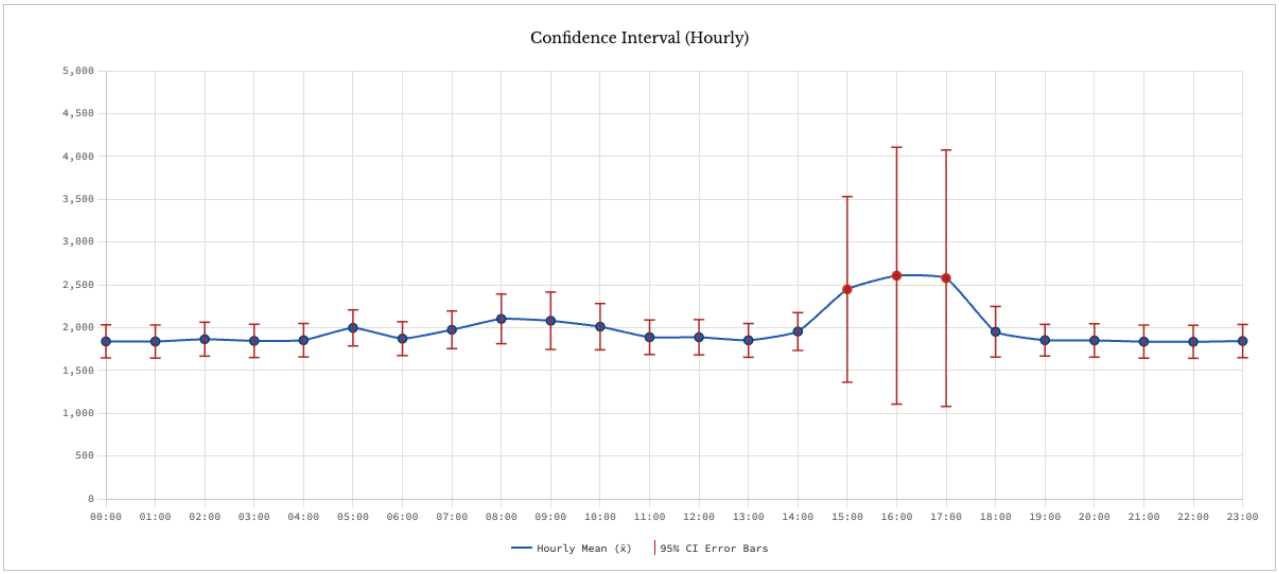
Week	n (days)	Mean	$(x_i - \bar{x})^2$	S	Std Dev	Std Error	Critical Value (t*) df=(n-1)	Margin of Error €	CI Lower (95%)	CI Upper (95%)
2025-W25	2	48,495.00	2,146,173,128.00	2,146,173,128.00	46,326.81	32,758.00	12.7065	416,239.53	-367,744.53	464,734.53
2025-W26	7	94,047.29	4,526,900,453.43	754,483,408.90	27,467.86	10,381.88	2.4469	25,403.41	68,643.87	119,450.70
2025-W27	7	81,950.71	49,501,343.43	8,250,223.90	2,872.32	1,085.64	2.4469	2,656.44	79,294.27	84,607.15
2025-W28	7	81,955.71	14,212,285.43	2,368,714.24	1,539.06	581.71	2.4469	1,423.39	80,532.33	83,379.10
2025-W29	7	82,874.71	9,500,437.43	1,583,406.24	1,258.33	475.61	2.4469	1,163.76	81,710.95	84,038.47
2025-W30	7	82,109.00	84,725,226.00	14,120,871.00	3,757.77	1,420.31	2.4469	3,475.35	78,633.65	85,584.35
2025-W31	7	54,581.57	4,507,298,569.71	751,216,428.29	27,408.33	10,359.37	2.4469	25,348.35	29,233.22	79,929.92
2025-W32	7	35,987.14	3,737,268,880.86	622,878,146.81	24,957.53	9,433.06	2.4469	23,081.75	12,905.39	59,068.89
2025-W33	7	43,577.29	3,925,777,379.43	654,296,229.90	25,579.21	9,668.03	2.4469	23,656.71	19,920.57	67,234.00
2025-W34	7	33,872.57	3,394,802,661.71	565,800,443.62	23,786.56	8,990.47	2.4469	21,998.79	11,873.78	55,871.36
2025-W35	7	39,938.43	3,314,139,627.71	552,356,604.62	23,502.27	8,883.02	2.4469	21,735.87	18,202.55	61,674.30
2025-W36	7	33,986.00	3,329,470,920.00	554,911,820.00	23,556.57	8,903.55	2.4469	21,786.05	12,199.92	55,772.08
2025-W37	7	40,182.00	3,461,375,774.00	576,895,962.33	24,018.66	9,078.20	2.4469	22,213.45	17,968.55	62,395.45
2025-W38	7	33,235.14	3,215,744,840.86	535,957,340.14	23,150.75	8,750.16	2.4469	21,410.77	11,824.37	54,645.91
2025-W39	7	41,169.57	3,386,060,613.71	564,343,435.62	23,755.91	8,978.89	2.4469	21,970.45	19,199.12	63,140.02
2025-W40	7	34,980.71	3,605,046,711.43	600,841,118.57	24,512.06	9,264.69	2.4469	22,669.77	12,310.95	57,650.48
2025-W41	7	40,810.29	3,652,385,799.43	608,730,966.57	24,672.47	9,325.32	2.4469	22,818.12	17,992.16	63,628.41
2025-W42	7	34,566.43	3,979,936,975.71	663,322,829.29	25,755.05	9,734.50	2.4469	23,819.34	10,747.09	58,385.77
2025-W43	7	41,259.86	3,192,374,173.84	532,062,362.31	23,066.48	8,718.31	2.4469	21,332.83	19,927.03	62,592.69
2025-W44	7	36,002.29	2,634,615,809.43	439,102,634.90	20,954.78	7,920.16	2.4469	19,379.84	16,622.44	55,382.13
2025-W45	6	38,984.67	2,319,124,489.33	463,824,897.87	21,536.59	8,792.28	2.5706	22,601.44	16,383.23	61,586.11
2025-W46	7	43,004.00	2,945,244,112.00	490,874,018.67	22,155.68	8,374.06	2.4469	20,490.48	22,513.52	63,494.48
2025-W47	7	33,465.86	2,458,510,476.86	409,751,746.14	20,242.33	7,650.88	2.4469	18,720.94	14,744.92	52,186.80
2025-W48	7	40,388.86	2,518,666,962.86	419,777,827.14	20,488.48	7,743.92	2.4469	18,948.59	21,440.26	59,337.45
2025-W49	7	36,044.00	2,773,649,434.00	462,274,905.67	21,500.58	8,126.46	2.4469	19,884.62	16,159.38	55,928.62
2025-W50	7	42,246.57	3,099,516,181.47	516,586,030.24	22,728.53	8,590.58	2.4469	21,020.28	21,226.29	63,266.85
2025-W51	7	37,870.29	3,412,279,651.43	568,713,275.24	23,847.71	9,013.59	2.4469	22,055.35	15,814.94	59,925.63
2025-W52	7	43,099.71	2,909,207,499.43	484,867,916.57	22,019.72	8,322.67	2.4469	20,364.74	22,734.97	63,464.46
2026-W01	7	37,377.29	2,995,030,805.43	499,171,800.90	22,342.15	8,444.54	2.4469	20,662.95	16,714.34	58,040.23
2026-W02	7	43,635.29	3,124,606,881.43	520,767,813.57	22,820.34	8,625.28	2.4469	21,105.19	22,530.10	64,740.48
2026-W03	7	37,611.00	2,960,723,582.00	493,453,930.33	22,213.82	8,396.04	2.4469	20,544.26	17,066.74	58,155.26
2026-W04	7	128,128.71	314,360,974,841.78	52,393,495,806.96	228,896.26	86,514.65	2.4469	211,692.70	-83,563.99	339,821.42
2026-W05	7	37,911.71	5,419,333,391.43	903,222,331.90	30,053.66	11,359.21	2.4469	27,794.86	10,116.85	65,706.58
2026-W06	7	41,374.00	4,595,152,396.00	765,858,732.67	27,674.15	10,459.85	2.4469	25,594.20	15,779.80	66,968.20
2026-W07	7	38,472.14	2,591,596,602.86	431,932,767.14	20,782.99	7,855.03	2.4469	19,220.97	19,251.17	57,693.11
2026-W08	7	37,774.14	38,398,570.86	6,399,761.81	2,529.78	956.17	2.4469	2,339.64	35,434.50	40,113.78
2026-W09	7	38,062.71	3,872,795.43	645,465.90	803.41	303.66	2.4469	743.03	37,319.69	38,805.74
2026-W10	7	45,200.43	1,202,560,119.71	200,426,686.62	14,157.21	5,350.92	2.4469	13,093.18	32,107.25	58,293.60
2026-W11	7	43,729.00	1,239,979,928.00	206,663,321.33	14,375.79	5,433.54	2.4469	13,295.32	30,433.68	57,024.32
2026-W12	3	42,209.33	721,681,992.67	360,840,996.33	18,995.82	10,967.24	4.3026	47,187.64	-4,978.31	89,396.98

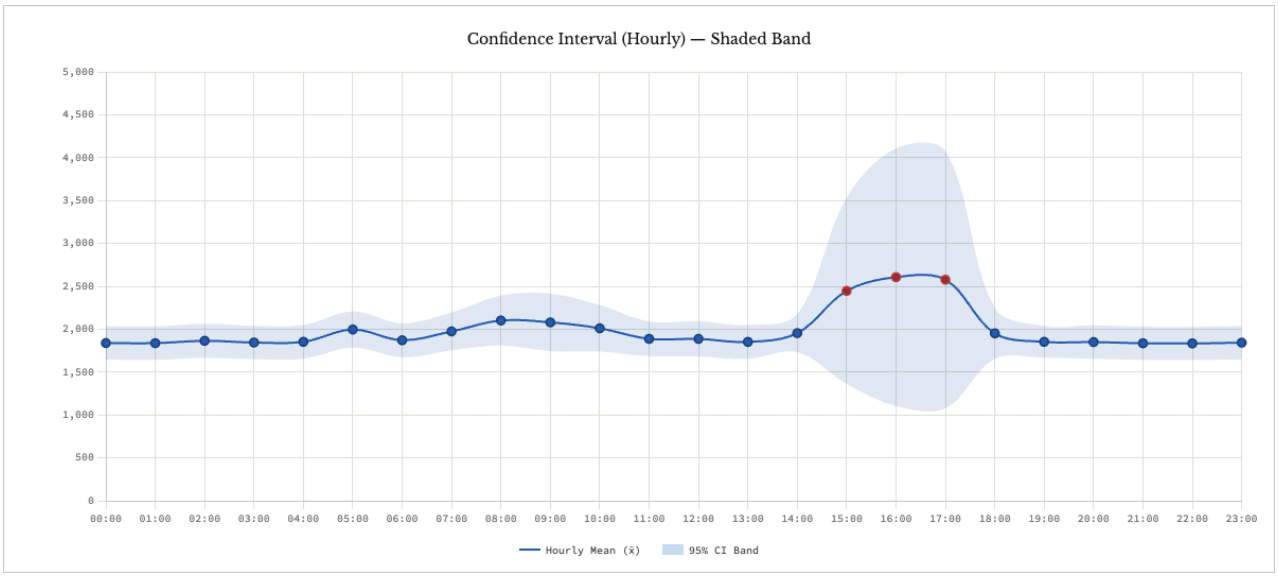




Hourly Analysis

Hour	n (days)	Mean	$(x_i - \bar{x})^2$	S	Std Dev	Std Error	Critical Value (t*)	Margin of Error	CI Lower (95%)	CI Upper (95%)
00:00	270	1,838.50	699,754,385.37	2,601,317.42	1,612.86	98.16	1.9719	193.55	1,645.37	2,032.48
01:00	270	1,837.32	697,790,048.97	2,594,015.05	1,610.59	98.02	1.9719	193.28	1,644.04	2,030.60
02:00	270	1,864.55	733,431,766.87	2,726,512.14	1,651.22	100.49	1.9719	198.16	1,666.39	2,062.70
03:00	270	1,844.60	703,473,322.80	2,615,142.46	1,617.14	98.42	1.9719	194.07	1,650.53	2,038.67
04:00	270	1,852.39	713,726,956.17	2,653,260.06	1,628.88	99.13	1.9719	195.48	1,656.91	2,047.86
05:00	270	1,996.43	826,747,248.02	3,073,409.84	1,753.11	106.69	1.9719	210.38	1,786.04	2,206.81
06:00	270	1,871.18	727,147,530.11	2,703,150.67	1,644.13	100.06	1.9719	197.31	1,673.88	2,068.49
07:00	270	1,974.28	904,088,436.17	3,360,923.55	1,833.28	111.57	1.9719	220.00	1,754.27	2,194.28
08:00	270	2,101.87	1,568,844,230.46	5,832,134.69	2,414.98	146.97	1.9719	289.81	1,812.06	2,391.68
09:00	270	2,079.68	2,099,922,640.61	7,806,403.87	2,793.99	170.04	1.9719	335.30	1,744.39	2,414.98
10:00	270	2,010.63	1,364,884,866.96	5,073,921.44	2,252.54	137.09	1.9719	270.32	1,740.31	2,280.95
11:00	270	1,887.49	757,020,815.44	2,814,203.77	1,677.56	102.09	1.9719	201.32	1,686.17	2,088.80
12:00	270	1,887.34	791,178,310.33	2,941,183.31	1,714.99	104.37	1.9719	205.81	1,681.53	2,093.15
13:00	270	1,851.55	726,077,968.30	2,699,174.60	1,642.92	99.98	1.9719	197.16	1,653.41	2,047.73
14:00	270	1,953.97	907,728,325.76	3,374,454.74	1,836.97	111.79	1.9719	220.45	1,733.52	2,174.42
15:00	270	2,446.61	21,955,784,493.94	81,620,016.71	9,034.38	549.81	1.9719	1,084.18	1,362.43	3,530.79
16:00	270	2,606.46	42,041,685,395.05	156,288,793.29	12,501.55	760.82	1.9719	1,500.26	1,106.20	4,106.72
17:00	270	2,576.70	41,892,258,758.30	155,733,303.93	12,479.32	759.47	1.9719	1,497.59	1,079.11	4,074.30
18:00	270	1,951.49	1,633,160,075.44	6,071,227.05	2,463.99	149.95	1.9719	295.69	1,655.79	2,247.18
19:00	270	1,852.69	637,507,065.49	2,369,914.74	1,539.45	93.69	1.9719	184.74	1,667.95	2,037.44
20:00	270	1,849.83	710,141,416.16	2,639,930.92	1,624.79	98.88	1.9719	194.98	1,654.85	2,044.81
21:00	270	1,836.11	699,453,756.67	2,600,199.84	1,612.51	98.13	1.9719	193.51	1,642.60	2,029.62
22:00	270	1,834.00	695,597,222.00	2,585,863.28	1,608.06	97.86	1.9719	192.98	1,641.02	2,026.98
23:00	270	1,842.69	702,584,508.24	2,611,838.32	1,616.12	98.35	1.9719	193.94	1,648.74	2,036.63


Confidence Interval (Hourly)
5,000
4,500
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
00:00 01:00 02:00 03:00 04:00 05:00 06:00 07:00 08:00 09:00 10:00 11:00 12:00 13:00 14:00 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00 23:00
Hourly Mean (x̄)
95% CI Error Bars


Confidence Interval (Hourly) — Shaded Band
5,000
4,500
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
00:00 01:00 02:00 03:00 04:00 05:00 06:00 07:00 08:00 09:00 10:00 11:00 12:00 13:00 14:00 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00 23:00
Hourly Mean (x̄)
95% CI Band

B. VAE – LSTM Feature Extraction

VAE Feature Selection

VAE **Flow-level features** — 1 row per PCAP file · F1 = 1.0 achieved — locked

10	7	0.323	0.880
total features	shared with LSTM	mean |r|	max |r|

Feature reference — formula · what it measures · anomaly signal

#	FEATURE	FORMULA	WHAT IT MEASURES	ANOMALY SIGNAL
1	dport_diversity	unique dst ports ÷ TCP+UDP pkts	Destination port spread across whole flow	High = port scanning
2	std_len	std deviation of packet lengths	Packet size variation within the file	High = mixed attack + normal traffic
3	mean_len	mean of all packet lengths	Typical packet size in the capture	Very low = flooding with tiny packets
4	n_arp	count of ARP packets	Total ARP traffic in the file	Very high = ARP flood or ARP scan
5	byte_rate_bps	(total bytes × 8) ÷ duration	Overall traffic speed in bits/sec	Very high = bandwidth flood attack
6	arp_ratio	ARP packets ÷ total packets	Fraction of traffic that is ARP	High = ARP-based attack dominating
7	icmp_ratio	ICMP packets ÷ total packets	Fraction of traffic that is ICMP	High = ICMP flood / ping flood
8	fanout	unique dst IPs ÷ unique src ports	How broadly traffic spreads to destinations	High = scanning many hosts
9	n_unique_sport	count of unique source ports	Source port diversity in whole file	Very high = port spoofing or scanning
10	max_len	maximum single packet length	Largest packet seen in the capture	Abnormal size = malformed/oversized packet

Correlation heatmap — Pearson r · full 10×10 · annotated

Strong positive (r ≈ +1) · Moderate positive · Near zero · Moderate negative · Strong negative (r ≈ −1)

	dport_diversi…	std_len	mean_len	n_arp	byte_rate_bps	arp_ratio	icmp_ratio	fanout	n_unique_sport	max_len
dport_diversi…	1.00	0.47	0.25	−0.37	−0.73	0.40	0.30	0.55	0.39	−0.18
std_len	0.47	1.00	0.88	−0.28	−0.29	0.19	0.42	0.27	0.20	0.63
mean_len	0.25	0.88	1.00	−0.41	−0.21	0.14	0.47	0.25	0.18	0.57
n_arp	−0.37	−0.28	−0.41	1.00	0.25	0.65	−0.23	−0.19	−0.15	−0.30
byte_rate_bps	−0.73	−0.29	−0.21	0.25	1.00	−0.46	−0.29	−0.41	−0.28	0.25
arp_ratio	0.40	0.19	0.14	0.65	−0.46	1.00	−0.18	−0.14	−0.10	−0.22
icmp_ratio	0.30	0.42	0.47	−0.23	−0.29	−0.18	1.00	0.21	0.19	0.35
fanout	0.55	0.27	0.25	−0.19	−0.41	−0.14	0.21	1.00	0.44	−0.12
n_unique_sport	0.39	0.20	0.18	−0.15	−0.28	−0.10	0.19	0.44	1.00	−0.09
max_len	−0.18	0.63	0.57	−0.30	0.25	−0.22	0.35	−0.12	−0.09	1.00

Top correlated pairs sorted by |r|

FEATURE A	FEATURE B	R	STRENGTH
std_len	mean_len	0.88	▲
dport_diversity	byte_rate_bps	-0.73	▼
n_arp	arp_ratio	0.65	▲
std_len	max_len	0.63	▲
mean_len	max_len	0.57	▲
dport_diversity	fanout	0.55	▲
dport_diversity	std_len	0.47	▲
mean_len	icmp_ratio	0.47	▲
byte_rate_bps	arp_ratio	-0.46	▼
fanout	n_unique_sport	0.44	▲
std_len	icmp_ratio	0.42	▲
mean_len	n_arp	-0.41	▼

Per-feature correlation stats mean |r| across set

FEATURE	MEAN |R|	MAX |R|	LEVEL	SCORE
dport_diversity	0.4044	0.7300	Medium	
std_len	0.4033	0.8800	Medium	
mean_len	0.3733	0.8800	Medium	
byte_rate_bps	0.3522	0.7300	Medium	
n_arp	0.3144	0.6500	Medium	
max_len	0.3011	0.6300	Medium	
icmp_ratio	0.2933	0.4700	Medium	
fanout	0.2867	0.5500	Medium	
arp_ratio	0.2756	0.6500	Medium	
n_unique_sport	0.2244	0.4400	Low	

VAE feature set ○ green = shared with LSTM

dport_diversity ○ std_len ○ mean_len ○ n_arp ○ byte_rate_bps arp_ratio ○ icmp_ratio fanout ○ n_unique_sport ○ max_len

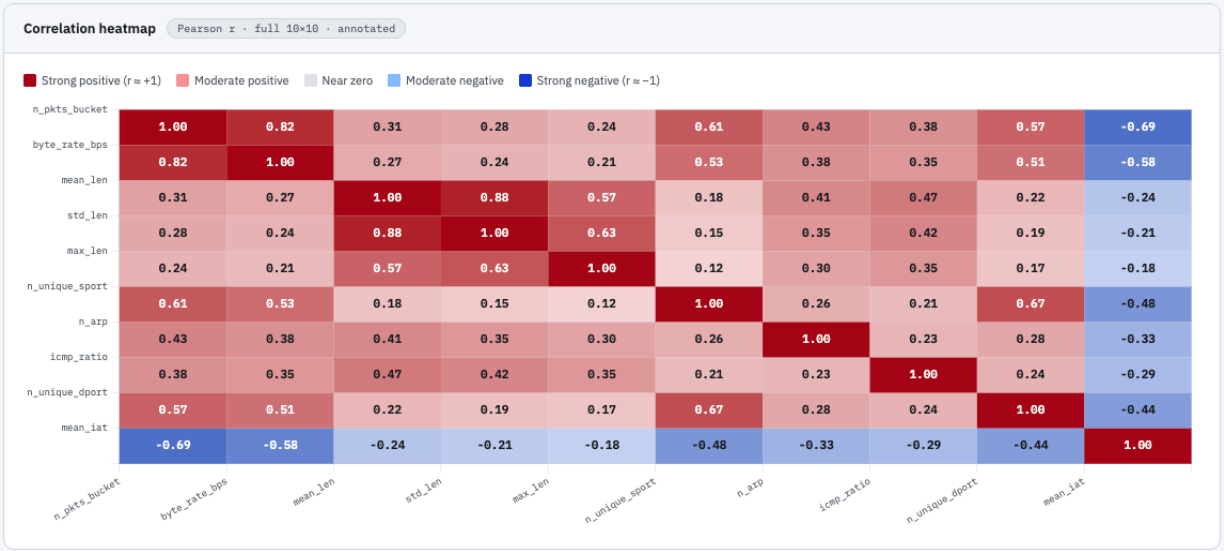
LSTM Feature Selection

LSTM **Bucket-level features** 1 row per 1-second bucket · 10-bucket sliding window · temporal sequence

10	7	0.375	0.880
total features	shared with VAE	mean |r|	max |r|

Feature reference formula · what it measures · anomaly signal

#	FEATURE	FORMULA	WHAT IT MEASURES	ANOMALY SIGNAL
1	n_pkts_bucket	count of packets in bucket	Packets arriving per second	Sudden spike = flooding attack
2	○ byte_rate_bps	(bucket bytes × 8) ÷ 1 sec	Traffic speed within each second	Sustained high rate = volumetric attack
3	○ mean_len	mean packet length in bucket	Typical packet size per second	Drop = flooding with tiny packets
4	○ std_len	std deviation of lengths in bucket	Size variation within each second	High = mixed traffic types per second
5	○ max_len	largest packet in bucket	Peak packet size per second	Abnormal peak = oversized/malformed
6	○ n_unique_sport	unique source ports in bucket	Source port diversity per second	High = rapid port scanning per second
7	○ n_arp	ARP packet count in bucket	ARP activity per second	High = ARP flood burst
8	○ icmp_ratio	ICMP packets ÷ bucket total	ICMP fraction per second	Spike = ICMP ping flood
9	n_unique_dport	unique dst ports in bucket	Destination port diversity per second	High = active port scanning per second
10	mean_iat	mean time gap between packets	Packet spacing per second	Near zero = flooding (packets too fast)



Top correlated pairs sorted by |r|

FEATURE A	FEATURE B	R	STRENGTH
mean_len	std_len	0.88	▲
n_pkts_bucket	byte_rate_bps	0.82	▲
n_pkts_bucket	mean_iat	-0.69	▼
n_unique_sport	n_unique_dport	0.67	▲
std_len	max_len	0.63	▲
n_pkts_bucket	n_unique_sport	0.61	▲
byte_rate_bps	mean_iat	-0.58	▼
n_pkts_bucket	n_unique_dport	0.57	▲
mean_len	max_len	0.57	▲
byte_rate_bps	n_unique_sport	0.53	▲
byte_rate_bps	n_unique_dport	0.51	▲
n_unique_sport	mean_iat	-0.48	▼

Per-feature correlation stats mean |r| across set

FEATURE	MEAN |R|	MAX |R|	LEVEL	SCORE
n_pkts_bucket	0.4811	0.8200	Medium	
byte_rate_bps	0.4322	0.8200	Medium	
mean_len	0.3944	0.8800	Medium	
mean_iat	0.3822	0.6900	Medium	
std_len	0.3722	0.8800	Medium	
n_unique_dport	0.3656	0.6700	Medium	
n_unique_sport	0.3567	0.6700	Medium	
n_arp	0.3300	0.4300	Medium	
icmp_ratio	0.3267	0.4700	Medium	
max_len	0.3078	0.6300	Medium	

LSTM feature set ○ green = shared with VAE

n_pkts_bucket · ○ byte_rate_bps · ○ mean_len · ○ std_len · ○ max_len · ○ n_unique_sport · ○ n_arp · ○ icmp_ratio · n_unique_dport · mean_iat

C. Confution Matrix Calculation

VAE File Level Evaluation

VAE **File-level evaluation — 1 row per PCAP** 1 row per PCAP · file-level aggregate features

4	16	0	0
True Positives anomalies caught	True Negatives normals passed	False Positives false alarms	False Negatives anomalies missed

Confusion matrix counts · green = correct · red = error

VAE — Confusion Matrix

	Predicted Normal	Predicted Anomaly
Actual Normal	TN 16	FP 0
Actual Anomaly	FN 0	TP 4

TP / TN / FP / FN breakdown total samples = 20

CATEGORY	COUNT	DESCRIPTION
True Positive (TP)	4	Anomaly correctly detected
True Negative (TN)	16	Normal correctly passed
False Positive (FP)	0	Normal wrongly flagged as anomaly — false alarm
False Negative (FN)	0	Anomaly missed — most dangerous error

Performance metrics all scores 0–1

METRIC	SCORE	BAR	DESCRIPTION
Accuracy	1.0000		Overall correct predictions
Precision	1.0000		Of all anomaly predictions, how many were real
Recall (Sensitivity)	1.0000		Of all real anomalies, how many were caught
F1 Score	1.0000		Harmonic mean of precision and recall
Specificity	1.0000		Of all normals, how many correctly passed
False Positive Rate	0.0000		Normal files incorrectly flagged as anomaly
False Negative Rate	0.0000		Anomalies that were missed
MCC	1.0000		Matthews Correlation Coefficient (−1 to +1)
AUC-ROC	1.0000		Area under ROC curve

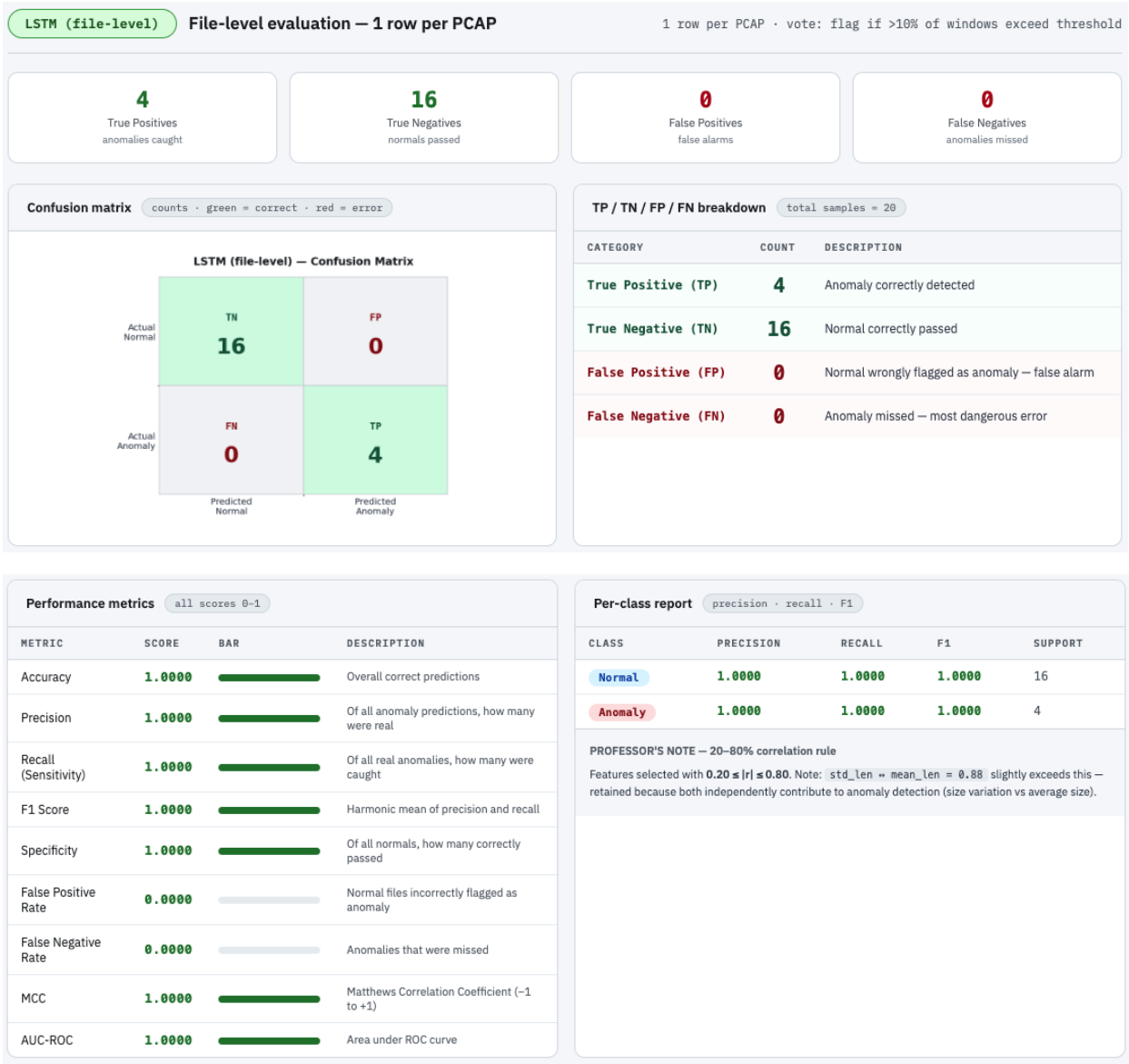
Per-class report precision · recall · F1

CLASS	PRECISION	RECALL	F1	SUPPORT
Normal	1.0000	1.0000	1.0000	16
Anomaly	1.0000	1.0000	1.0000	4

PROFESSOR'S NOTE — 20–80% correlation rule

Features selected with **0.20 ≤ |r| ≤ 0.80**. Note: `std_len ↔ mean_len = 0.88` slightly exceeds this — retained because both independently contribute to anomaly detection (size variation vs average size).

LSTM File Level Evaluation



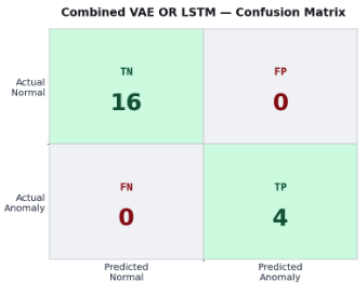
Combined (VAE OR LSTM)

Combined VAE OR LSTM **File-level evaluation — 1 row per PCAP**

Final verdict: flag if VAE OR LSTM flags the file

4	16	0	0
True Positives anomalies caught	True Negatives normals passed	False Positives false alarms	False Negatives anomalies missed

Confusion matrix counts · green = correct · red = error



TP / TN / FP / FN breakdown total samples = 20

CATEGORY	COUNT	DESCRIPTION
True Positive (TP)	4	Anomaly correctly detected
True Negative (TN)	16	Normal correctly passed
False Positive (FP)	0	Normal wrongly flagged as anomaly — false alarm
False Negative (FN)	0	Anomaly missed — most dangerous error

Performance metrics all scores 0–1

METRIC	SCORE	BAR	DESCRIPTION
Accuracy	1.0000		Overall correct predictions
Precision	1.0000		Of all anomaly predictions, how many were real
Recall (Sensitivity)	1.0000		Of all real anomalies, how many were caught
F1 Score	1.0000		Harmonic mean of precision and recall
Specificity	1.0000		Of all normals, how many correctly passed
False Positive Rate	0.0000		Normal files incorrectly flagged as anomaly
False Negative Rate	0.0000		Anomalies that were missed
MCC	1.0000		Matthews Correlation Coefficient (−1 to +1)
AUC-ROC	1.0000		Area under ROC curve

Per-class report precision · recall · F1

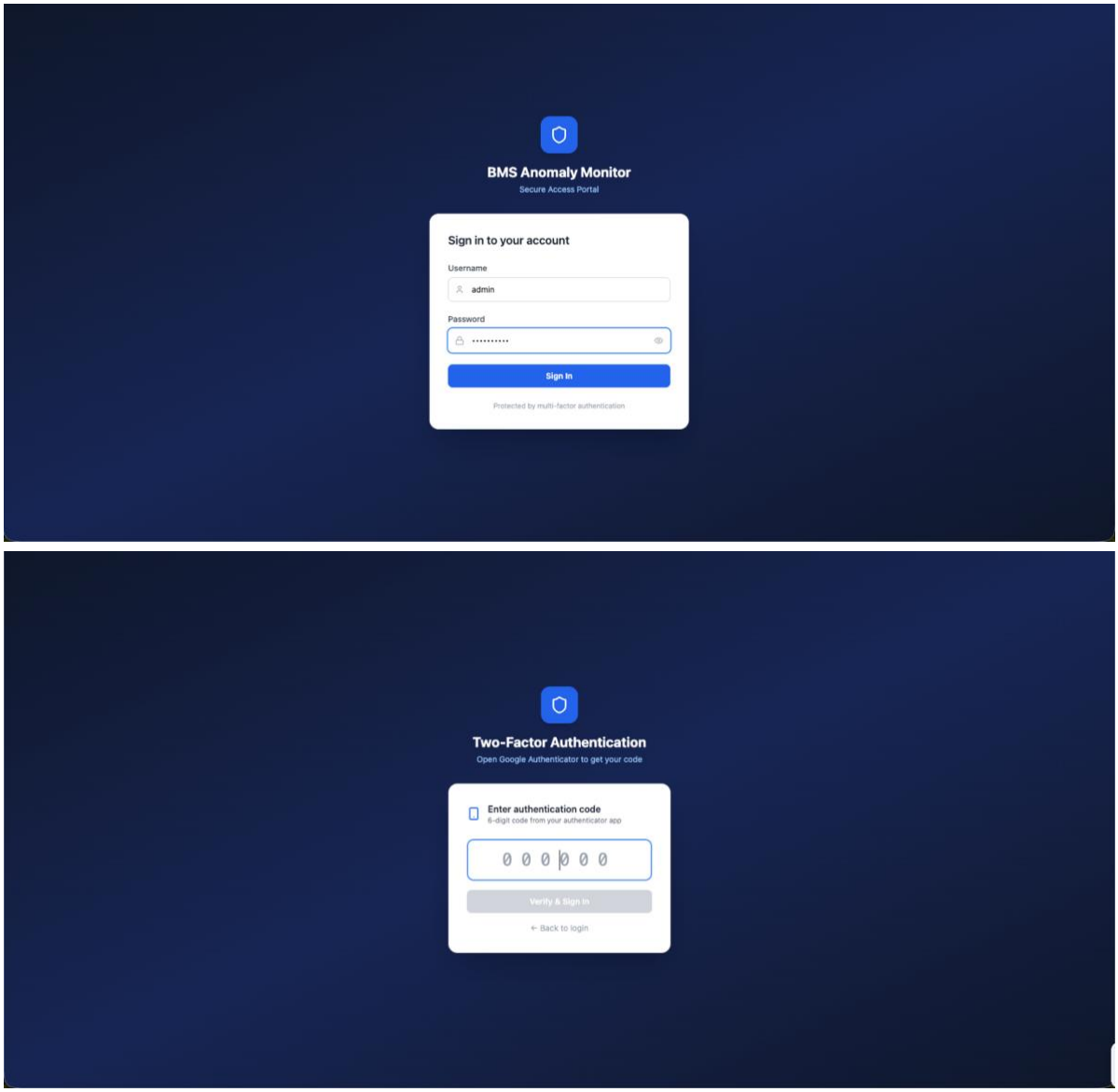
CLASS	PRECISION	RECALL	F1	SUPPORT
Normal	1.0000	1.0000	1.0000	16
Anomaly	1.0000	1.0000	1.0000	4

PROFESSOR'S NOTE — 20–80% correlation rule

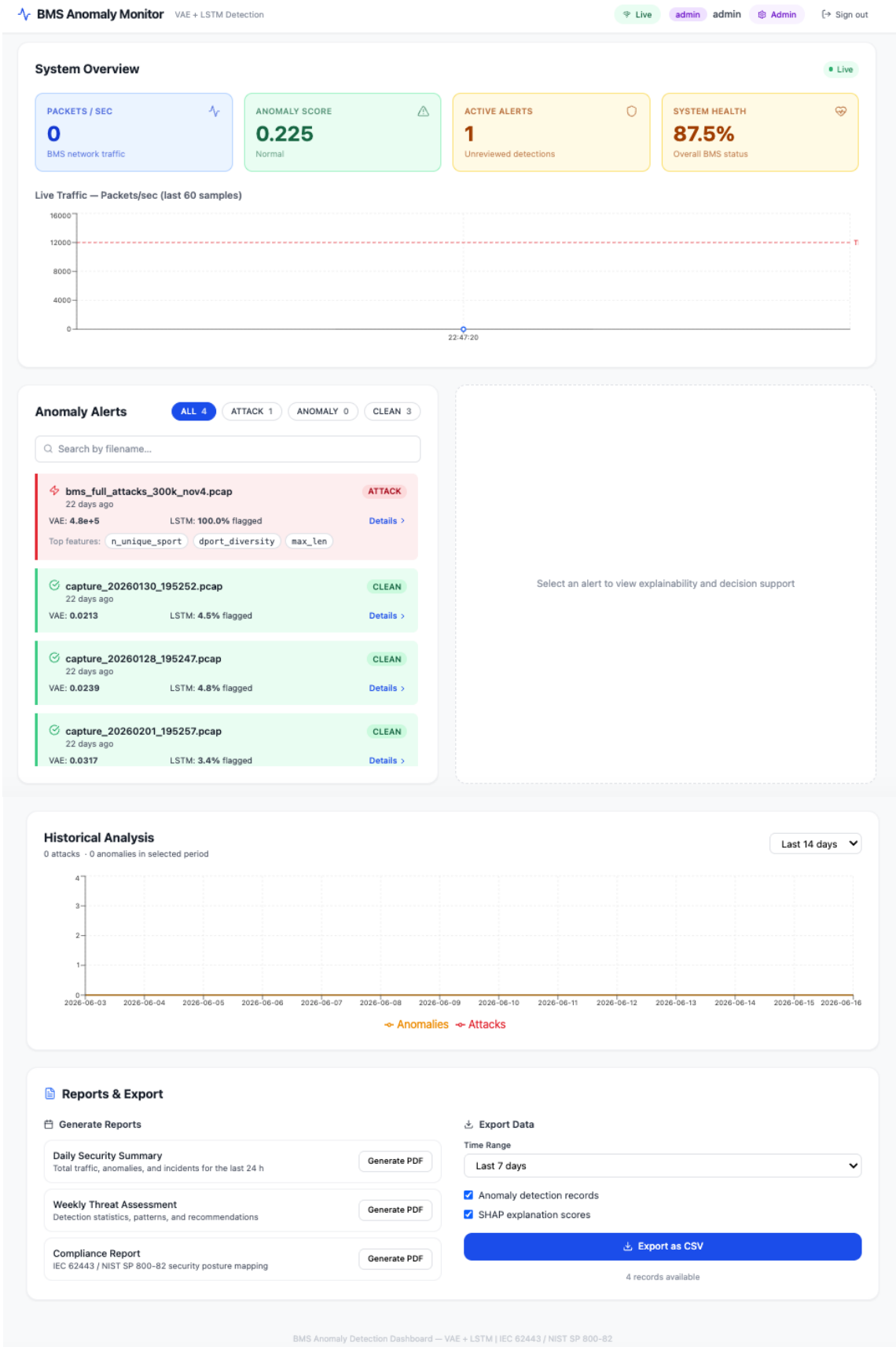
Features selected with **0.20 ≤ |r| ≤ 0.80**. Note: `std_len ↔ mean_len = 0.88` slightly exceeds this — retained because both independently contribute to anomaly detection (size variation vs average size).

D. Dashboard

Login page integrated with two-factor authentication



Dashboard overview



Attack Explainability

